

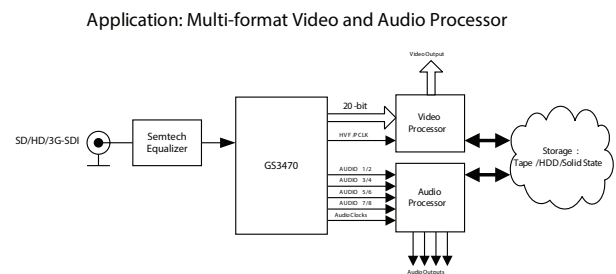
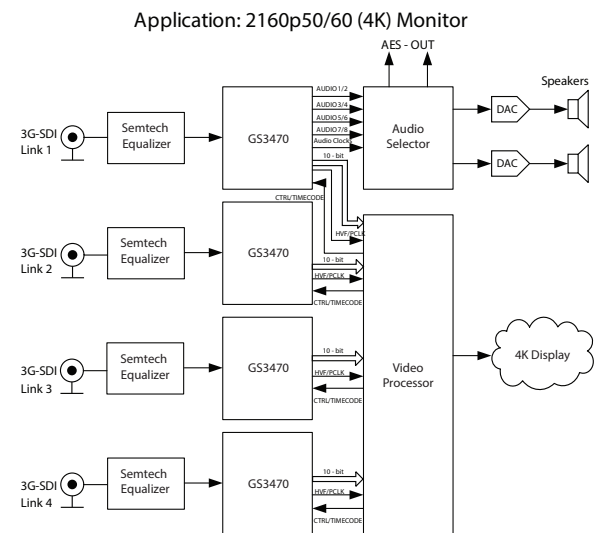
Key Features

- Operation at 2.970Gb/s, 2.970/1.001Gb/s, 1.485Gb/s, 1.485/1.001Gb/s, and 270Mb/s
- Supports SMPTE ST 425 (Level A and Level B), SMPTE ST 424, SMPTE 292, SMPTE ST 259-C, and DVB-ASI
- 2K and Multi-link UHD support
- Configurable Power-down modes
- Integrated Retimer
- Serial digital reclocked or non-reclocked loop-through output
- Integrated audio de-embedder for 8 channels of 48kHz audio and audio clock generation
- Ancillary data extraction
- Parallel data bus selectable as either 20-bit or 10-bit, SDR or DDR rate
- Comprehensive error detection and correction features
- Dual serial digital input buffer with 2x2 MUX
- Serial Loopback independently configurable to select either input
 - ♦ Performance optimized for 270Mb/s, 1.485Gb/s, and 2.97Gb/s.
- Dual/Quad Link 3G-SDI support with multiple GS3470 devices
- Output H, V, F, or CEA 861 timing signals
- GSPI host interface
- +1.2V digital core power supply, +1.2V and +1.8V analog power supplies, and selectable +1.8V or +2.5V I/O power supply
- -20°C to +85°C operating temperature range
- Low power operation — typically 220mW
- Small 9mm x 9mm 100-ball BGA package (0.80mm Ball Pitch)
- Pb-free/Halogen-free/RoHS & WEEE-compliant

Applications

SDI Interfaces for:

- Monitors
- DVRs
- Video Switchers
- Editing Systems
- Cameras
- Medical Imaging
- Aviation, Military, and Vehicular video systems
- LED Wall and Digital Signage Applications



The GS3470 is a multi-rate SDI Receiver which includes complete SMPTE processing. The SMPTE processing features can be bypassed to support signals with other coding schemes. Multi-link UHD can be supported when multiple GS3470 devices are used.

The device features a dual input buffer with a 2x2 MUX. The 2x2 MUX can select between either input for de-serialization and can route either of the two inputs to the serial loopback independently (relocked or non-relocked). In addition, the integrated Retimer with an internal VCO provides a wide Input Jitter Tolerance (IJT).

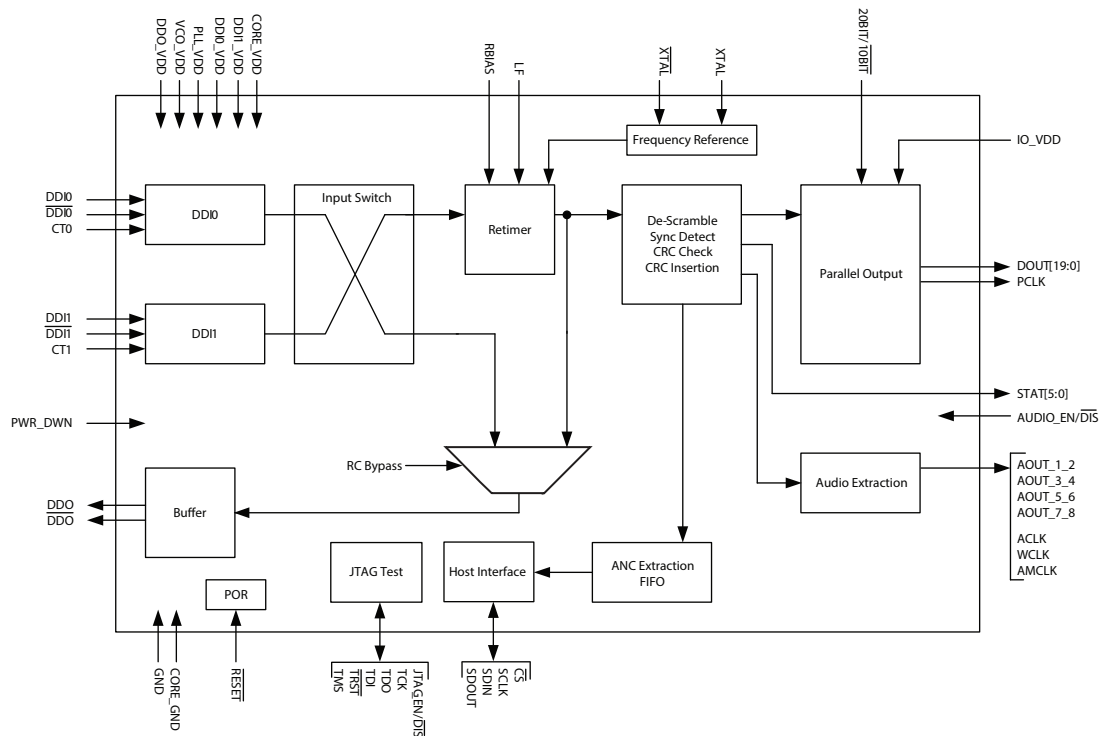
Configurable Power-down modes are available and allows for increased flexibility. Each Power-down mode enables power savings to a varying degree by selectively enabling or disabling key features. Some of the options available in

Power-down mode are CSR access, PCLK, retimed DDO loop-through output, and non-retimed DDO loop-through output. Enabling or disabling each of these options will offer power consumption levels to suit the application's requirements.

The device has three other basic modes of operation which include:

- SMPTE mode
- DVB-ASI mode
- Data-Through mode

The GS3470 includes an audio de-embedder and audio clocks are internally generated. Up to eight channels (two audio groups) of serial digital audio may be extracted from the video data stream, in accordance with SMPTE ST 272-C and SMPTE ST 299.



GS3470 Functional Block Diagram

Revision History

Version	ECO	Date	Changes and/or Modifications
10	047413	June 2019	Changes made to Ordering Information
9	043185	August 2018	Updated Table 1-1 , Table 2-3 , and Section 4 . Added Table 5-12 , and updated Section 5 .
8	038819	September 2017	Updated to latest corporate template.
7	038315	September 2017	Updated several register and parameter names throughout Section 4 . Updated Figure 4-1 , Figure 6-1 .
6	037007	May 2017	Updated Table 2-2 , Table 2-3 , Table 2-4 , Table 4-27 , Table 5-8 . Changed all instances of DBUS to DOUT, and VSS/VEE to A_GND.
5	035144	March 2017	Added Figure 4-14 through Figure 4-19 . Updated Section 4.16.1 . Updated Table 2-2 , Table 2-3 , Table 2-4 , Table 4-27 .
4	033598	October 2016	Updated data sheet to reflect GS3471 modifications.
3	029850	March 2016	Updates to 1.1 Pin Assignment and 2.2 Recommended Operating Conditions .
2	029341	February 2016	Initial release changes.
1	028179	October 2015	Initial release changes.
0	020778	July 2014	New document.

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1. Pin Out

1.1 Pin Assignment

Figure 1-1: Pin Assignment

	1	2	3	4	5	6	7	8	9	10
A	$\overline{\text{DDI0}}$	DDI0	CT0	RBIAS	XTAL	$\overline{\text{XTAL}}$	RSVD	PCLK	DOUT18	DOUT17
B	DDI0_VDD	DDI0_VDD	RSVD	RSVD	STAT0	STAT1	IO_VDD	DOUT19	DOUT16	DOUT15
C	PLL_VDD	PLL_VDD	LF	VCO_VDD	STAT2	STAT3	CORE_GND	DOUT12	DOUT14	DOUT13
D	DDI1_VDD	PLL_VDD	A_GND	VCO_VDD	STAT4	STAT5	$\overline{\text{TRST}}$	TDI	CORE_GND	IO_VDD
E	CT1	DDI1_VDD	A_GND	CORE_GND	CORE_VDD	CORE_VDD	TDO	TCK	DOUT10	DOUT11
F	DDI1	A_GND	A_GND	CORE_GND	CORE_GND	CORE_VDD	TMS	SDIN	DOUT8	DOUT9
G	$\overline{\text{DDI1}}$	A_GND	A_GND	CORE_GND	CORE_GND	CORE_VDD	SDOUT	SCLK	CORE_GND	IO_VDD
H	NC	NC	JTAG_EN/DIS	WCLK	$\overline{\text{RESET}}$	BIT20/BIT10	$\overline{\text{CS}}$	CORE_GND	DOUT6	DOUT7
J	DDO_VDD	DDO_VDD	PWR_DWN	AOUT_1_2	ACLK	AOUT_5_6	CORE_GND	DOUT1	DOUT4	DOUT5
K	$\overline{\text{DDO}}$	DDO	AUDIO_EN/DIS	AOUT_3_4	AMCLK	AOUT_7_8	IO_VDD	DOUT0	DOUT2	DOUT3

1.2 Pin Descriptions

Table 1-1: Pin Descriptions

Pin Number	Name	Type	Description
A1, A2 G1, F1	$\overline{\text{DDI0}}$, DDI0 $\overline{\text{DDI1}}$, DDI1	SDI Input	Serial digital differential input. It is possible to DC-couple to upstream Semtech devices supporting 1.2V outputs. Additionally, devices with 1.8 and 2.5V outputs are supported through a 4.7 μ F capacitor in series with the $\overline{\text{DDI}}$ /DDI input. Connect unused inputs to DDI_VDD through 1k Ω resistors.
A4	RBIAS	Analog Input	External resistor for the bias circuit. Connect to ground through 777 Ω resistor.
A5, A6	XTAL, $\overline{\text{XTAL}}$	Analog Input	Input connection for 27MHz crystal. When a reference clock input is used on $\overline{\text{XTAL}}$, do not connect XTAL.

Table 1-1: Pin Descriptions (Continued)

Pin Number	Name	Type	Description
A8	PCLK	Output	Parallel data bus clock. Please refer to the Output Logic parameters in Table 2-3: DC Electrical Characteristics for logic level threshold and compatibility. Please refer to Table 4-5: GS3470 Output Data Formats for PCLK output rates.
A7, B3, B4	RSVD	—	These pins are reserved, do not connect.
B7, D10, G10, K7	IO_VDD	Power	Power connection for digital I/O. Connect to 1.8V or 2.5V DC digital.
B8, A9, A10, B9, B10, C9, C10, C8, E10, E9, F10, F9, H10, H9, J10, J9, K10, K9, J8, K8	DOUT[19:0]	Output	Parallel data bus. Please refer to the Output Logic parameters in Table 2-3: DC Electrical Characteristics for logic level threshold and compatibility.
			20-bit mode 20BIT/10BIT = HIGH SMPTE mode ($\overline{\text{SMPTE_BYPASS}}$ = HIGH and DVB_ASI = LOW): DOUT[19:10] — Luma data output for SD and HD data rates; Data Stream 1 for 3G data rate DOUT[9:0] — Chroma data output for SD and HD data rates; Data Stream 2 for 3G data rate Data-Through mode ($\overline{\text{SMPTE_BYPASS}}$ = LOW and DVB_ASI = LOW): Data output
			10-bit mode 20BIT/10BIT = LOW (DOUT[19:10]) SMPTE mode ($\overline{\text{SMPTE_BYPASS}}$ = HIGH and DVB_ASI = LOW): Multiplexed Luma/Chroma data output for SD and HD data rates; Multiplexed Data Stream 1&2 for 3G data rate DVB-ASI mode ($\overline{\text{SMPTE_BYPASS}}$ = LOW and DVB_ASI = HIGH): 8/10bit decoded DVB-ASI data for SD data rates Data-Through mode ($\overline{\text{SMPTE_BYPASS}}$ = LOW and DVB_ASI = LOW): Data output Note 1: When in 10-bit mode, DOUT[9:0] are set to 0. Note 2: When in 10-bit mode, leave unused output pins unconnected.
C1, C2, D2	PLL_VDD	Power	Power pins for the Retimer PLL. Connect to 1.2V DC analog.
C3	LF	Analog Input	Loop Filter component connection. Connect as per Typical Application Circuit .

Table 1-1: Pin Descriptions (Continued)

Pin Number	Name	Type	Description
C4, D4	VCO_VDD	Power	Power pin for the VCO. Connect to RC filter as per Typical Application Circuit . Connect to a 1.2V $\pm 5\%$ analog supply through a 24.9 Ω resistor. Additionally, connect to ground through a 10 μ F capacitor.
C7, D9, E4, F4, F5, G4, G5, G9, J7, H8	CORE_GND	Power	Ground pins for digital circuitry. Connect to digital ground.
D3, E3, F2, F3, G2, G3	A_GND	Power	Ground pins for analog circuitry. Connect to analog ground.
D1, E2 B1, B2	DDI1_VDD DDIO_VDD	Power	Power pins for SDI buffer. Connect to 1.2V DC analog.
D6, D5, C6, C5, B6, B5	STAT[5:0]	Digital Output	Multi-function status outputs. See Section 4.11 for more details on assigning signals to STAT pins. Please refer to the Output Logic parameters in Table 2-3: DC Electrical Characteristics for logic level threshold and compatibility. Each of the STAT[5:0] pins can be configured individually to output one of the following signals. See Table 4-7: Output Signals Available on Programmable Multifunction Pins for Status Signal Selection Codes and Default Output Pins.
D7	$\overline{\text{TRST}}$	Digital Input, Internal Pull-down	JTAG interface reset. Digital active-low reset input. Used to reset the JTAG test sequence. When LOW, the JTAG test sequence is reset. When HIGH, normal operation of the JTAG test sequence resumes. When not in use, $\overline{\text{TRST}}$ can be left unconnected.
D8	TDI	Digital Input, Internal Pull-up	JTAG interface Test Data Input. Serial instructions and data are received on this pin. When not in use, TDI can be left unconnected.
E1, A3	CT[1:0]	Analog Input	Decoupling for internal SDI termination resistors. Connect as per Typical Application Circuit . When an input is not used, its corresponding CT pin can be left unconnected.
E5, E6, F6, G6	CORE_VDD	Power	Power connection for device core. Connect to 1.2V DC digital.
E7	TDO	Digital Output	JTAG interface Test Data Output. TDO is the serial output for test instructions and data.
E8	TCK	Digital Input	JTAG interface Test Clock input. The test clock input provides the clock for the test logic of this device.
F7	TMS	Digital Input, Internal Pull-up	JTAG interface Test Mode Select input. This signal is decoded by the internal TAP controller to control test operations. When not in use, TMS can be left unconnected.
F8	SDIN	Digital Input	Serial Digital Data Input for the Gennum Serial Peripheral Interface (GSPI) host control/status port. When GSPI is not used, SDIN should be tied HIGH or LOW to minimize noise.
G7	SDOUT	Digital Output	Serial Digital Data Output for the Gennum Serial Peripheral Interface (GSPI) host control/status port. Active-high output. When GSPI is not used, leave unconnected.

Table 1-1: Pin Descriptions (Continued)

Pin Number	Name	Type	Description
G8	SCLK	Digital Input	Serial Data Clock input. Burst-mode clock input for the Gennum Serial Peripheral Interface (GSPI) host control/status port. When GSPI is not used, SCLK should be tied HIGH or LOW to minimize noise.
H1, H2	NC	—	No connect. Pins are not connected internally.
H3	JTAG_EN/ $\overline{\text{DIS}}$	Digital Input, Internal Pull-down	JTAG enable. Digital active-high to enable JTAG communications. When HIGH, JTAG operational mode is enabled. When LOW, JTAG operational mode is disabled.
H4	WCLK	Output	48kHz word clock for audio. When not used, leave unconnected.
H5	$\overline{\text{RESET}}$	Digital Input, Internal Pull-up	Device reset signal. When LOW, the device will be set to default conditions.
H6	BIT20/ $\overline{\text{BIT10}}$	Digital Input, Internal Pull-up	Control signal input. Used to select the output bus width. HIGH = 20-bit, LOW = 10-bit. Please refer to the Input Logic parameters in Table 2-3: DC Electrical Characteristics for logic level threshold and compatibility.
H7	$\overline{\text{CS}}$	Digital Input	Chip Select input for the Gennum Serial Peripheral Interface (GSPI) host control/status port. Active-low input. When GSPI is not used, connect $\overline{\text{CS}}$ to IO_VDD.
J1, J2	DDO_VDD	Power	Power pin for the serial digital output 50Ω buffer. Connect to 1.2V or 1.8V DC analog.
J3	PWR_DWN	Digital Input, Internal Pull-down	When HIGH, places the device in a power-down state.
J4, K4, J6, K6	AOUT_1_2, AOUT_3_4, AOUT_5_6, AOUT_7_8	Output	Serial Audio Outputs. When not in use, leave unconnected.
J5	ACLK	Output	64fs sample clock for audio. When not in use, leave unconnected.
K1, K2	$\overline{\text{DDO}}$, DDO	Digital Output	Differential serial digital outputs. When not in use, leave unconnected.
K3	AUDIO_EN/ $\overline{\text{DIS}}$	Digital Input, Internal Pull-up	Control signal input. When HIGH, enables audio extraction. When LOW, disables audio extraction. Please refer to the Input Logic parameters in Table 2-3: DC Electrical Characteristics for logic level threshold and compatibility.
K5	AMCLK	Output	Oversampled master clock for audio (128fs, 256fs, 512fs selectable). When not in use, leave unconnected.

2. Electrical Characteristics

2.1 Absolute Maximum Ratings

Table 2-1: Absolute Maximum Ratings

Parameter	Value
Supply Voltage, Digital Core (CORE_VDD)	-0.3V to +1.5V
Supply Voltage, Digital I/O (IO_VDD)	-0.3V to +2.8V
Supply Voltage, Analog 1.2V (PLL_VDD, VCO_VDD, DDI_VDD)	-0.3V to +1.5V
Supply Voltage, Analog 1.8V (DDO_VDD)	-0.3V to +2.0V
Input Voltage Range (Digital Inputs)	-0.3V to IO_VDD + 0.3V
Ambient Operating Temperature (T _A)	-20°C to +85°C
Storage Temperature (T _{STG})	-50°C to +125°C
Peak Reflow Temperature (JEDEC J-STD-020C)	260°C
ESD Sensitivity, HBM (JESD22-A114)	3kV

Note: Absolute Maximum Ratings are those values beyond which damage may occur. Functional operation under these conditions or at any other condition beyond those indicated in the AC/DC Electrical Characteristics sections is not implied.

2.2 Recommended Operating Conditions

Table 2-2: Recommended Operating Conditions

Parameter	Symbol	Conditions	Min	Typ	Max	Units
Operating Temperature Range, Ambient	T _A	—	-20	—	+85	°C
Supply Voltage, Digital Core	CORE_VDD	—	1.14	1.2	1.26	V
Supply Voltage, Digital I/O	IO_VDD	1.8V mode	1.71	1.8	1.89	V
		2.5V mode	2.38	2.5	2.63	V
Supply Voltage, PLL	PLL_VDD	—	1.14	1.2	1.26	V
Supply Voltage, VCO	VCO_VDD	—	1.14	1.2	1.26	V
Supply Voltage, Serial Digital Input	DDI0_VDD, DDI1_VDD	—	1.14	1.2	1.26	V
Supply Voltage, CD Buffer	DDO_VDD	1.2V mode	1.14	1.2	1.26	V
		1.8V mode	1.71	1.8	1.89	V
Serial Input Data Rate	—	—	270	—	2970	Mb/s

2.3 DC Electrical Characteristics

Table 2-3: DC Electrical Characteristics

Guaranteed over recommended operating conditions unless otherwise noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
System							
DDI0_VDD, DDI1_VDD Supply Current	I _{DDI}	1.2V	0.02	0.07	0.12	mA	—
IO_VDD Supply Current	I _{IO}	1.8V	2.52	10.31	21.53	mA	—
		2.5V	6.62	21.7	44.86	mA	—
DDO_VDD Supply Current	I _{DDO}	1.2V	8.21	8.92	9.71	mA	—
		1.8V	8.30	9.05	9.85	mA	—
PLL_VDD Supply Current	I _{PLL}	1.2V	59.8	67.8	74.4	mA	—
CORE_VDD Supply Current	I _{CORE}	1.2V	17.7	20.3	22.2	mA	—
Total Device Power DDO_VDD = 1.2V IO_VDD = 1.8V (Audio Enabled)	P	10-bit 3GA	160	191	227	mW	—
		10-bit 3GB	135	158	192	mW	—
		20-bit 3GA	137	161	193	mW	—
		20-bit 3GB	154	184	230	mW	—
		10-bit HD	125	148	179	mW	—
		20-bit HD	109	129	170	mW	—
		10/20-bit SD	97	109	141	mW	—
		DVB-ASI	—	103	—	mW	—
		Sleep	3.3	5	11.3	mW	—
		Standby with DDO Retimed	82	105	121	mW	—
Total Device Power DDO_VDD = 1.8V IO_VDD = 2.5V (Audio Enabled)	P	10-bit 3GA	275	287	300	mW	—
		10-bit 3GB	224	230	237	mW	—
		20-bit 3GA	219	228	241	mW	—
		20-bit 3GB	276	286	325	mW	—
		10-bit HD	211	218	221	mW	—
		20-bit HD	165	174	209	mW	—
		10/20-bit SD	125	132	164	mW	—
		DVB-ASI	—	121	—	mW	—
		Sleep	7.1	10.1	16.9	mW	—
		Standby with DDO Retimed	108	127	163	mW	—

Table 2-3: DC Electrical Characteristics (Continued)

Guaranteed over recommended operating conditions unless otherwise noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
Digital I/O							
Input Logic LOW	V _{IL}	2.5V or 1.8V operation	—	—	0.3 x IO_VDD	V	—
Input Logic HIGH	V _{IH}	2.5V or 1.8V operation	0.7 x IO_VDD	—	—	V	—
Output Logic LOW	V _{OL}	I _{OL} = 8mA, 1.8V operation	—	—	0.41	V	—
		I _{OL} = 8mA, 2.5V operation	—	—	0.29	V	—
Output Logic HIGH	V _{OH}	I _{OL} = 8mA, 1.8V operation	1.49	—	—	V	—
		I _{OL} = 8mA, 2.5V operation	2.27	—	—	V	—
Serial Input							
Serial Input Common Mode Voltage	—	AC or DC-coupled	0.90	0.96	1.06	V	—
Serial Output							
Serial Output Common Mode Voltage	—	50Ω load	—	DDO_VDD - V _{swing} /2	—	V	1

Note:

1. Serial output swing limited when using $DDO_VDD = 1.2V$.

2.4 AC Electrical Characteristics

Table 2-4: AC Electrical Characteristics

Guaranteed over recommended operating conditions unless otherwise noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
System							
Device Latency: AUDIO_EN = 1, SMPTE mode, IOPROC_EN = 1	—	3G (Level A)	65	67	69	PCLK	—
		3G (Level B)	141	144	147	PCLK	—
		HD	65	67	69	PCLK	—
		SD	37	39	41	PCLK	—
Device Latency: AUDIO_EN = 0, SMPTE mode, IOPROC_EN = 1	—	3G (Level A)	28	30	32	PCLK	—
		3G (Level B)	63	65	67	PCLK	—
		HD	25	27	29	PCLK	—
		SD	25	27	29	PCLK	—
Device Latency: AUDIO_EN = 0, SMPTE mode, IOPROC_EN = 0	—	3G (Level A)	20	22	24	PCLK	—
		3G (Level B)	47	50	53	PCLK	—
		HD	21	23	25	PCLK	—
		SD	19	21	23	PCLK	—
Device Latency: AUDIO_EN = 0, SMPTE bypass, IOPROC_EN = 0	—	3G (Level A)	11	13	15	PCLK	—
		3G (Level B)	11	13	15	PCLK	—
		HD	11	13	15	PCLK	—
		SD	11	13	15	PCLK	—
Device Latency: DVB-ASI	—	—	12	14	16	PCLK	—
Reset Time	t _{reset}	—	1	—	—	ms	—
Parallel Output							
Parallel Clock Frequency	f _{PCLK}	3G/HD (10-bit)	—	148.5 or 148.5/ 1.001	—	MHz	—
		HD (20-bit), 10-bit DDR	—	74.25 or 74.25/ 1.001	—	MHz	—
		SD (20-bit), 10-bit DDR	—	13.5	—	MHz	—
		SD (10-bit)	—	27	—	MHz	—

Table 2-4: AC Electrical Characteristics (Continued)

Guaranteed over recommended operating conditions unless otherwise noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
Parallel Clock Duty Cycle	DC _{PCLK}	—	—	50	—	%	—
Output Data Hold Time (1.8V)	6pF C _{load}	SPI	1.5	—	—	ns	—
		AUDIO	1.5	—	—	ns	—
	3G 10-bit 6pF C _{load}	DOUT	0.3	—	—	ns	—
		STAT	0.3	—	—	ns	—
	3G 20-bit 6pF C _{load}	DOUT	0.5	—	—	ns	—
		STAT	0.5	—	—	ns	—
	HD 10-bit 6pF C _{load}	DOUT	1.5	—	—	ns	—
		STAT	1.5	—	—	ns	—
	HD 20-bit 6pF C _{load}	DOUT	5.0	—	—	ns	—
		STAT	5.0	—	—	ns	—
	SD 10-bit 6pF C _{load}	DOUT	15.0	—	—	ns	—
		STAT	15.0	—	—	ns	—
	SD 20-bit 6pF C _{load}	DOUT	30.0	—	—	ns	—
		STAT	30.0	—	—	ns	—
Output Data Hold Time (2.5V)	6pF C _{load}	SPI	1.5	—	—	ns	—
		AUDIO	1.5	—	—	ns	—
	3G 10-bit 6pF C _{load}	DOUT	0.3	—	—	ns	—
		STAT	0.3	—	—	ns	—
	3G 20-bit 6pF C _{load}	DOUT	0.5	—	—	ns	—
		STAT	0.5	—	—	ns	—
	HD 10-bit 6pF C _{load}	DOUT	1.5	—	—	ns	—
		STAT	1.5	—	—	ns	—
	HD 20-bit 6pF C _{load}	DOUT	4.0	—	—	ns	—
		STAT	4.0	—	—	ns	—
	SD 10-bit 6pF C _{load}	DOUT	15.0	—	—	ns	—
		STAT	15.0	—	—	ns	—
	SD 20-bit 6pF C _{load}	DOUT	30.0	—	—	ns	—
		STAT	30.0	—	—	ns	—

Table 2-4: AC Electrical Characteristics (Continued)

Guaranteed over recommended operating conditions unless otherwise noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
Output Data Delay Time (1.8V)	t_{od}	15pF C_{load}	SPI	—	28.0	ns	—
			AUDIO	—	10.0	ns	—
		3G 10-bit 15pF C_{load}	DOUT	—	2.4	ns	—
			STAT	—	2.8	ns	—
		3G 20-bit 15pF C_{load}	DOUT	—	6.0	ns	—
			STAT	—	6.3	ns	—
		HD 10-bit 15pF C_{load}	DOUT	—	4.0	ns	—
			STAT	—	4.2	ns	—
		HD 20-bit 15pF C_{load}	DOUT	—	14.2	ns	—
			STAT	—	14.4	ns	—
		SD 10-bit 15pF C_{load}	DOUT	—	21.0	ns	—
			STAT	—	21.0	ns	—
		SD 20-bit 15pF C_{load}	DOUT	—	40.0	ns	—
			STAT	—	40.0	ns	—
Output Data Delay Time (2.5V)	t_{od}	15pF C_{load}	SPI	—	28.0	ns	—
			AUDIO	—	10.0	ns	—
		3G 10-bit 15pF C_{load}	DOUT	—	2.3	ns	—
			STAT	—	2.8	ns	—
		3G 20-bit 15pF C_{load}	DOUT	—	6.0	ns	—
			STAT	—	6.3	ns	—
		HD 10-bit 15pF C_{load}	DOUT	—	3.8	ns	—
			STAT	—	4.2	ns	—
		HD 20-bit 15pF C_{load}	DOUT	—	13.0	ns	—
			STAT	—	13.5	ns	—
		SD 10-bit 15pF C_{load}	DOUT	—	21.0	ns	—
			STAT	—	21.0	ns	—
		SD 20-bit 15pF C_{load}	DOUT	—	40.0	ns	—
			STAT	—	40.0	ns	—
Output Data Rise/Fall Time (1.8V)	t_r/t_f	6pF C_{load}	STAT	—	3.1	ns	—
			DOUT	—	3.1	ns	—
			AUDIO	—	3.3	ns	—

Table 2-4: AC Electrical Characteristics (Continued)

Guaranteed over recommended operating conditions unless otherwise noted.

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes	
Output Data Rise/Fall Time (2.5V)	t_r/t_f	6pF C_{load}	STAT	—	—	2.1	ns	—
			DOUT	—	—	2.1	ns	—
			AUDIO	—	—	2.2	ns	—
Serial Digital Input								
Serial Input Data Rate	DR_{SDI}	—	0.27	—	2.97	Gb/s	—	
Serial Input Swing	ΔV_{DDI}	Differential with 100Ω load	200	400	1000	mV _{ppd}	—	
Serial Input Jitter Tolerance	SIJT	Nominal loop bandwidth Square wave mod.	0.8	—	—	UI	—	
Serial Digital Output								
Serial Output Data Rate	DR_{DDO}	—	0.27	—	2.97	Gb/s	—	
Serial Output Swing	ΔV_{DDO}	Differential with 100Ω load	—	400	—	mV _{ppd}	2, 4	
Serial Output Rise Time 20% ~ 80%	t_{rDDO}	—	—	112	135	ps	—	
Serial Output Fall Time 20% ~ 80%	t_{fDDO}	—	—	114	135	ps	—	
Rise/ Fall Mismatch		—	—	2	8	ps	—	
Serial Output Intrinsic Jitter	t_{OJ}	3G PRBS	0.05	0.06	0.08	UI	3	
		HD PRBS	0.03	0.04	0.05	UI	3	
		SD PRBS	0.01	0.02	0.03	UI	3	
Serial Output Duty Cycle Distortion	DCD_{SDD}	3G	3	5	10	ps	—	
		HD	1	5	7	ps	—	
		SD	1	2	5	ps	—	
Asynchronous Lock Time	—	—	—	—	750	μs	—	
Lock Time from Power-up	—	After 20 minutes at -20°C	—	725	—	ms	—	

Notes:

1. Serial output swing limited when using DDO_VDD = 1.2V
2. Serial output swing can be adjusted through GSPI.
3. Retiming enabled.
4. Default setting and typical condition.

3. Input/Output Circuits

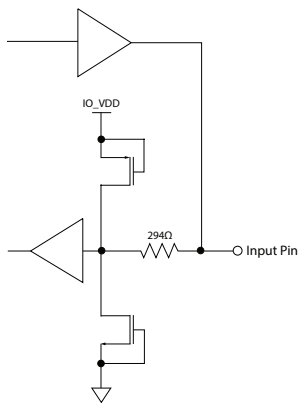


Figure 3-1: Bidirectional Digital Input/Output Pin Configured as an Input (SDIN, CS, SCLK)

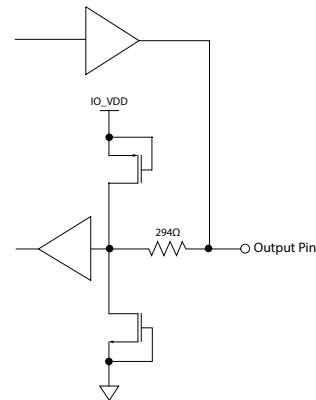


Figure 3-2: Bidirectional Digital Input/Output Pin Configured as an Output (AMCLK, TDO, SDOUT, WCLK, AOUT_1_2, AOUT_3_4, AOUT_5_6, AOUT_7_8, ACLK)

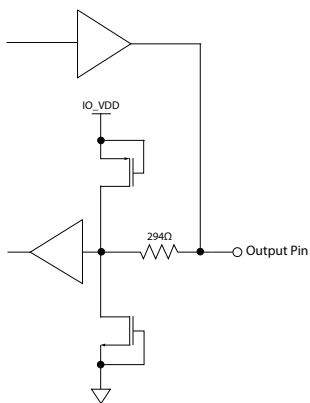


Figure 3-3: Bidirectional Digital Input/Output Pin Configured as an Output with Programmable Drive Strength (DOUT[19:0], PCLK, STAT[5:0])

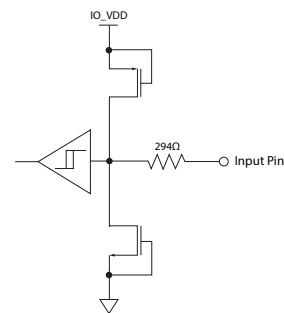


Figure 3-4: Digital Input with Schmitt Trigger and 100kΩ Internal Pull-Up (AUDIO_EN/DIS, TDI, TMS, RESET, BIT20/BIT10)

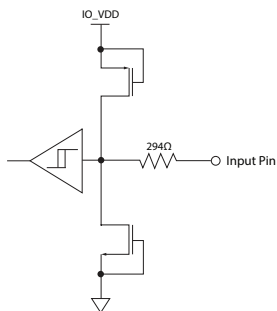


Figure 3-5: Digital Input with Schmitt Trigger and 100kΩ Internal Pull-Down (TRST, JTAG_EN/DIS, PWR_DWN)

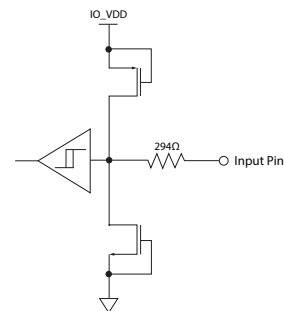


Figure 3-6: Digital Input with Schmitt Trigger (TCK)

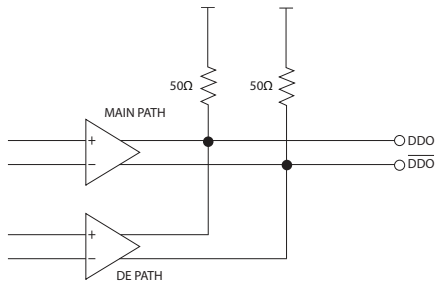


Figure 3-7: DDO/DDO

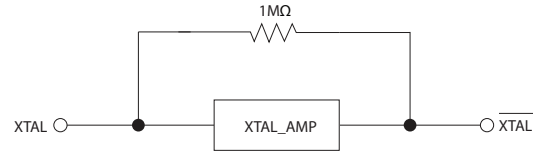


Figure 3-8: XTAL/XTAL

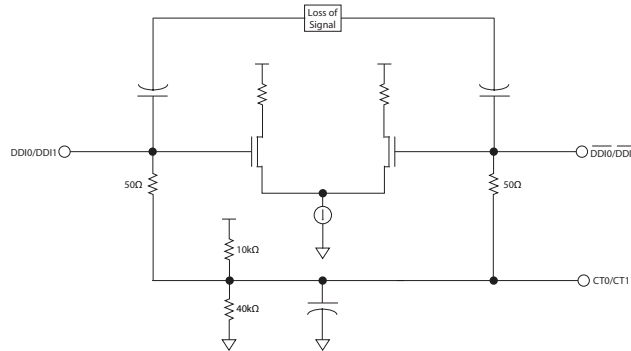


Figure 3-9: DDI0/DDI0, DDI1/DDI1, CT[1:0]

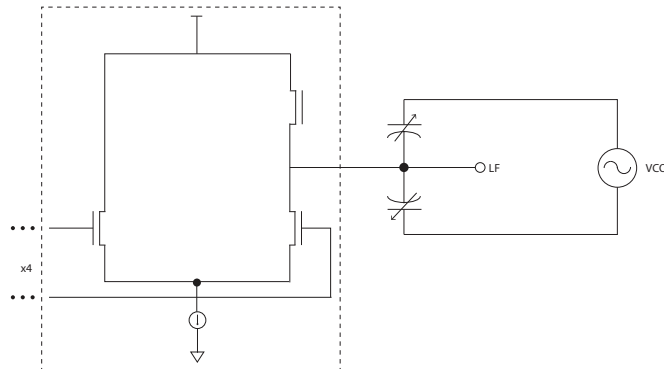


Figure 3-10: LF

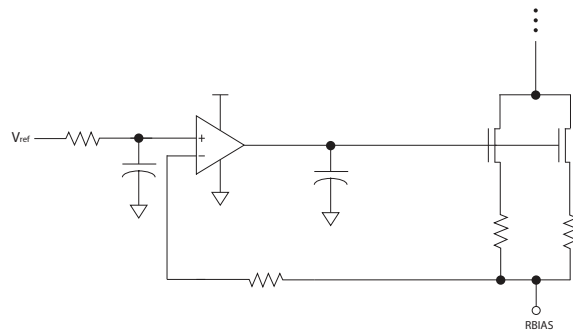


Figure 3-11: RBIAS

4. Detailed Description

4.1 Functional Overview

The GS3470 includes a dual serial digital input buffer with 2x2 MUX, an integrated retimer, serial data loop through output, robust serial-to-parallel conversion, integrated SMPTE video processing, and additional processing functions such as audio extraction, ancillary data extraction, EDH support, and DVB-ASI decoding.

The serial digital input buffer with 2x2 MUX offers a lot of flexibility for use in default and various Power-down modes.

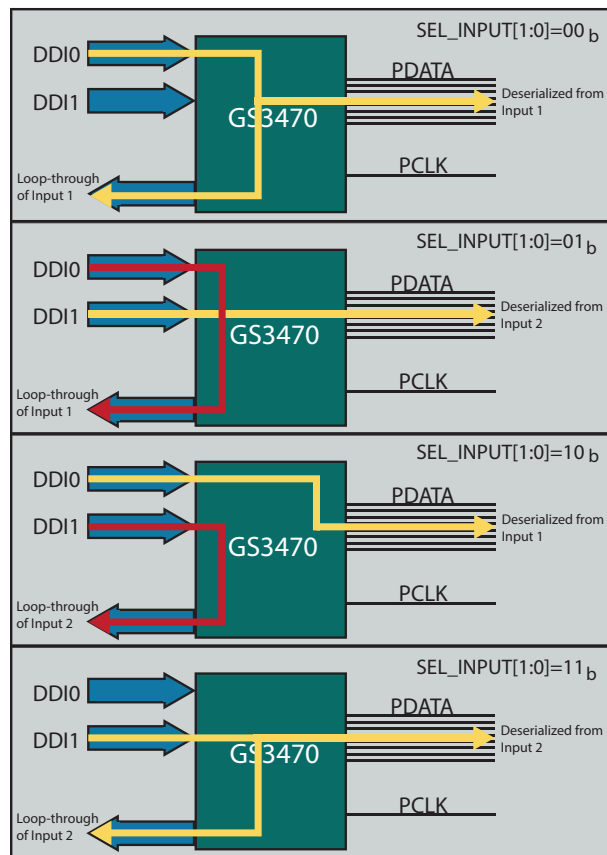


Figure 4-1: Flexible Input Loopback

Expanded and configurable Power-down modes offer increased flexibility by selectively enabling or disabling key features (such as CSR access, PCLK, retimed DDO loop-through output, and non-retimed DDO loop-through output). [Figure 4-2](#) show the various Power-down modes.

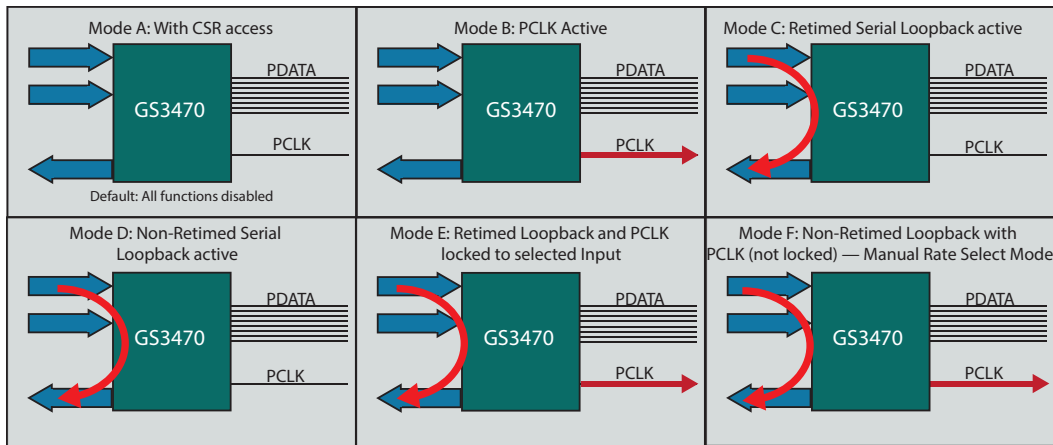


Figure 4-2: Flexible Power Down Modes

The device has three other primary modes of operation which include SMPTE mode, DVB-ASI mode, and Data-Through mode. In SMPTE mode, when receiving a SMPTE compliant SDI input, the GS3470 performs full SMPTE processing, and features a number of data integrity checks and measurement capabilities. The device also supports ancillary data extraction, and can provide up to 1024 x 16-bit ancillary data words through host-accessible registers. Packet detection and error handling features are also offered. All processing features are optional, and may be individually enabled or disabled through register programming. In DVB-ASI mode, sync word detection, alignment, and 8/10bit decoding is applied to the received data stream. While in Data-Through mode, all forms of SMPTE and DVB-ASI processing are disabled, and the device can be used as a simple serial to parallel converter.

The GS3470 includes an audio de-embedder and audio clocks are internally generated. Up to eight channels (two audio groups) of serial digital audio may be extracted from the video data stream, in accordance with SMPTE ST 272-C and SMPTE ST 299. The output audio formats supported by the device include AES/EBU and I²S. A variety of audio processing features are provided to ease implementation.

4.2 Device Power-Up

The GS3470 is designed to operate in a multi-voltage environment which allows any power-up sequence to be used. Supply pins can all be powered up in any order.

4.2.1 Power-Down Mode

The *PWR_DWN* pin reduces power to a minimum by disabling various device features. When the *PWR_DWN* pin is de-asserted, the device returns to its previous operating condition within 1 second, without requiring input from the host interface. There are several power-down options which can be configured through GSPI prior to the device going into power-down. Table 4-1 provides a summary of the supported power-down options by accessing the **POWER_DOWN** register.

Table 4-1: Power-down Mode

Power-down Mode	CSR Access	DDO Loop-through Mode	PCLK Mode
Power-down PD_PCLK_ENABLE = 0 SERIAL_LOOPBACK_EN = 0 PD_CSR_ACCESS = 0 RC_BYP = X	No	DDO Disabled	PCLK Disabled
Power-down with CSR Access PD_PCLK_ENABLE = 0 SERIAL_LOOPBACK_EN = 0 PD_CSR_ACCESS = 1 RC_BYP = X	Yes	DDO Disabled	PCLK Disabled
Power-down with PCLK PD_PCLK_ENABLE = 1 SERIAL_LOOPBACK_EN = 0 PD_CSR_ACCESS = X RC_BYP = X	Yes	DDO Disabled	PCLK Enabled
Power-down with DDO PD_PCLK_ENABLE = 0 SERIAL_LOOPBACK_EN = 1 PD_CSR_ACCESS = X RC_BYP = 1	No	DDO Enabled Non-retimed	PCLK Disabled
Power-down with DDO retimed PD_PCLK_ENABLE = 0 SERIAL_LOOPBACK_EN = 1 PD_CSR_ACCESS = X RC_BYP = 0	Yes	DDO Enabled Retimed	PCLK Disabled

Table 4-1: Power-down Mode (Continued)

Power-down Mode	CSR Access	DDO Loop-through Mode	PCLK Mode
Power-down with DDO/PCLK			
PD_PCLK_ENABLE = 1 SERIAL_LOOPBACK_EN = 1 PD_CSR_ACCESS = X RC_BYP = 1	Yes	DDO Enabled Non-retimed	PCLK Enabled
Power-down with DDO/PCLK retimed			
PD_PCLK_ENABLE = 1 SERIAL_LOOPBACK_EN = 1 PD_CSR_ACCESS = X RC_BYP = 0	Yes	DDO Enabled Retimed	PCLK Enabled

Table 4-2: Status Output Support in Power Down Modes

Mode	Rate Detect	Carrier Detect	Lock	All Other Status Outputs
Power-down	N/A	N/A	N/A	N/A
Power-down with DDO	N/A	N/A	N/A	N/A
Power-down with DDO retimed	Available in automatic or manual modes	Analog detect only	Locked status available on STAT outputs	N/A
Power-down with PCLK	Available in manual mode only, rate must be set	Analog detect only	N/A	N/A
Power-down with DDO/PCLK retimed	Available in automatic or manual modes	Analog detect only	Locked status available on STAT outputs	N/A
Power-down with DDO/PCLK	Available in manual mode only, rate must be set	Analog detect only	N/A	N/A
Power-down with CSR access	N/A	Analog detect only	N/A	N/A

4.2.2 Device Reset

Note: On power-up, the device must be reset to operate correctly.

In order to initialize all internal operating conditions to their default states, hold the $\overline{RESET}$ signal LOW for a minimum of $t_{reset} = 1\text{ ms}$ after all power supplies are stable. There are no requirements for power supply sequencing.

When held in reset, all device outputs are driven to a high-impedance state, with the exception of $SDOUT$. $SDOUT$ continues normal operation during reset.

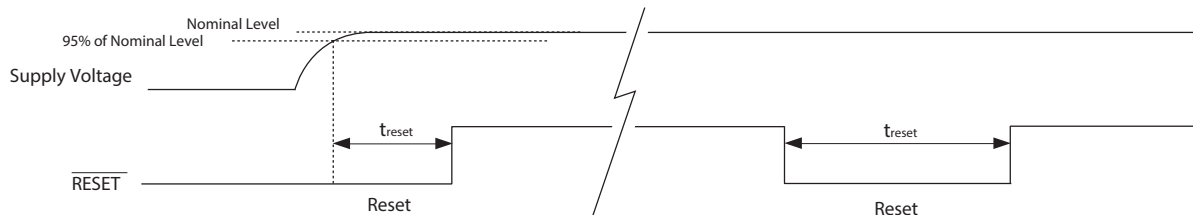


Figure 4-3: Reset Pulse

4.3 Modes of Operation

4.3.1 Auto and Manual Mode

The lock detection algorithm is a continuous process, beginning at device power-up or after a device reset. It continues until the device is powered down or held in reset.

The device first determines if a valid serial digital input signal has been presented to the device. If no valid serial data stream has been detected, the serial data into the device is considered invalid, and the LOCKED signal is LOW.

Once a valid input signal has been detected, the device attempts to detect the presence of either TRS words or DVB-ASI sync words.

By default, the device powers up in Auto mode (**AUTO_** $\overline{\text{MAN}}$ (Address 835_h, bit 2) in the host interface is set HIGH). In this mode, the device operating frequency toggles between 3G, HD, and SD rates as it attempts to lock to the incoming data rate. As it searches through rates, PCLK output cycles through 148.5MHz, 74.25MHz, 27MHz, and 13.5MHz. The PCLK output pin can be set to be high-impedance when not locked through GSPI using **CSR_PCLK_HIZ** (Address 878_h, bit 9).

When the device is operating in Manual mode (**AUTO_** $\overline{\text{MAN}}$ bit in the host interface register is LOW), the operating data rate needs to be set through **RATE_SEL_TOP** (Address 835_h, bit[1:0]) where **RATE_SEL_TOP[0]** = **SD/** $\overline{\text{HD}}$ and **RATE_SEL_TOP[1]** = **3G/** $\overline{\text{HD}}$.

Note: The **SD/** $\overline{\text{HD}}$ bit takes precedence over the **3G/** $\overline{\text{HD}}$ bit, so if the **SD/** $\overline{\text{HD}}$ bit is HIGH, the **3G/** $\overline{\text{HD}}$ bit is ignored.

4.3.2 Low Latency Video Path

The GS3470 has a low latency mode of operation for audio and ancillary data extraction.

Audio can be extracted without incurring any associated delay if the error correction feature and audio packet delete feature are not required. The device will automatically select low latency mode if **ALL_DEL** (Address B01_h, bit 13) is set LOW (SD) or **ALL_DEL** (Address A02_h, bit 2) is set LOW and **ECC_OFF** (Address A02_h, bit 3) is set HIGH (HD/3G). This means that in low latency mode for audio, ECC errors in the HD/3G audio data packets will not be corrected and no audio packets will be deleted from the data stream after extraction. If either of these features are desired, then a delay will be incurred through the audio extraction blocks. To maintain consistent delay independent of selected features, **LOW_LATENCY_BYPASS** (Address 800_h, bit 14) must be set HIGH.

Ancillary data will automatically be extracted without incurring any associated delay if **ANC_DATA_DEL** (Address 817_h, bit 2) is set LOW.

4.3.3 SMPTE and SMPTE Bypass Mode

The GS3470 has the ability to run either in SMPTE mode or SMPTE Bypass mode.

In SMPTE mode (**SMPTE_BYPASS** (Address 835_h, bit 4) = HIGH), the timing signal generator becomes operational, video signals, error detection and SMPTE processing functions are available, and the retimer PLL locks to valid SMPTE video.

In SMPTE Bypass mode (**SMPTE_BYPASS** = LOW), the GS3470 operates either in DVB-ASI mode or Data-Through mode. When operating in SMPTE Bypass mode, none of the SMPTE detection and processing functions are available.

4.3.3.1 Descrambling and Word Alignment

The GS3470 performs NRZI (Non Return to Zero Invert) to NRZ (Non Return to Zero) decoding and data descrambling according to SMPTE ST 424/SMPTE ST 292/SMPTE ST 259-C and word aligns the data to TRS sync words.

When operating in SMPTE mode (**SMPTE_BYPASS** = HIGH and **DVB_ASI** (Address 835_h, bit 3) = LOW), the GS3470 carries out descrambling and word alignment to enable the detection of TRS sync words. When two consecutive valid TRS words (SAV and EAV), with the same bit alignment have been detected, the device word-aligns the data to the TRS ID words.

TRS ID word detection is a continuous process. The device remains in SMPTE mode until TRS ID words fail to be detected.

Note 1: Both 8-bit and 10-bit TRS headers are identified by the device.

Note 2: In 3G Level B mode, the device only supports Data Stream 1 and Data Stream 2 having the same bit width (i.e. both data streams contain 8-bit data, or both data streams contain 10-bit data). If the bit widths between the two data streams are different, the GS3470 cannot word align the input stream. When **SMPTE_BYPASS** is HIGH and the device is set to Auto mode, it will continuously try to lock.

4.3.4 DVB-ASI Mode

When in DVB-ASI mode ($\overline{\text{SMPTE_BYPASS}}$ = LOW and **DVB_ASI** = HIGH), the retimer PLL locks to a DVB-ASI stream. In DVB-ASI mode, the parallel outputs are configured appropriately as described in [4.9.3 Parallel Output in DVB-ASI Mode](#). None of the SMPTE detection and processing functions are available in this mode.

4.4 Digital Differential Input (DDI/ $\overline{\text{DDI}}$)

The GS3470 contains a 100 Ω differential input buffer which can be DC-coupled to Semtech equalizers, but only if equalizer output stage is connected to 1.2V. Otherwise must be AC-coupled.

The GS3470 can accept two serial digital inputs however, only one of the input serial data streams can be retimed.

See [Figure 4-1](#) for a visualization of the Flex Input Loopback.

Using **SEL_INPUT** of **INPUT_CONFIG[3:2]** (Address 84D_h, bit[3:2]) allows for selection of which input DDI0 or DDI1 goes into the parallel retimed output and which goes to the loopback DDO output.

LOS_SEL (Address 86F_h, bit 8) allows for selection of DDI0 or DDI1 for LOS sensing.

4.5 Serial Digital Loop-Through Output

The GS3470 contains a differential serial digital output buffer. This output provides an active loop-through of the input signal. It can be a reclocked or non-reclocked version of the input used for processing or a non-reclocked version of the other input. Moreover, selection of the loop-through output is independent of the selection of the signal going into the de-serializer block.

[Table 4-3](#) provides a summary of all the options available for the serial digital output.

The $\text{DDO}/\overline{\text{DDO}}$ differential signal is capable of driving a Semtech Cable Driver through up to 150mm of 100 Ω differential FR4 trace, such that the Cable Driver output conforms to the relevant SMPTE specification for the data rate, with the exception of the jitter specifications.

The output can be DC-coupled into Semtech Cable Drivers that support 1.2V, 1.8V and 2.5V inputs.

The output buffer may be disabled to achieve power savings. This can be done using **SERIAL_LOOPBACK_EN** (Address 811_h, bit 1) through the GSPI interface.

Table 4-3: Serial Digital Output

SERIAL_LOOPBACK_EN	RC_BYP	DDO/ $\overline{\text{DDO}}$
0	X	Disabled
1	0	Re-timed
1	1	Not Re-timed

4.6 Serial Digital Retimer

The retimer operates at three frequencies: 2.97Gb/s, 1.485Gb/s, and 270Mb/s.

Note: The $\text{SD}/\overline{\text{HD}}$ bit takes precedence over the $\text{3G}/\overline{\text{HD}}$ bit, so if the $\text{SD}/\overline{\text{HD}}$ bit is HIGH, the $\text{3G}/\overline{\text{HD}}$ bit is ignored.

The retimer can automatically determine the supported rate based on the input signal, or the rate can be set manually. For more detail on these modes, please refer to [Section 4.3.1](#).

4.7 External Crystal/Reference Clock

The GS3470 requires an external 27MHz reference clock for correct operation. This reference clock is generated by connecting a crystal to the XTAL and $\overline{\text{XTAL}}$ pins of the device. Refer to [Typical Application Circuit](#).

A crystal with a maximum frequency variation of $\pm 100\text{ppm}$ and a maximum equivalent resistance of 50Ω should be selected. The external crystal is used in the frequency acquisition process. It has no impact on the output jitter performance of the device when the device is locked to incoming data.

Alternately, a 27MHz external clock source can be connected to the $\overline{\text{XTAL}}$ pin of the device. It is recommended to DC-couple the reference clock input and to ensure the reference clock does not exceed 1.2V.

4.8 Lock Detect

The LOCKED output signal is set HIGH by the Lock Detect block under the following conditions:

Table 4-4: Lock Detect Conditions

Mode of Operation	Mode Setting	Condition for Locked
SMPTE Mode	$\overline{\text{SMPTE_BYPASS}} = \text{HIGH}$ $\text{DVB_ASI} = \text{LOW}$	Retimer PLL is locked to valid SMPTE video.
DVB-ASI Mode	$\overline{\text{SMPTE_BYPASS}} = \text{LOW}$ $\text{DVB_ASI} = \text{HIGH}$	Retimer PLL is locked to a DVB-ASI stream.
Data-Through Mode	$\overline{\text{SMPTE_BYPASS}} = \text{LOW}$ $\text{DVB_ASI} = \text{LOW}$	Retimer PLL is locked.

The LOCKED output signal is available by default on the *STAT3* output pin, but can be programmed to be output through any one of the six programmable multi-functional pins of the device, *STAT[5:0]*.

Note: In Power-down mode with **RC_BYP** disabled, the PLL unlocks. However, the LOCKED signal retains whatever state it previously held. For instance, if before power-down assertion the LOCKED signal is HIGH, during power-down it will remain HIGH regardless of the status of the PLL.

4.9 Parallel Data Outputs

A 20-bit parallel bus is available which can be configured in 10-bit or 20-bit mode. The parallel data outputs are aligned to the rising edge of the PCLK.

4.9.1 Parallel Data Bus Output Levels

The parallel data bus supports 1.8V or 2.5V (LVTTTL and LVCMOS levels) supplied at the *IO_VDD* pins.

4.9.2 Parallel Output in SMPTE Mode

When the device is operating in SMPTE mode ($\overline{\text{SMPTE_BYPASS}} = \text{HIGH}$), data is output in either multiplexed or demultiplexed form depending on the setting of the $20\text{BIT}/10\text{BIT}$ pin or **PIN_CSR_SELECT** register (877_h).

When operating in 20-bit mode ($20\text{BIT}/10\text{BIT} = \text{HIGH}$), the output data is demultiplexed Luma ($\text{DOUT}[19:10]$) and Chroma ($\text{DOUT}[9:0]$) data for SD and HD data rates. For 3G data rate, Data Stream 1 is output on the $\text{DOUT}[19:10]$ pins and Data Stream 2 is output on the $\text{DOUT}[9:0]$ pins.

When operating in 10-bit mode ($20\text{BIT}/10\text{BIT} = \text{LOW}$), the output data format is multiplexed Luma and Chroma data. In this mode, the data is presented on the $\text{DOUT}[19:10]$ pins, with $\text{DOUT}[9:0]$ being forced LOW. For SD/HD data rates, the clock is either at the 10-bit word rate or at half of this rate (DDR mode). For 3G data rates, the clock is always at half the 10-bit word rate (DDR mode).

4.9.3 Parallel Output in DVB-ASI Mode

The DVB-ASI mode of the GS3470 is enabled when the $\overline{\text{SMPTE_BYPASS}}$ is LOW and the **DVB_ASI** is HIGH. It is required to set $20\text{BIT}/10\text{BIT} = \text{LOW}$.

The extracted 8-bit data is presented on $\text{DOUT}[17:10]$ such that $\text{DOUT}[17:10] = \text{HOUT} \sim \text{AOUT}$, where *AOUT* is the least significant bit of the decoded transport stream data.

In addition, the $\text{DOUT}19$ and $\text{DOUT}18$ pins are configured as DVB-ASI status signals **WORDERR** and **SYNCOUT** respectively.

SYNCOUT is HIGH whenever a K28.5 sync character is output from the device.

WORDERR is HIGH whenever the device has detected a running disparity error or illegal code word.

$\text{DOUT}[9:0]$ is forced LOW, when the GS3470 is operating in DVB-ASI mode.

The clock is either at the 10-bit word rate or at half of this rate (DDR mode).

4.9.4 Parallel Output in Data-Through Mode

This mode is enabled when the $\overline{\text{SMPTE_BYPASS}}$ and **DVB_ASI** are LOW.

In this mode, data is passed to the output bus without any decoding, descrambling, or word-alignment.

GSPI can be used to set the output data width to either 10-bit or 20-bit, adjust the drive strength of the outputs and enable DDR mode.

The output data width (10-bit or 20-bit) can also be controlled through the $20\text{BIT}/10\text{BIT}$ pin.

4.9.5 Parallel Output Data Format Clock/PCLK Settings

The PCLK output frequency of the GS3470 is determined by the output data format. [Table 4-5](#) lists the output signal formats according to the external selection pins for the GS3470.

Table 4-5: GS3470 Output Data Formats

Pin/CSR Bit Settings						Output Data Format	PCLK Rate
20BIT/ 10BIT	SD/HD	3G/HD	SMPTE BYPASS	DVB-ASI	SD_HD_ DDR_SEL		
HIGH	LOW	HIGH	HIGH	LOW	LOW	20-bit 3G format	148.5 or 148.5/1.001MHz
HIGH	LOW	HIGH	LOW	LOW	LOW	20-bit data output	148.5 or 148.5/1.001MHz
HIGH	LOW	LOW	HIGH	LOW	LOW	20-bit HD format	74.25 or 74.25/1.001MHz
HIGH	LOW	LOW	LOW	LOW	LOW	20-bit data output	74.25 or 74.25/1.001MHz
HIGH	HIGH	X	HIGH	LOW	LOW	20-bit SD format	13.5MHz
HIGH	HIGH	X	LOW	LOW	LOW	20-bit data output	13.5MHz
LOW	LOW	HIGH	HIGH	LOW	X	10-bit multiplexed 3G DDR format	148.5 or 148.5/1.001MHz
LOW	LOW	HIGH	LOW	LOW	X	10-bit data output DDR format	148.5 or 148.5/1.001MHz
LOW	LOW	LOW	HIGH	LOW	LOW	10-bit multiplexed HD format	148.5 or 148.5/1.001MHz
LOW	LOW	LOW	LOW	LOW	LOW	10-bit data output	148.5 or 148.5/1.001MHz
LOW	LOW	LOW	HIGH	LOW	HIGH	10-bit multiplexed HD DDR format	74.25 or 74.25/1.001MHz
LOW	LOW	LOW	LOW	LOW	HIGH	10-bit data DDR format	74.25 or 74.25/1.001MHz
LOW	HIGH	X	HIGH	LOW	LOW	10-bit multiplexed SD format	27MHz
LOW	HIGH	X	LOW	LOW	LOW	10-bit data output	27MHz
LOW	HIGH	X	LOW	HIGH	LOW	10-bit ASI output	27MHz
LOW	HIGH	X	HIGH	LOW	HIGH	10-bit multiplexed SD DDR format	13.5MHz
LOW	HIGH	X	LOW	LOW	HIGH	10-bit data output DDR format	13.5MHz
LOW	HIGH	X	LOW	HIGH	HIGH	10-bit ASI output DDR format	13.5MHz

4.9.5.1 Delay Line

The GS3470 has the ability to shift the Setup/Hold window on the receive interface, by using an on-chip delay line to shift the phase of PCLK with respect to the data bus. The timing of the PCLK output, relative to the data, can be adjusted through the host interface registers. Each data rate has its own 5-bit delay line offset setting as well as a PCLK invert option.

The delay adjustment range is defined in Table 4-6. The PCLK output can be delayed by up to 0.5UI using the rate dependent **PCLK_DELAY_XX** (Address 870_h) parameters and it can be advanced by 0.5UI by using the **PCLK_INVERT_XX** (Address 871_h) parameters.

Table 4-6: Delay Adjustment Range

Data Rate	Delay Line Control Parameter	Delay Range (UI)
SD	PCLK_DELAY_SD[14:10]	0.1
HD	PCLK_DELAY_HD[9:5]	0.5
3G	PCLK_DELAY_3G[4:0]	0.5

4.9.6 DDR Parallel Clock Timing

The GS3470 has the ability to transmit 10-bit parallel video data with a DDR (Dual Data Rate) pixel clock over a single-ended interface.

Figure 4-4 and Figure 4-5 show how the DDR interface operates. Data is sampled at the receiver on both clock edges.

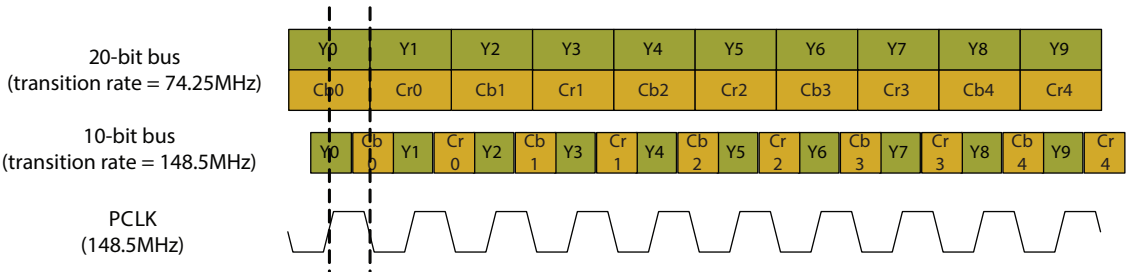


Figure 4-4: DDR Video Interface - 3G Level A

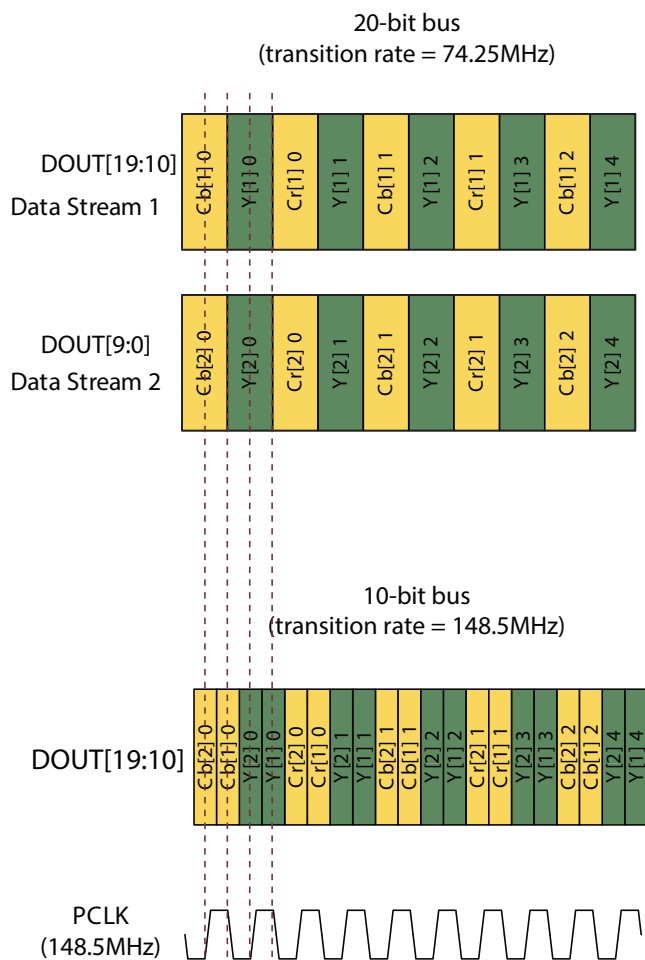


Figure 4-5: DDR Video Interface - 3G Level B

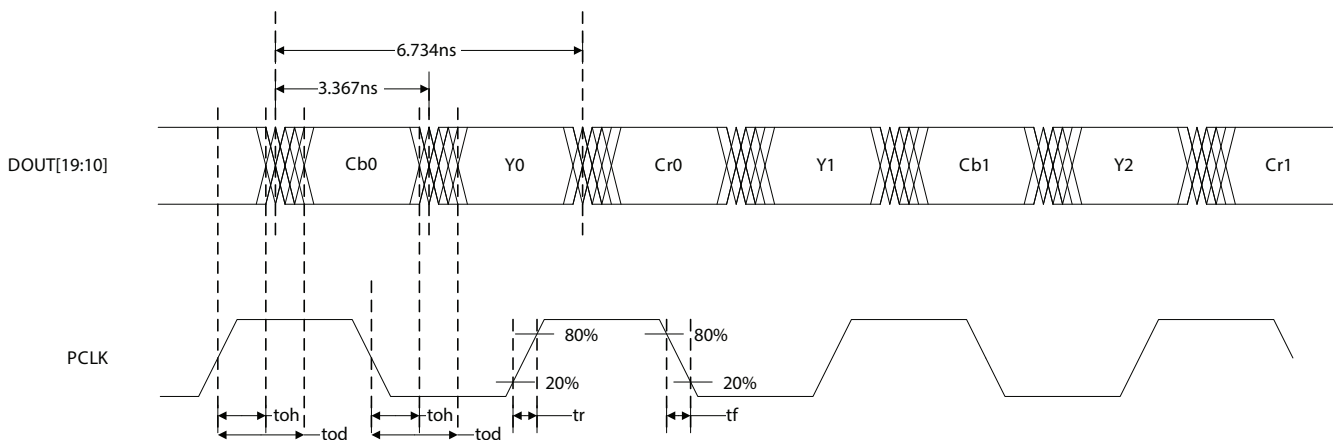


Figure 4-6: DDR Mode Timing Diagram

Note: For output data hold times, please refer to [Table 2-4: AC Electrical Characteristics](#).

4.10 Timing Signal Extraction

The GS3470 extracts timing information from the input data stream and provides FVH timing reference signals.

Video timing signals are only operational in SMPTE mode (**SMPTE_BYPASS** = HIGH).

It takes one video frame to obtain full synchronization of the received video standard.

Note: Both 8-bit and 10-bit TRS words are identified. Once synchronization is achieved, the device continues to monitor the received TRS timing information to maintain synchronization.

4.10.1 Automatic Switch Line Lock Handling

The principle of switch line lock handling is that the switching of synchronous video sources will only disturb the horizontal timing and alignment, whereas the vertical timing remains in synchronization – i.e. switching between video sources of the same format. Automatic switch line lock handling is only available in SMPTE mode.

The synchronous switch point is defined for all major video standards in SMPTE RP168-2002. The device automatically re-synchronizes the word alignment block and timing signal generator at the switch point, based on the detected video standard.

The switch line is defined as follows:

- For 525 line interlaced systems:
resynchronization takes place at the end of lines 10 & 273
- For 525 line progressive systems:
resynchronization takes place at the end of line 10
- For 625 line interlaced systems:
resynchronization takes place at the end of lines 6 & 319
- For 625 line progressive systems:
resynchronization takes place at the end of line 6
- For 750 line progressive systems:
resynchronization takes place at the end of line 7
- For 1125 line interlaced systems:
resynchronization takes place at the end of lines 7 & 569
- For 1125 line progressive systems:
resynchronization takes place at the end of line 7

Note: Unless indicated by SMPTE ST 352 payload identifier packets, the GS3470 does not distinguish between 1125-line progressive segmented-frame (PsF) video and 1125-line interlaced video operating at 25 or 30fps. However, PsF video operating at 24fps is detected by the device.

A full list of all major video standards and switching lines can be found in SMPTE RP168-2002.

4.10.2 Manual Switch Line Lock Handling

The automatic switch point can be reconfigured using GSPI. The switch line is programmed by the user via the host interface. The user may program two lines, one for Field One and one for Field Two of an interlaced standard. For progressive formats, only the first number is used. This enables the user to force immediate lock-up on any line, if the switch point is non-standard. If the numbers are set to zero, then the switch lines used are those defined in RP168-2002.

Note: Please contact a Semtech FAE for an explanation of how to set CSRs.

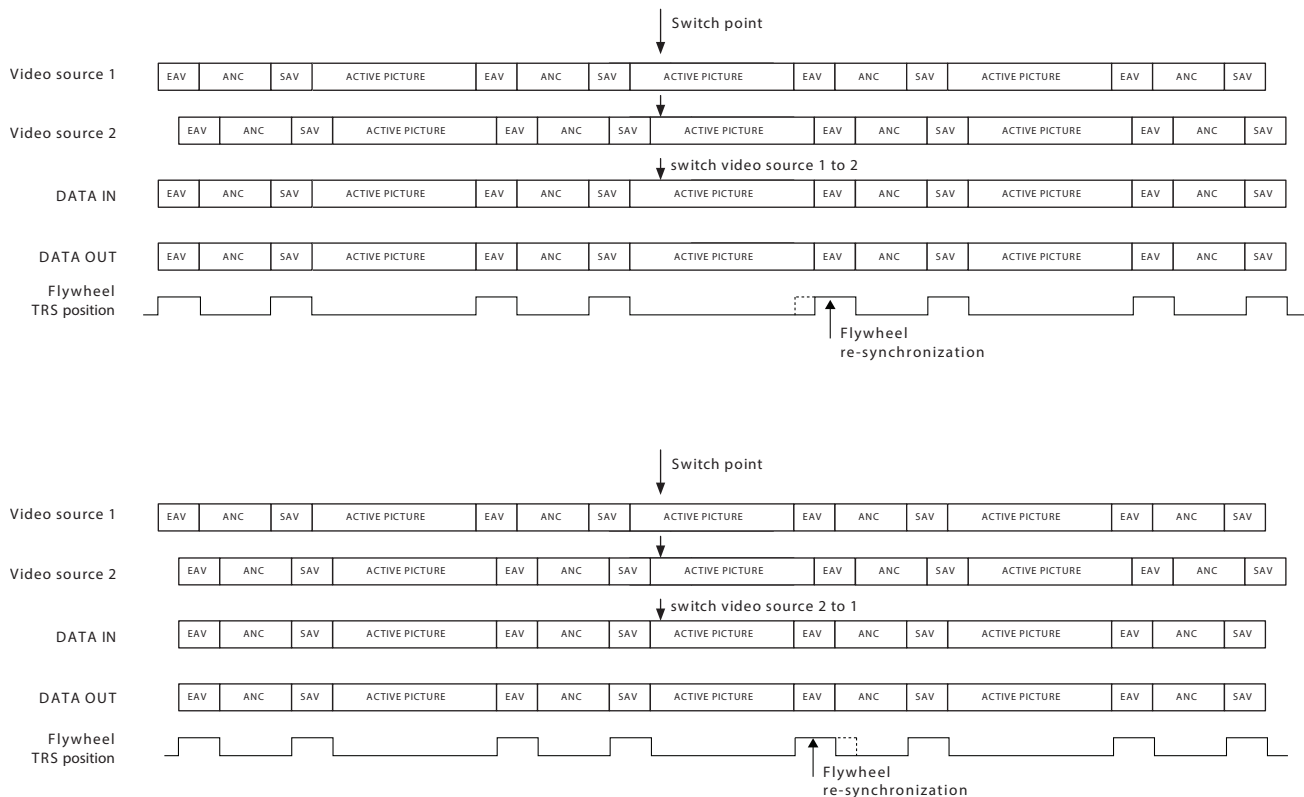


Figure 4-7: Switch Line Locking on a Non-Standard Switch Line

4.11 Programmable Multifunction Outputs

The GS3470 has 6 multifunction output pins, *STAT[5:0]*, which are programmable via the host interface register **STAT[5:0]_CONFIG** (Addresses 812_h and 813_h) to output one of the following signals:

Table 4-7: Output Signals Available on Programmable Multifunction Pins

Status Signal	Selection Code	Default Output Pin
H/HSYNC (according to TIM_861 register) Section 4.12	00000	STAT0
V/VSNC (according to TIM_861 register) Section 4.12	00001	STAT1
F/DE (according to TIM_861 register) Section 4.12	00010	STAT2
LOCKED Section 4.8	00011	STAT3
Y/1ANC Section 4.17	00100	—
C/2ANC Section 4.17	00101	—
DATA ERROR	00110	STAT5
VIDEO ERROR	00111	—
AUDIO ERROR	01000	—
EDH DETECTED	01001	—
CARRIER DETECT	01010	—
SD/H $\overline{\text{D}}$	01011	STAT4
3G/H $\overline{\text{D}}$	01100	—
SMPTE BYPASS	11101	—
DVB_ASI	11110	—

Note: Unused digital output pins can be left unconnected.

4.12 H:V:F Timing Signal Extraction

The GS3470 extracts critical timing parameters from the received TRS words.

Horizontal blanking (H), Vertical blanking (V), and Field odd/even (F) timing are output on the *STAT[2:0]* pins by default.

The H signal timing can be selected through GSPI using **H_CONFIG** (Address 800_H, bit 9). By default: **H_CONFIG** = LOW, and the H signal timing is set to active line blanking including EAV TRS words. This can be changed to TRS based blanking by setting **H_CONFIG** to HIGH.

The timing of these signals is shown in [Figure 4-8](#) through [Figure 4-13](#).

Note: Both 8-bit and 10-bit TRS words are identified by the device.

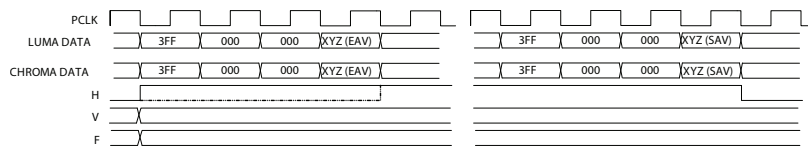


Figure 4-8: H:V:F Output Timing - 3G Level A and HD 20-bit Mode

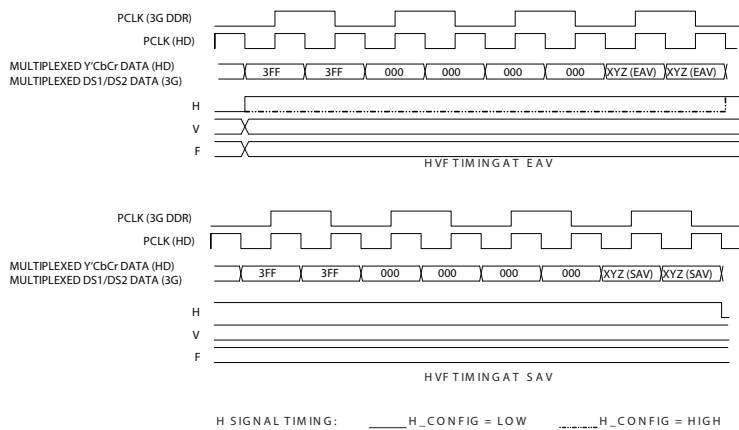


Figure 4-9: H:V:F Output Timing - 3G Level A and HD 10-bit Mode

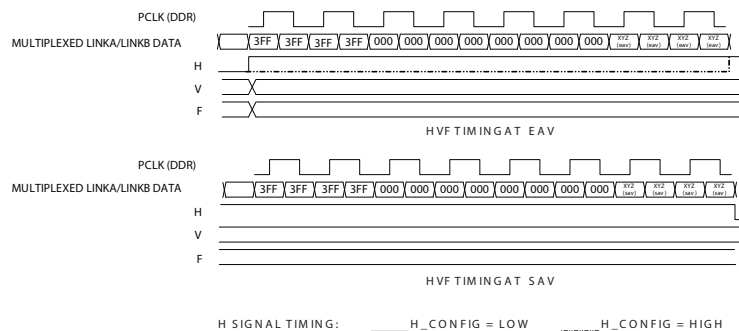


Figure 4-10: H:V:F Output Timing - 3G Level B 10-bit Mode

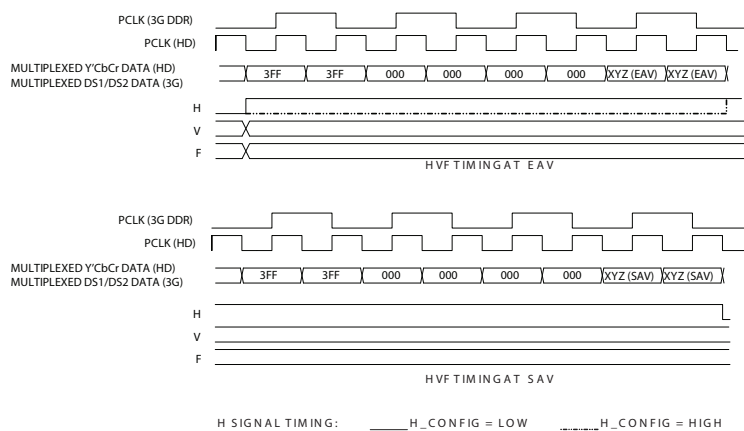


Figure 4-11: H:V:F Output Timing - 3G Level B 20-bit Mode, each 10-bit Stream

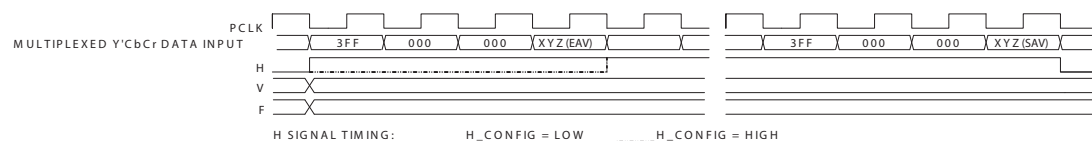


Figure 4-12: H:V:F Output Timing - SD 10-bit Mode

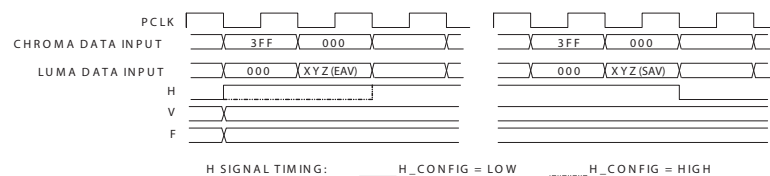


Figure 4-13: H:V:F Output Timing - SD 20-bit Mode

4.12.1 CEA-861 Timing Generation

The GS3470 is capable of generating CEA861 timing for all of the supported video formats.

Table 4-8: Supported CEA-861 Formats

Format	CEA-861	VD_STD[5:0]
720 (1440) x 480i @ 59.94/60Hz	6 & 7	16 _h , 17 _h , 19 _h , 1B _h
720(1440) x 576i @ 50Hz	21 & 22	18 _h , 1A _h
1280 x 720p @ 59.94/60Hz	4	20 _h , 00 _h
1280 x 720p @ 50Hz	19	24 _h , 04 _h
1280x720p @ 29.97/30Hz	62	27 _h , 02 _h
1280x720p @ 25Hz	61	26 _h , 06 _h
1920 x 1080i @ 59.94/60Hz	5	2A _h , 0A _h
1920 x 1080i @ 50Hz	20	2C _h , 0C _h
1920 x 1080p @ 29.97/30Hz	34 ¹	2E _h , 0B _h
1920 x 1080p @ 25Hz	33 ²	2F _h , 0D _h
1920 x 1080p @ 23.98/24Hz	32	30 _h , 10 _h
1920 x 1080p @ 59.94/60Hz	16 ¹	2B _h
1920 x 1080p @ 50Hz	31 ²	2D _h
2048x1080p @ 30/25/24/48/50/60Hz	Undefined ³	21 _h , 22 _h , 23 _h , 37 _h , 38 _h , 39 _h , 3A _h , 3B _h , 3C _h
2048x1080i @ 48/50/59.94/60Hz	Undefined ³	29 _h , 32 _h , 33 _h , 34 _h , 35 _h , 36 _h

Notes:

1,2: Timing is identical for the corresponding formats

3: Derived from the standard. Timing diagram provided.

4.12.1.1 Vertical Timing

When CEA 861 timing is selected, the device outputs standards compliant CEA 861 timing signals as shown in the figures below, for example 240 active lines per field for SMPTE ST 125.

The timing of the CEA 861 timing reference signals can be found in the CEA 861 specifications.

Table 4-9: CEA861 Timing Formats

Format	Parameters
4	H:V:DE Input Timing 1280 x720p @ 59.94/60Hz
5	H:V:DE Input Timing 1920 x1080i @ 59.94/60Hz
6 & 7	H:V:DE Input Timing 720(1440)x480i @ 59.94/60Hz
16	H:V:DE Input Timing 1920 x1080p @ 59.94/60Hz
19	H:V:DE Input Timing 1280 x 720p @ 50Hz
20	H:V:DE Input Timing 1920 x1080i @ 50Hz
21 & 22	H:V:DE Input Timing 720 (1440) x576 @ 50Hz
31	H:V:DE Input Timing 1920 x1080p @ 50Hz
32	H:V:DE Input Timing 1920 x1080p @ 23.976/24Hz
33	H:V:DE Input Timing 1920 x1080p @ 25Hz
34	H:V:DE Input Timing 1920 x1080p @ 29.97/30Hz
61	H:V:DE Input Timing 1280 x720p @ 25Hz
62	H:V:DE Input Timing 1280 x720p @ 29.97/30Hz
Undefined	H:V:DE Input Timing 2048x1080p @ 30/60Hz
Undefined	H:V:DE Input Timing 2048x1080p @ 25/50Hz
Undefined	H:V:DE Input Timing 2048x1080p @ 24/48Hz
Undefined	H:V:DE Input Timing 2048x1080i @ 30/60Hz
Undefined	H:V:DE Input Timing 2048x1080i @ 25/50Hz
Undefined	H:V:DE Input Timing 2048x1080i @ 24/48Hz

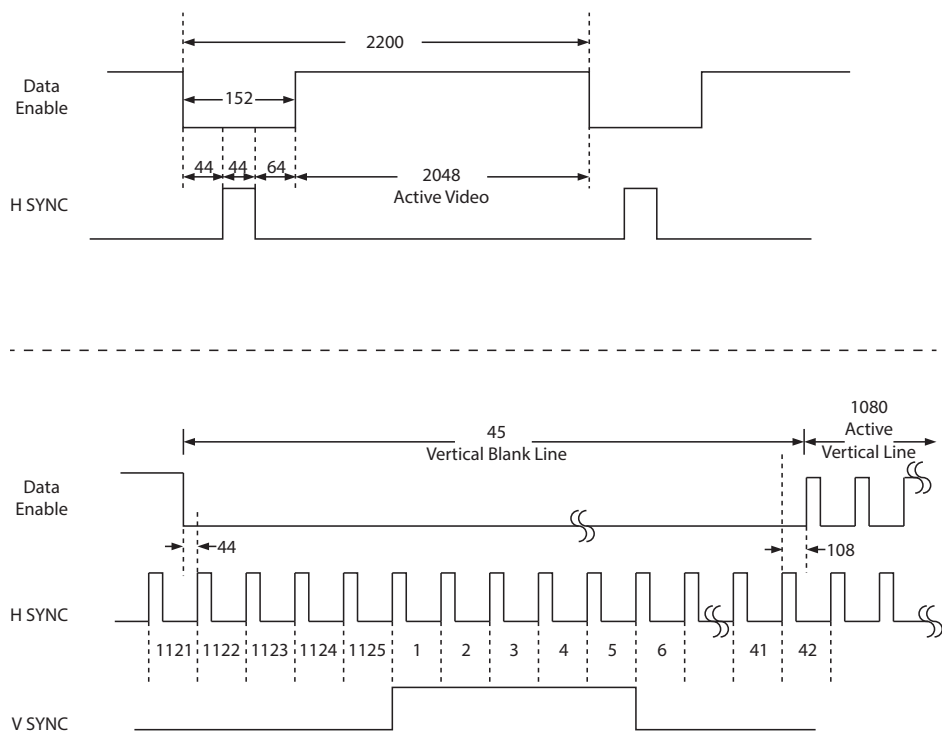


Figure 4-14: H:V:DE Output Timing 2048 x 1080p @ 30/60

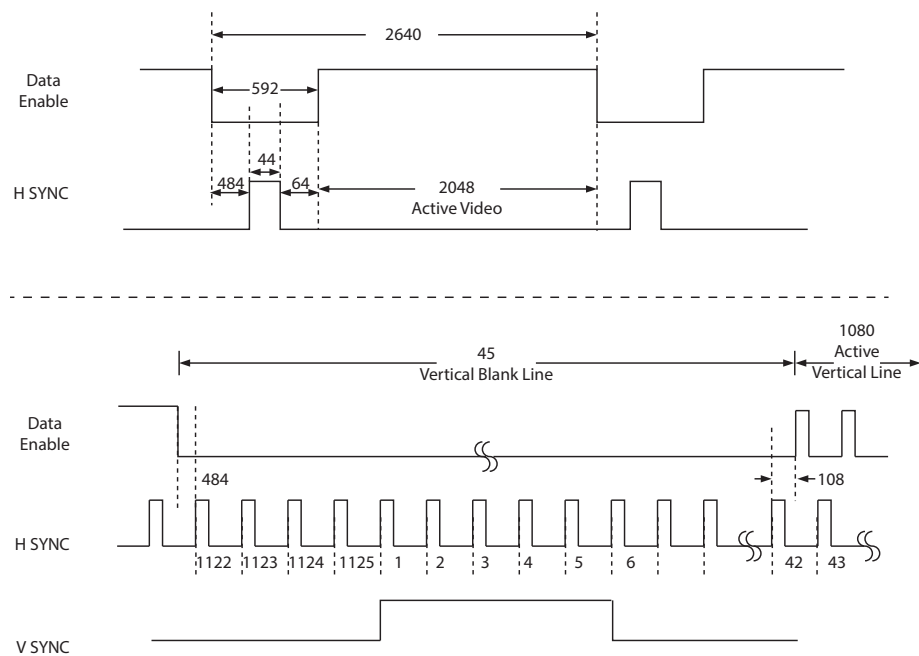


Figure 4-15: H:V:DE Output Timing 2048 x 1080p @ 25/50

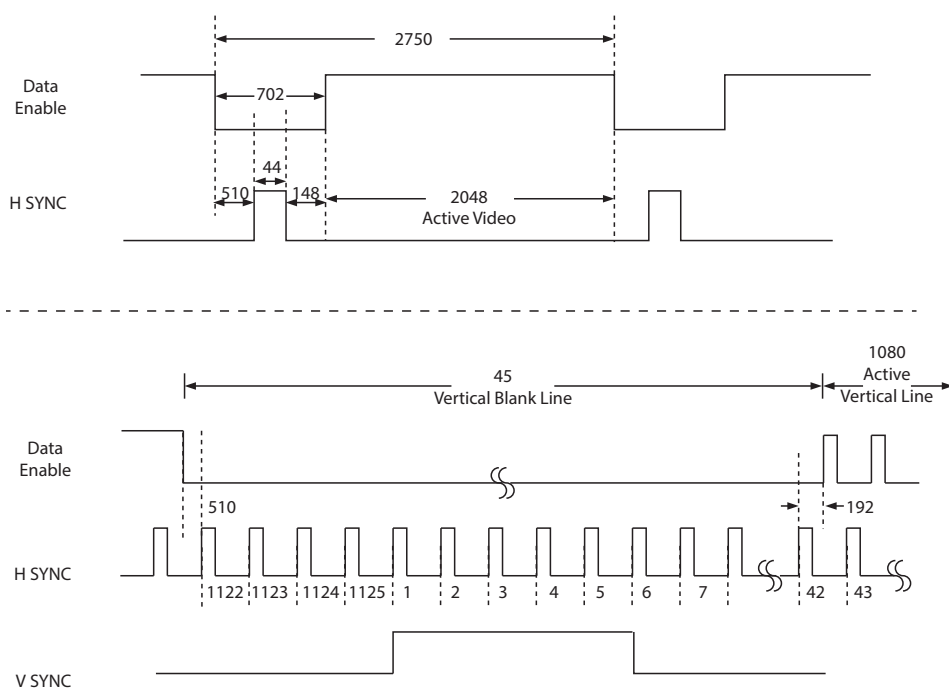


Figure 4-16: H:V:DE Output Timing 2048 x 1080p @ 24/48

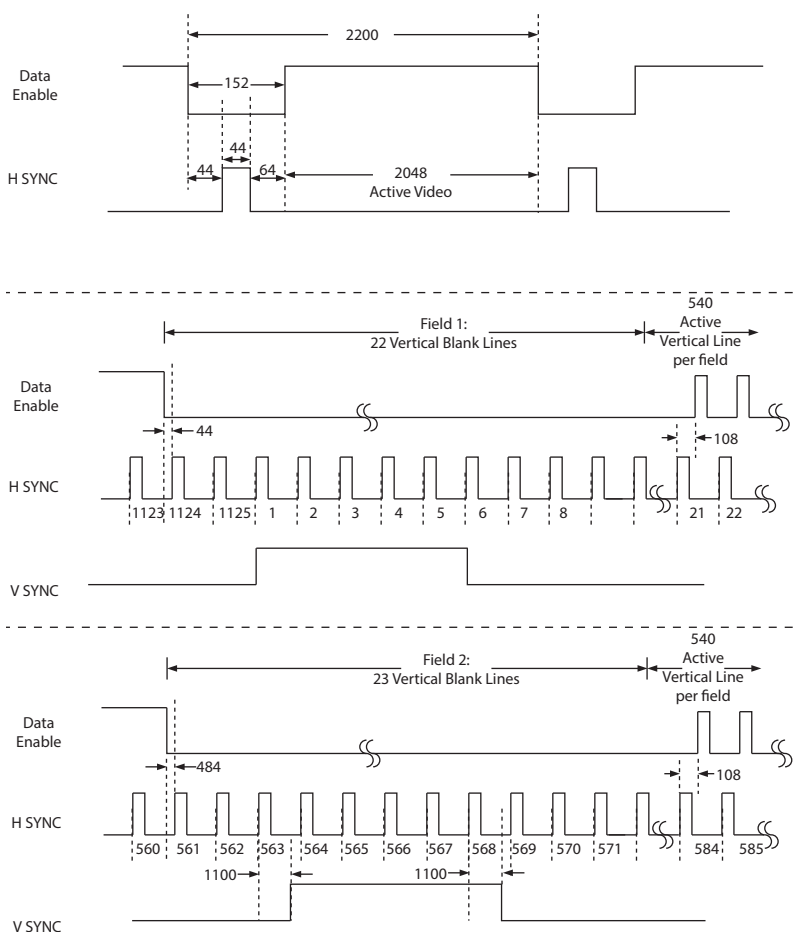


Figure 4-17: H:V:DE Output Timing 2048 x 1080i @ 30/60

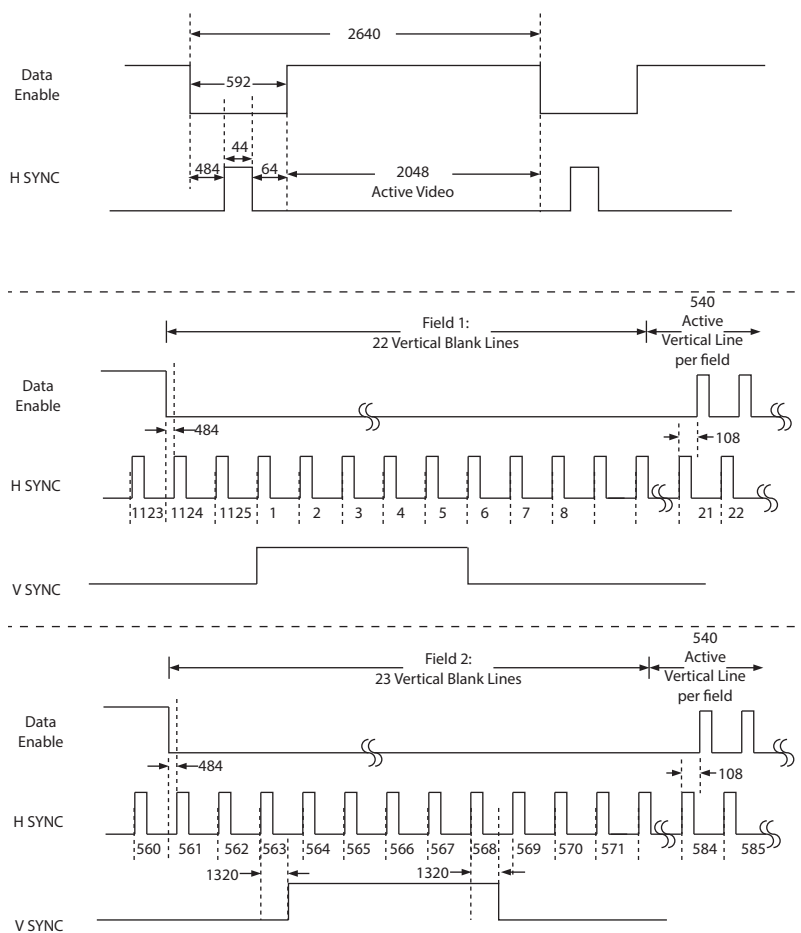


Figure 4-18: H:V:DE Output Timing 2048 x 1080i @ 25/50

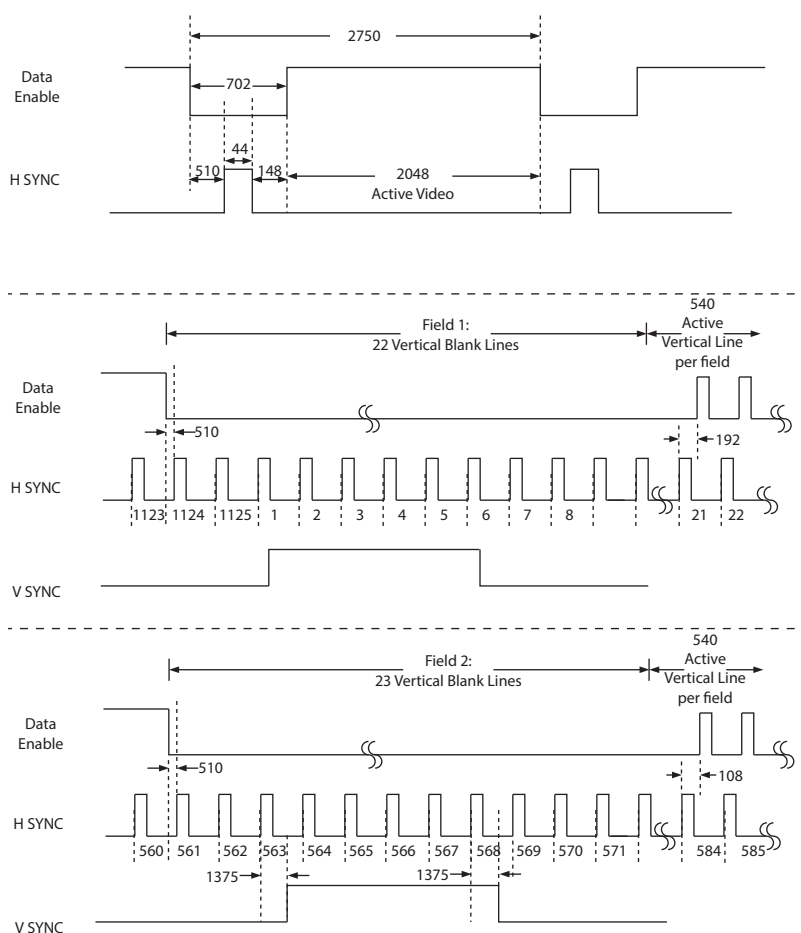


Figure 4-19: H:V:DE Output Timing 2048 x 1080i @ 24/48

4.13 Automatic Video Standards Detection

The GS3470 is able to identify the received video standard. It will also indicate whether the 3G input signal is Level A or Level B. The **VID_STD** (Address 80D_h) register can be used to access this information. [Table 4-10](#) describes the 5-bit codes for the recognized video standards.

Additionally registers are provided to allow the host to read the video standard information from the device such as total words per line, active words per line, total lines per field/frame, and active lines per field/frame. This information can be accessed through the **RASTER_STRUC_4[1]_DS1** (Address 830_h to 833_h) and **RASTER_STRUC_4[1]_DS2** (Address 8CF_h to 8D2_h) registers.

The **RASTER_STRUC_4_DS[2:1]** registers also contain three status bits: **STD_LOCK_DS[2:1]**, **INT_PROG_DS[2:1]**, and **M_DS[2:1]**. The **STD_LOCK_DS[2:1]** bit is set HIGH whenever the timing signal generator is fully synchronized to the incoming standard, and detects it as one of the supported formats. The **INT_PROG_DS[2:1]** bit is set HIGH if the detected video standard is interlaced and LOW if the detected video standard is progressive. **M_DS[2:1]** is set HIGH if the clock frequency includes the "1000/1001" factor denoting a 23.98Hz, 29.97Hz, or 59.94Hz frame rate.

Note: In certain systems, due to greater ppm offsets in the crystal, the 'M' bit may not assert properly. In such cases, the **M_DETECTION_TOLERANCE_DS[2:1]** value can be increased through GSPI.

Note: By default (after power up or after systems reset), the **RASTER_STRUC [4:1]_DS[2:1]**, **VD_STD_DS[2:1]**, **STD_LOCK_DS[2:1]**, and **INT_PROG_DS[2:1]** are set to zero until a valid input is available. These fields are cleared when the **SMPTE_BYPASS** register is LOW.

Table 4-10: Supported Video Standard Codes

SMPTE Standard	Active Video Area	3G/HD	SD/HD	Lines per Frame	Active Lines per Field	Words per Active Line	Words per Line	VD_STD [5:0]	3G Level B VD_STD [5:0]
2048 – 2 4:2:2	2048x1080/60(1:1)	1	0	1125	1080	2048	2200	37 _h	29 _h
	2048x1080/50(1:1)	1	0	1125	1080	2048	2640	38 _h	32 _h
	2048x1080/48(1:1)	1	0	1125	1080	2048	2750	39 _h	33 _h
	2048x1080/30(1:1)	0	0	1125	1080	2048	2200	21 _h	N/A
	2048x1080/25(1:1)	0	0	1125	1080	2048	2640	22 _h	N/A
	2048x1080/24(1:1)	0	0	1125	1080	2048	2750	23 _h	N/A
	2048x1080/60(2:1) or 2048x1080/30(PsF)	0	0	1125	540	2048	2200	29 _h	N/A
	2048x1080/50(2:1) or 2048x1080/25(PsF)	0	0	1125	540	2048	2640	32 _h	N/A
	2048x1080/48(2:1) or 2048x1080/24(PsF)	0	0	1125	540	2048	2750	33 _h	N/A
2048 – 2 4:4:4	2048x1080/60 (2:1) or 2048x1080/30 (PsF)	1	0	1125	540	4096	4400	34 _h	29 _h
	2048x1080/50 (2:1) or 2048x1080/25 (PsF)	1	0	1125	540	4096	5280	35 _h	32 _h
	2048x1080/48 (2:1) or 2048x1080/24 (PsF)	1	0	1125	540	4096	5500	36 _h	33 _h
	2048x1080/30 (1:1)	1	0	1125	1080	4096 ¹	4400	3A _h	21 _h
	2048x1080/25 (1:1)	1	0	1125	1080	4096 ¹	5280	3B _h	22 _h
	2048x1080/24 (1:1)	1	0	1125	1080	4096 ¹	5500	3C _h	23 _h

Table 4-10: Supported Video Standard Codes (Continued)

SMPTE Standard	Active Video Area	3G/HD	SD/HD	Lines per Frame	Active Lines per Field	Words per Active Line	Words per Line	VD STD [5:0]	3G Level B VD STD [5:0]
425M (3G) 4:2:2	1920x1080/60 (1:1)	1	0	1125	1080	1920	2200	2B _h	0A _h
	1920x1080/50 (1:1)	1	0	1125	1080	1920	2640	2D _h	0C _h
425M (3G) 4:4:4	1920x1080/60 (2:1) or 1920x1080/30 (PsF)	1	0	1125	540	3840 ¹	4400	2A _h	0A _h
	1920x1080/50 (2:1) or 1920x1080/25 (PsF)	1	0	1125	540	3840 ¹	5280	2C _h	0C _h
	1280x720/60 (1:1)	1	0	750	720	2560 ¹	3300	20 _h	Not Supported
	1280x720/50 (1:1)	1	0	750	720	2560 ¹	3960	24 _h	Not Supported
	1920x1080/30 (1:1)	1	0	1125	1080	3840 ¹	4400	2E _h	0B _h
	1920x1080/25 (1:1)	1	0	1125	1080	3840 ¹	5280	2F _h	0D _h
	1280x720/25 (1:1)	1	0	750	720	2560 ¹	7920	26 _h	Not Supported
	1920x1080/24 (1:1)	1	0	1125	1080	3840 ¹	5500	30 _h	10 _h
	1920x1080/24 (PsF)	1	0	1125	540	3840 ¹	5500	31 _h	11 _h
	1280x720/24 (1:1)	1	0	750	720	2560 ¹	8250	28 _h	Not Supported
260M (HD)	1920x1035/60 (2:1)	0	0	1125	517 or 518	1920	2200	15 _h	N/A
	1920x1080/50 (2:1)	0	0	1250	540	1920	2376	14 _h	N/A

Table 4-10: Supported Video Standard Codes (Continued)

SMPTE Standard	Active Video Area	3G/HD	SD/HD	Lines per Frame	Active Lines per Field	Words per Active Line	Words per Line	VD STD [5:0]	3G Level B VD STD [5:0]
274M (HD)	1920x1080/60 (2:1) or 1920x1080/30 (PsF)	0	0	1125	540	1920	2200	0A _h	N/A
	1920x1080/50 (2:1) or 1920x1080/25 (PsF)	0	0	1125	540	1920	2640	0C _h	N/A
	1920x1080/30 (1:1)	0	0	1125	1080	1920	2200	0B _h	N/A
	1920x1080/25 (1:1)	0	0	1125	1080	1920	2640	0D _h	N/A
	1920x1080/24 (1:1)	0	0	1125	1080	1920	2750	10 _h	N/A
	1920x1080/24 (PsF)	0	0	1125	540	1920	2750	11 _h	N/A
	1920x1080/25 (1:1)	0	0	1125	1080	2304	2640	0E _h	N/A
	1920x1080/25 (PsF) – EM	0	0	1125	540	2304	2640	0F _h	N/A
	1920x1080/24 (1:1)	0	0	1125	1080	2400	2750	12 _h	N/A
	1920x1080/24 (PsF) – EM	0	0	1125	540	2400	2750	13 _h	N/A
296M (HD)	1280x720/30 (1:1)	0	0	750	720	1280	3300	02 _h	N/A
	1280x720/30 (1:1) – EM	0	0	750	720	2880	3300	03 _h	N/A
	1280x720/50 (1:1)	0	0	750	720	1280	1980	04 _h	N/A
	1280x720/50 (1:1) – EM	0	0	750	720	1728	1980	05 _h	N/A
	1280x720/25 (1:1)	0	0	750	720	1280	3960	06 _h	N/A
	1280x720/25 (1:1) – EM	0	0	750	720	3456	3960	07 _h	N/A
	1280x720/24 (1:1)	0	0	750	720	1280	4125	08 _h	N/A
	1280x720/24 (1:1) – EM	0	0	750	720	3600	4125	09 _h	N/A
	1280x720/60 (1:1)	0	0	750	720	1280	1650	00 _h	N/A
	1280x720/60 (1:1) – EM	0	0	750	720	1440	1650	01 _h	N/A

Table 4-10: Supported Video Standard Codes (Continued)

SMPTE Standard	Active Video Area	3G/HD	SD/HD	Lines per Frame	Active Lines per Field	Words per Active Line	Words per Line	VD STD [5:0]	3G Level B VD STD [5:0]
125M (SD)	1440x487/60 (2:1)	x	1	525	244 or 243	1440	1716	16 _h	N/A
	1440x507/60	x	1	525	254 or 253	1440	1716	17 _h	N/A
	525-line 487 generic	x	1	525	244 or 243	not (1440)	1716	19 _h	N/A
	525-line 507 generic	x	1	525	254 or 253	not (1440)	1716	1B _h	N/A
ITU-R BT.656 (SD)	1440x576/50 (2:1) or dual link progressive	x	1	625	288	1440	1728	18 _h	N/A
	625-line generic	x	1	625	288	not (1440)	1728	1A _h	N/A
Unknown HD	SD/HD = 0	0	0	-	-	-	-	1D _h	N/A
Unknown SD	SD/HD = 1	x	1	-	-	-	-	1E _h	N/A
Unknown 3G	SD/HD = 0	1	0	-	-	-	-	3F _h	N/A

Notes:

1. The 4:4:4 standards have 2 clocks per sample at the data stream level.

4.14 Data Format Detection & Indication

In addition to detecting the video standard, the GS3470 also detects the data format. This information can be found in the **DATA_FORMAT_DS[2:1]** (Addresses 80C_h and 80E_h) registers. Data format information is only accessible while the device is locked. By default, at power-up, after reset or while the device is not locked, the **DATA_FORMAT_DS[2:1]** registers are set to F_h.

Table 4-11: Data Format Register Codes

YDATA_FORMAT[3:0] or CDATA_FORMAT[3:0]	Data Format	Remarks
0 _h to 05 _h	SDTI	SMPTE ST 321, SMPTE ST 322, SMPTE ST 326
6 _h	SDI	—
7 _h	Reserved	—
8 _h	TDM	SMPTE ST 346
9 _h	HD-SDTI	—
A _h to E _h	Reserved	—
F _h	Non-SMPTE data format	Detected data format is not SMPTE. LOCKED = LOW. Note: This Data Format register is invalid in SMPTE_BYPASS mode.

The data format is determined based on the presence of TRS ID words, SDTI header and TDM header.

Note: In SD video mode only the Y data format register contains the data, and the C register is set to F_h (undefined format).

4.14.1 SMPTE ST 425 Mapping - 3G Level A and Level B Formats

4.14.1.1 Level A Mapping

Direct image format mapping - the mapping structure used to define 1080p/50/59.94/60 4:2:2 YCbCr 10 bit data, as supported by the GS3470. See [Figure 4-20](#).

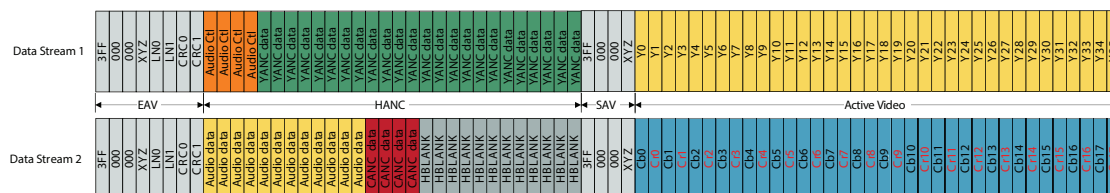


Figure 4-20: Level A Mapping for 1080p/50/59.94/60 4:2:2 YCbCr 10 bit data

4.14.1.2 Level B Mapping

Level B defines 2 mappings:

1. SMPTE ST 372 Dual-Link Mapping
2. 2 x SMPTE ST292-1 (HD-SDI) Dual Stream Mapping

For 1080p/50/59.94/60 4:2:2 video formats, each link should be line-interleaved as per SMPTE ST 372 (See [Figure 4-21](#)).

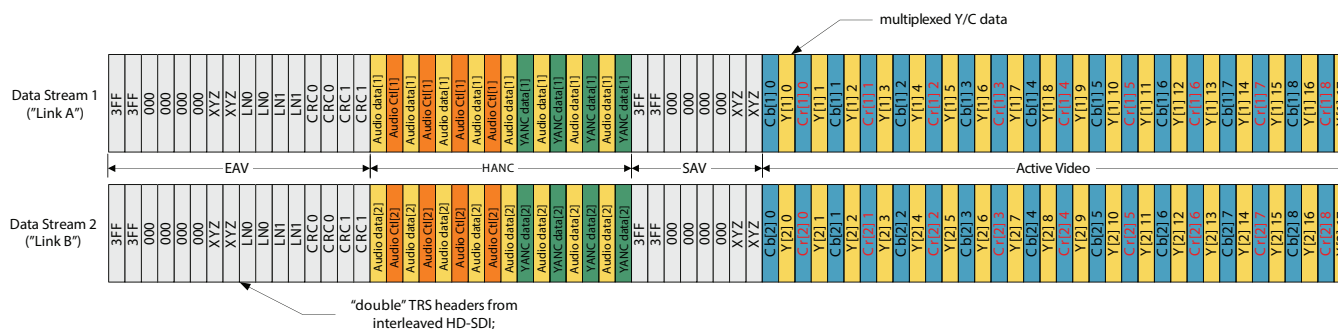


Figure 4-21: Level B Mapping

The GS3470 distinguishes between Level A and Level B mappings at 3Gb/s. When Level B data is detected, each 10-bit link is demultiplexed into its individual component streams, and most video processing features, including error detection and correction are enabled separately for Data Stream 1 and Data Stream 2 (Link A and Link B, respectively).

Note: Audio demultiplexing and ancillary data extraction can only be enabled for one link for 3Gb/s Level B data. Data Stream 1 or Data Stream 2 can be selected via the host interface.

4.15 EDH Detection

4.15.1 EDH Packet Detection (SD Only)

The GS3470 determines if EDH packets are present in the incoming video data and asserts the **EDH_DETECT** (Address 80A_h, bit 15) status according to the SMPTE standard.

EDH_DETECT is set HIGH when EDH packets have been detected and remains HIGH until EDH packets are no longer present. It is set LOW at the end of the vertical blanking (falling edge of V) if an EDH packet has not been detected during vertical blanking.

EDH_DETECT can be programmed to be an output on the *STAT[5:0]* pins through GSPI.

4.15.2 EDH Flag Detection

The EDH flags for ancillary data, active picture, and full field regions are extracted from the detected EDH packets and placed in the **EDH_FLAG_IN** (Address 80A_h) register.

When **EDH_FLAG_UPDATE_MASK** (Address 800_h, bit 11) in the host interface is set HIGH, the GS3470 updates the Ancillary Data, Full Field, and Active Picture EDH flags according to SMPTE RP165. The updated EDH flags are available in **EDH_FLAG_OUT** (Address 80B_h). The EDH packet output from the device contains these updated flags.

Flags are provided for both fields 1 and 2. The field 1 flag data is overwritten by the field 2 flag data.

When EDH packets are not detected, the UES flags in the **EDH_FLAG_OUT** register are set HIGH to signify that the received signal does not support Error Detection and Handling. In addition, the **EDH_DETECT** bit is set LOW. These flags are set regardless of the setting of the **EDH_FLAG_UPDATE_MASK** bit.

EDH_FLAG_OUT and **EDH_FLAG_IN** may be read via the host interface at any time during the received frame except on the lines defined in SMPTE RP165, when these flags are updated.

The GS3470 indicates the CRC validity for both active picture and full field CRCs.

AP_CRC_V (Address 80C_h, bit 8) in the host interface indicates the active picture CRC validity, and **FF_CRC_V** (Address 80C_h, bit 9) indicates the full field CRC validity. When **EDH_DETECT** = LOW, these bits are cleared.

The **EDH_FLAG_OUT** and **EDH_FLAG_IN** register values remain set until overwritten by the decoded flags in the next received EDH packet. When an EDH packet is not detected during vertical blanking, the flag registers are cleared at the end of the vertical blanking period.

4.16 Video Signal Error Detection & Indication

The GS3470 includes a number of video signal error detection functions. These are provided when operating in SMPTE mode ($\overline{\text{SMPTE_BYPASS}} = \text{HIGH}$).

Signal errors that can be detected include:

1. TRS errors
2. HD line based CRC errors
3. EDH errors
4. HD line number errors
5. Device lock error
6. Ancillary data checksum errors

The GS3470 has two different registers for each set of error flags, **ERROR_STAT_[2:1]** (Addresses 807_h and 809_h) and **ERROR_STAT_[2:1]_STICKY** (Addresses 806_h and 808_h). **ERROR_STAT_[2:1]** registers are cleared on write, when not locked, on a change of video standard, and once per frame. **ERROR_STAT_[2:1]_STICKY** registers are read only and only cleared on read or device reset.

ERROR_MASK_[3:1] (Addresses 848_h to 84A_h) are also provided, allowing the user to select which error conditions are reported through **GLOBAL_ERROR**. Each bit of the **ERROR_MASK** register corresponds to a unique error type. Please refer to Table 4-12 for a description of the **ERROR_MASK** register bits.

Separate interrupt enable (**AUDIO_INT_ENABLE**) registers for audio errors are also provided, allowing select error conditions to be reported. Each bit of each **ERROR_MASK** register corresponds to a unique error type.

By default (at power-up or after system reset), all bits of the **ERROR_MASK** registers are zero, enabling all errors to be reported. Individual error detection may be disabled by setting the corresponding bit HIGH in the mask registers.

Error conditions are indicated by a **VIDEO_ERROR** signal and an **AUDIO_ERROR** signal, which are available for output on the multifunction output pins, *STAT[5:0]*. The two signals are also combined into a **GLOBAL_ERROR** signal, which is also available on the multifunction output pins. These signals are normally HIGH, but are set LOW by the device when an error condition has been detected.

These signals are a logical 'NOR' of the appropriate error status flags stored in the **ERROR_STAT_[2:1]** register, which are gated by the bit settings in the **ERROR_MASK_[3:1]** registers. When an error status bit is HIGH and the corresponding error mask bit is LOW, the corresponding **DATA_ERROR** signal is set LOW by the device.

All bits of the error status register, including the **LOCK_ERR** bit, is set LOW during system reset.

Table 4-12 shows the **ERROR_STAT_[2:1]** register.

Table 4-12: Error Status Register and Error Mask Register

Video Error Status Register		Video Error Mask Register	Link A (Data Stream 1) or Link B (Data Stream 2)
EAV_ERR_[2:1]	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h),	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
	ERROR_STAT_2_STICKY (808 _h), ERROR_STAT_2 (809 _h)	ERROR_MASK_3 (84A _h) (sticky and non - sticky)	Data Stream 2
SAV_ERR_[2:1]	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h),	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
	ERROR_STAT_2_STICKY (808 _h), ERROR_STAT_2 (809 _h)	ERROR_MASK_3 (84A _h) (sticky and non - sticky)	Data Stream 2
YCRC_ERR_[2:1]	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h),	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
	ERROR_STAT_2_STICKY (808 _h), ERROR_STAT_2 (809 _h)	ERROR_MASK_3 (84A _h) (sticky and non - sticky)	Data Stream 2
CCRC_ERR_[2:1]	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h),	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
	ERROR_STAT_2_STICKY (808 _h), ERROR_STAT_2 (809 _h)	ERROR_MASK_3 (84A _h) (sticky and non - sticky)	Data Stream 2
LNUM_ERR_[2:1]	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h),	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
	ERROR_STAT_2_STICKY (808 _h), ERROR_STAT_2 (809 _h)	ERROR_MASK_3 (84A _h) (sticky and non - sticky)	Data Stream 2
YCS_ERR_[2:1]	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h),	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
	ERROR_STAT_2_STICKY (808 _h), ERROR_STAT_2 (809 _h)	ERROR_MASK_3 (84A _h) (sticky and non - sticky)	Data Stream 2
CCS_ERR_[2:1]	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h),	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
	ERROR_STAT_2_STICKY (808 _h), ERROR_STAT_2 (809 _h)	ERROR_MASK_3 (84A _h) (sticky and non - sticky)	Data Stream 2
AP_CRC_ERR	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h)	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1
FF_CRC_ERR	ERROR_STAT_1_STICKY (806 _h),	ERROR_MASK_1 (848 _h) (sticky)	Data Stream 1
	ERROR_STAT_1 (807 _h)	ERROR_MASK_2 (849 _h) (non - sticky)	Data Stream 1

Note 1: See [Section 4.19.4](#) for Audio Error Status.

Note 2: In 3G Level B mode, separate Video Error Mask registers exist for Link A and Link B.

4.16.1 TRS Error Detection

TRS error flags are generated by the GS3470 under the following two conditions:

1. A phase shift in received TRS timing is observed.
2. The received TRS Hamming codes are incorrect

Both SAV and EAV TRS words are checked for timing and data integrity errors.

For HD mode, only the Y channel TRS codes are checked for errors.

For 3G mode Level A signals, only Data Stream 1 TRS codes are checked for errors. For 3G Level B signals, the Y channel TRS codes of both Link A and Link B are checked for errors.

Both 8-bit and 10-bit TRS code words are checked for errors.

4.16.2 Line Based CRC Error Detection

The GS3470 calculates line based CRCs for HD and 3G video signals. CRC calculations are performed for each 10-bit channel (Y and C for HD video, DS1 and DS2 for 3G video).

If a mismatch in the calculated and received CRC values is detected for Y channel data (Data Stream 1 for 3G video), the **YCRC_ERR** bit in the **ERROR_STAT_[2:1]** register is set HIGH.

If a mismatch in the calculated and received CRC values is detected for C channel data (Data Stream 2 for 3G video), the **CCRC_ERR** bit in the **ERROR_STAT_[2:1]** register is set HIGH.

Y or C CRC errors are also generated if CRC values are not embedded.

3G Level B signals all consist of two data streams. Each data stream is a multiplex of a C channel and a Y channel. Each channel of each data stream has CRC.

For 3G Level B formats, YCRC errors are detected for both of the two Y channels, and CCRC errors are detected for both of the two C channels. Data stream 1 (Link A) YCRC/CCRC errors are reported through Address 806_h/807_h. Data stream 2 (Link B) YCRC/CCRC errors are reported through Address 808_h/809_h.

Note: By default, 8-bit to 10-bit TRS remapping is enabled. If an 8-bit input is used, the HD CRC check is based on the 10-bit remapped value, not the 8-bit value, so the CRC Error Flag is incorrectly asserted and should be ignored. If 8-bit to 10-bit remapping is enabled, then CRC insertion should be enabled by setting the **CRC_INS_DS[2:1]_MASK** bit LOW in the **IOPROC_1** or **IOPROC_2** register. This ensures that the CRC values are updated.

4.16.3 EDH CRC Error Detection

The GS3470 also calculates Full Field (FF) and Active Picture (AP) CRC's according to SMPTE RP165 in support of Error Detection and Handling packets in SD signals.

These calculated CRC values are compared with the received CRC values.

Error flags for AP and FF CRC errors are provided and each error flag is a logical OR of field 1 and field 2 error conditions.

The **AP_CRC_ERR** bit in the **ERROR_STAT_1** register is set HIGH when an Active Picture CRC mismatch has been detected in field 1 or 2.

The **FF_CRC_ERR** bit in the **ERROR_STAT_1** register is set HIGH when a Full Field CRC mismatch has been detected in field 1 or 2.

EDH CRC errors are only indicated when the device is operating in SD mode and when the device has correctly received EDH packets.

4.16.4 HD & 3G Line Number Error Detection

If a mismatch in the calculated and received line numbers is detected, the **LNUM_ERR** bit in the **ERROR_STAT_2[1]** register is set HIGH.

4.17 Ancillary Data Detection & Indication

The GS3470 detects ancillary data in both the vertical and horizontal ancillary data spaces. The status signal outputs Y/1ANC and C/2ANC are provided to indicate the position of ancillary data in the output data streams. These signals may be selected for output on the multifunction output pins, *STAT[5:0]*.

The GS3470 indicates the presence of all types of ancillary data by detecting the 000_h, 3FF_h, 3FF_h (00_h, FF_h, FF_h for 8-bit video) ancillary data preamble.

By default (at power up or after system reset) the GS3470 indicates all types of ancillary data. Up to 5 types of ancillary data can be specifically programmed for recognition.

For HD video signals, ancillary data may be placed in both the Y and Cb/Cr video data streams separately. For SD video signals, the ancillary data is multiplexed and combined into the YCbCr data space.

For 3G signals, ancillary data may exist in either or both of the virtual interface data streams. Both data streams are examined for ancillary data.

For a 3G data stream formatted as per Level A mapping:

- The ancillary data is placed in Data Stream 1 first, with overflow into Data Stream 2
- SMPTE ST 352 packets are mapped into both data streams

For a 3G data stream formatted as per Level B mapping:

- Each multiplexed data stream forming the 3G signal contains ancillary data embedded according to SMPTE ST 291
- Each multiplexed data stream forming the 3G signal contains SMPTE ST 352 packets mapped into the "Y" channel of Link A and Link B according to SMPTE ST 425

For Y/1ANC and C/2ANC assertion and de-assertion while operating in SD, HD and 3G Level A, please refer to [Figure 4-22: Y/1ANC and C/2ANC Signal Timing](#).

When detecting ancillary data in 3G Level B data, the Y/1ANC status output is HIGH when Data Stream 1 ancillary data is detected on either Y or C channels and the C/2ANC status output is HIGH whenever Data Stream 2 ancillary data is detected on either Y or C channels.

These status signal outputs are synchronous with PCLK and may be used as clock-enables for external logic, or as write-enables for an external FIFO or other memory devices.

Note 1: When I/O processing is disabled, the Y/1ANC and C/2ANC flags may toggle, but they are invalid and should be ignored.

Note 2: For 3G Level B 20-bit data, the Y/1ANC flag identifies all ANC data on Data Stream 1 (Link A), whether it is embedded in the Y or C component – ANC data is not identified separately for each component. Similarly, the C/2ANC flag identifies all ANC data on Data Stream 2 (Link B), whether it is embedded in the Y or C component.

Note 3: For 3G level B 10-bit data, the Y/1 ANC flag and C/2 ANC flag outputs are dependent on the setting of **ANC_EXT_SEL_DS2_DS1** and **ANC_EXT_SEL_DS1_Y_DS2_Y** as shown in [Figure 4-24](#).

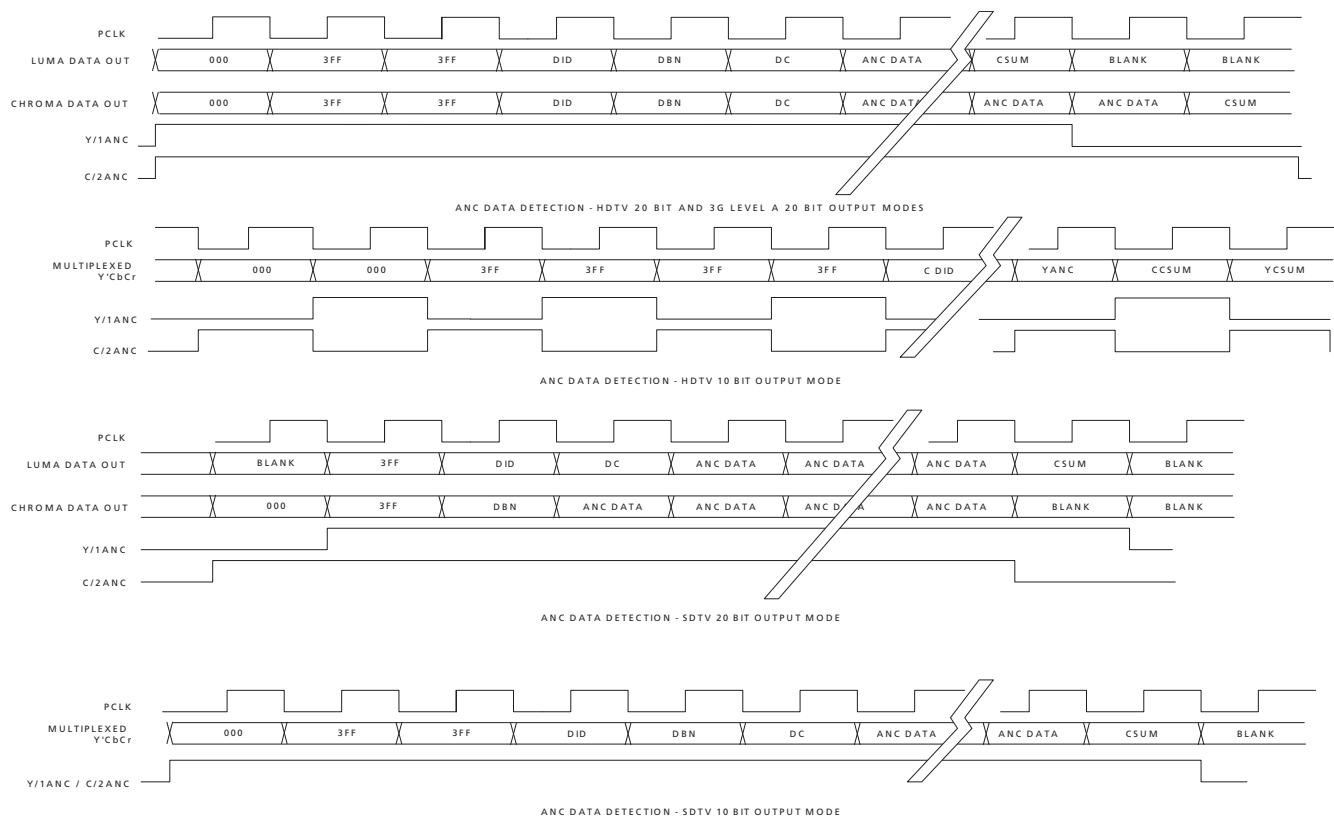


Figure 4-22: Y/1ANC and C/2ANC Signal Timing

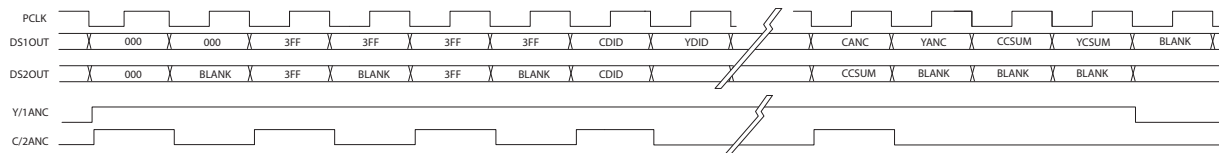


Figure 4-23: 3G Level B 20-bit Mode

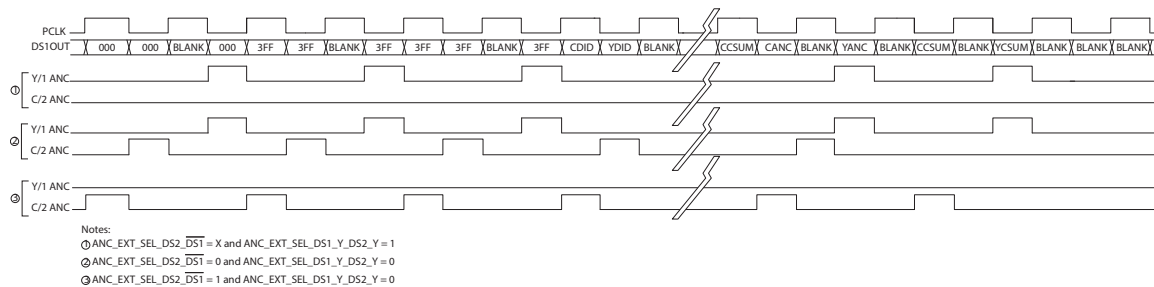


Figure 4-24: 3G Level B 10-bit Mode

4.17.1 Programmable Ancillary Data Detection

As described above in [Section 4.17](#), the GS3470 detects and indicates all ancillary data types by default.

Up to 5 different ancillary data types may be programmed for detection in the **ANC_TYPE[5:1]_DS[2:1]** register for SD, HD, and 3G Level A. When these are programmed, the GS3470 only indicates the presence of the specified ancillary data types, ignoring all other types. For each data type to be detected, the user must program the DID and/or SDID of that ancillary data type. If no DID or SDID values are programmed, the GS3470 indicates the presence of all ancillary data.

For any DID or SDID value set to zero, no comparison or match is made. For example, if the DID is programmed and the SDID is not programmed, the GS3470 only detects a match to the DID value.

If both DID and SDID values are non-zero, then the received ancillary data type must match both the DID and SDID before Y/1ANC and/or C/2ANC is set HIGH.

Note 1: For 3G Level B data, the **ANC_TYPE[5:1]_DS1** registers are valid for Data Stream 1, and a second set of **ANC_TYPE[5:1]_DS2** is provided for detection of specific ancillary data in Data Stream 2.

Note 2: SMPTE ST 352 Payload Identifier packets and Error Detection and Handling (EDH) Packets are always detected by the GS3470, regardless of the settings of the **ANC_TYPE[5:1]_DS1** registers and cannot be overridden.

4.17.1.1 Programmable Ancillary Data Checksum Calculation

As described above, the GS3470 calculates and compares checksum values for all ancillary data types by default. It is possible to program which ancillary data types are checked as described in [Section 4.17.1](#).

When so programmed, the GS3470 only checks ancillary data checksums for the specified data types, ignoring all other ancillary data.

The **YCS_ERR [2:1]** and/or **CCS_ERR [2:1]** bits in the **ERROR_STAT [2:1]** register are only set HIGH if an error condition is detected for the programmed ancillary data types.

4.17.2 SMPTE ST 352 Payload Identifier

The SMPTE ST 352 Payload Identifier is used to confirm the video format identified by the Automatic Video Standards Detection block.

Information contained in the packet is outlined in [Table 4-13](#).

Table 4-13: SMPTE ST 352 Packet Data

Bit Name	Bit	Name	Description	R/W	Default
VIDEO_FORMAT_352_A [2:1] Address: 828 _h , 82C _h	15:8	SMPTE ST 352 Byte 2	Data is available in this register when Video Payload Identification Packets are detected in the data stream.	R	0
	7:0	SMPTE ST 352 Byte 1		R	0
VIDEO_FORMAT_352_B [2:1] Address: 829 _h , 82D _h	15:8	SMPTE ST 352 Byte 4		R	0
	7:0	SMPTE ST 352 Byte 3		R	0
VIDEO_FORMAT_352_C [2:1] Address: 82A _h , 82E _h	15:8	SMPTE ST 352 Byte 2		R	0
	7:0	SMPTE ST 352 Byte 1		R	0
VIDEO_FORMAT_352_D [2:1] Address: 82B _h , 82F _h	15:8	SMPTE ST 352 Byte 4		R	0
	7:0	SMPTE ST 352 Byte 3		R	0

The GS3470 automatically extracts the SMPTE ST 352 payload identifier present in the input Luma data stream. For SD, HD, and 3G Level A, the bytes are written to

VIDEO_FORMAT_352_X_1. For 3G Level B, they are also written to **VIDEO_FORMAT_352_X_2**.

The device also indicates the version of the payload packet in **VERSION_352M** (bit-7 of byte 1) of the **VIDEO_FORMAT_352_X_X** register as per SMPTE 352M. When the SMPTE ST 352 packet is formatted as a “version 1” packet, the **VERSION_352M** bit is set LOW, when the packet is formatted as a “version 2” packet, this bit is set HIGH.

The **VIDEO_FORMAT_352_X_1** and **VIDEO_FORMAT_352_X_2** registers are only updated if checksum errors are not present.

By default (at power up or after system reset), the **VIDEO_FORMAT_352_X_1** and **VIDEO_FORMAT_352_X_2** bits are set to 0, indicating an undefined format.

4.17.2.1 Extension for UHD Multi-Link 3G Identification

UHD formats that are transported in multi-link 3G formats will have a link identifier present in bits [7:4] of byte 4 of the 352M packet. This information is saved to 4 bits in the **VID_STD** register.

4.17.3 Ancillary Data Checksum Error

The GS3470 calculates checksums for all received ancillary data. These calculated checksums are compared with the received ancillary data checksum words.

If a mismatch in the calculated and received checksums is detected, then a checksum error is indicated.

When operating in HD mode, the device makes comparisons on both the Y and C channels separately. If an error condition in the Y channel is detected, the **YCS_ERR_1** bit in the **ERROR_STAT_1** register is set HIGH. If an error condition in the C channel is detected, the **CCS_ERR_1** bit in the **ERROR_STAT_1** register is set HIGH.

When operating in 3G Level A mode, the device makes comparisons on both the Y (Data Stream 1) and C (Data Stream 2) channels separately. If an error condition in the Y channel is detected, the **YCS_ERR_1** bit in the **ERROR_STAT_1** register is set HIGH. If an error condition in the C channel is detected, the **CCS_ERR_1** bit in the **ERROR_STAT_1** register is set HIGH.

When operating in 3G Level B mode, the device makes comparisons on both the Y channel and the C channel of both Link A and Link B. For Link A, if an error condition in the Y channel is detected, the **YCS_ERR_1** bit in the **ERROR_STAT_1** register is set HIGH. If an error condition in the C channel is detected, the **CCS_ERR_1** bit in the **ERROR_STAT_1** register is set LOW. For Link B, if an error condition in the Y channel is detected, the **YCS_ERR_2** bit in the **ERROR_STAT_2** register is set HIGH. If an error condition in the C channel is detected, the **CCS_ERR_2** bit in the **ERROR_STAT_2** register is set LOW.

When operating in SD mode, only the **YCS_ERR_1** bit in **ERROR_STAT_1** is set HIGH when checksum errors are detected.

4.18 Signal Processing

The GS3470 can correct errors by inserting corrected code words, checksums, and CRC values into the data stream. The following processing can be performed by the GS3470:

1. TRS insertion
2. HD line based CRC insertion
3. EDH CRC insertion
4. HD line number insertion
5. Illegal code re-mapping
6. Ancillary data checksum insertion
7. Audio extraction
8. Ancillary data extraction
9. PID regeneration

All of the above features are only available in SMPTE mode ($\overline{\text{SMPTE_BYPASS}} = \text{HIGH}$). To enable these features, the **IOPROC_EN** bit must be set HIGH, and the individual feature must be enabled via bits in the **IOPROC_1** and/or **IOPROC_2** (depending on the data stream) register(s).

The **IOPROC_1** and **IOPROC_2** registers contains one bit for each processing feature allowing each one to be enabled/disabled individually.

By default (at power-up or after device reset), all of the **IOPROC_1** and **IOPROC_2** register bits described in Table 4-14 below are set to zero (0), which enables all of the processing features.

To disable an individual processing feature, set the corresponding bit to one (1) in the **IOPROC_1** and/or **IOPROC_2** register(s).

Table 4-14: IOPROC_1 and IOPROC_2 Register Bits

Processing Feature	IOPROC_1 Register Bit	IOPROC_2 Register Bit
TRS insertion	TRS_INS_DS1_MASK	TRS_INS_DS2_MASK
Y and C line based CRC insertion	CRC_INS_DS1_MASK	CRC_INS_DS2_MASK
Y and C line number insertion	LNUM_INS_DS1_MASK	LNUM_INS_DS2_MASK
Ancillary data check sum insertion	ANC_CHECKSUM_INSERTION_DS1_MASK	ANC_CHECKSUM_INSERTION_DS2_MASK
EDH CRC insertion	EDH_CRC_INS_MASK	N/A
Illegal code re-mapping	ILLEGAL_WORD_REMAP_DS1_MASK	ILLEGAL_WORD_REMAP_DS2_MASK
H timing signal configuration	H_CONFIG	N/A
Update EDH Flags	EDH_FLAG_UPDATE_MASK	N/A
Audio Data Extraction	AUD_EXT_MASK	AUDIO_SEL_DS2_DS1
Ancillary Data Extraction	ANC_DATA_EXT_MASK	ANC_EXT_SEL_DS2_DS1, ANC_EXT_SEL_DS1_Y_DS2_Y

Table 4-14: IOPROC_1 and IOPROC_2 Register Bits (Continued)

Processing Feature	IOPROC_1 Register Bit	IOPROC_2 Register Bit
Regeneration of ST 352 packets	N/A	REGEN_352_MASK

Note: IOPROC_2 for Level B - 3G only. See [IOPROC_1](#) and [IOPROC_2](#) for description of ancillary data extraction.

4.18.1 Audio De-Embedding Mode

The GS3470 includes an integrated audio de-embedder which is active when the deserializer is configured for SMPTE mode unless disabled using the *AUDIO_EN/DIS* pin. In non-SMPTE modes, the audio de-embedder is powered down to reduce power consumption. All output pins are LOW when the de-embedder is powered down.

For detailed description of this feature, refer to [Section 4.19](#).

4.18.2 TRS Insertion

When TRS Insertion is enabled, the GS3470 generates and overwrites TRS code words as required.

TRS Word Generation and Insertion is performed using the timing generated by the Timing Signal Generator, providing an element of noise immunity over using just the received TRS information.

This feature is enabled when **IOPROC_EN** (Address 877_h, bit 0) is HIGH and the **TRS_INS_DS[2:1]_MASK** bits in the **IOPROC_[2:1]** registers are set LOW. The **TRS_INS_DS1_MASK** bit is in the **IOPROC_1** register and is used to enable/disable TRS insertion for SD, HD, 3G-A data streams, and data stream 1 of 3G Level B. The **TRS_INS_DS2_MASK** bit is in the **IOPROC_2** register and is used to enable/disable TRS insertion for 3G-B Data Stream 2 only.

For 3G Level A signals, TRS insertion occurs in both Data Stream 1 and Data Stream 2.

For 3G Level B signals, TRS insertion occurs in both Data Stream 1 and Data Stream 2 of both Link A and Link B.

Note: Inserted TRS code words are always 10-bit compliant, regardless of the bit depth of the incoming video stream.

4.18.3 Line Based CRC Insertion

When CRC Insertion is enabled, the GS3470 generates and inserts line based CRC words into both the Y and C channels of the data stream.

Line based CRC word generation and insertion only occurs in HD and 3G modes, and is enabled in when the **IOPROC_EN** bit is HIGH and the **CRC_INS_DS[2:1]_MASK** bit in the **ICPROC_[2:1]** register is set LOW.

For 3G Level A signals, line based CRC word generation and insertion occurs in both Data Stream 1 and Data Stream 2.

For 3G Level B signals, line based CRC word generation and insertion occurs in both Data Stream 1 and Data Stream 2 of both Link A and Link B.

The **CRC_INS_DS1_MASK** bit is used to enable/disable CRC insertion for HD, 3G-A data streams and data stream 1 of 3G-B. The **CRC_INS_DS2_MASK** bit is used to enable/disable CRC insertion for data stream 2 of 3G Level B.

4.18.4 Line Number Insertion

When Line Number Insertion is enabled, the GS3470 calculates and inserts line numbers into the output data stream. Re-calculated line numbers are inserted into both the Y and C channels.

Line number generation is in accordance with the relevant HD or 3G video standard as determined by the Automatic Standards Detection block.

This feature is enabled when the device is operating in HD or 3G modes, the **IOPROC_EN** bit is HIGH and the **LNUM_INS_DS[2:1]_MASK** bit in the **IOPROC_[2:1]** register is set LOW.

For 3G Level A signals, line number insertion occurs in both Data Stream 1 and Data Stream 2.

For 3G Level B signals, line number insertion occurs in both Data Stream 1 and Data Stream 2 of both Link A and Link B.

The **LNUM_INS_DS1_MASK** bit is used to enable/disable line number insertion for HD, 3G-A data streams and Data Stream 1 of 3G Level B. The **LNUM_INS_DS2_MASK** bit is used to enable/disable line number insertion for Data Stream 2 of 3G Level B.

4.18.5 ANC Data Checksum Insertion

When ANC data Checksum Insertion is enabled, the GS3470 generates and inserts ancillary data checksums for all ancillary data words by default.

Where user specified ancillary data has been programmed (see [Section 4.17.1](#)), only the checksums for the programmed ancillary data are corrected.

This feature is enabled when the **IOPROC_EN** bit is HIGH and the **ANC_CHECKSUM_INSERTION_DS[2:1]_MASK** bit in the **IOPROC_[2:1]** register is set LOW.

For 3G Level A signals, ANC data checksum insertion occurs in both Data Stream 1 and Data Stream 2.

For 3G Level B signals, ANC data checksum insertion occurs in both Y and C channels of both Data Stream 1 and Data Stream 2.

The **ANC_CHECKSUM_INSERTION_DS1_MASK** bit is used to enable/disable ANC data checksum insertion for SD, HD, 3G-A data streams and Data Stream 1 of 3G Level B. The **ANC_CHECKSUM_INSERTION_DS2_MASK** bit is used to enable/disable ANC data checksum insertion for Data Stream 2 of 3G Level B.

4.18.6 EDH CRC Insertion

When EDH CRC Insertion is enabled, the GS3470 generates and overwrites full field and active picture CRC check-words. Additionally, the device sets the active picture and full field CRC 'V' bits HIGH in the EDH packet. The **AP_CRC_V** and **FF_CRC_V** register bits only report the received EDH validity flags.

Although the GS3470 modifies and inserts EDH CRC's and EDH packet checksums, EDH error flags are only updated when the **EDH_FLAG_UPDATE_MASK** bit is LOW.

This feature is enabled in SD mode, when the **IOPROC_EN** bit is HIGH and the **EDH_CRC_INS_MASK** bit in the **IOPROC_1** register is set LOW.

4.18.7 Illegal Word Re-mapping

All words within the active picture (outside the horizontal and vertical blanking periods), between the values of 3FC_h and 3FF_h are re-mapped to 3FB_h. All words within the active picture area between the values of 000_h and 003_h are remapped to 004_h.

This feature is enabled when the **IOPROC_EN** bit is HIGH and the **ILLEGAL_WORD_REMAP_DS[2:1]_MASK** bit in the **IOPROC_[2:1]** register is set LOW.

For 3G Level A signals, illegal code remapping occurs in both Data Stream 1 and Data Stream 2.

For 3G Level B signals, illegal code remapping occurs in both Data Stream 1 and Data Stream 2 of both Link A and Link B.

The **ILLEGAL_WORD_REMAP_DS1_MASK** bit is used to enable/disable illegal word remapping for SD, HD, 3G-A data streams and Data Stream 1 of 3G Level B. The **ILLEGAL_WORD_REMAP_DS2_MASK** bit is used to enable/disable illegal word remapping for Data Stream 2 of 3G Level B.

4.18.8 TRS and Ancillary Data Preamble Remapping

8-bit TRS and ancillary data preambles are re-mapped to 10-bit values. 8-bit to 10-bit mapping of TRS headers is only supported if the TRS values are 3FC 000 000_h. Other values such as 3FD_h, 3FE_h, 001_h, 002_h, and 003_h are not supported.

This feature is enabled by default, and can be disabled via the **IOPROC_[2:1]** register using **TRS_WORD_REMAP_FE[2:1]_DISABLE_MASK**.

4.18.9 Ancillary Data Extraction

The GS3470 includes a FIFO, which extracts ancillary data using read access via the host interface to ease system implementation. The FIFO stores up to 2048 x 16 bit words of ancillary data in two separate 1024 word memory banks.

As an alternative, ancillary data may be extracted externally from the GS3470 output stream using the Y/1ANC and C/2ANC signals, and external logic.

Data is accessed from both memory banks using the same host interface addresses, C00_h to FFF_h.

The device writes the contents of ANC packets into the FIFO, starting with the first Ancillary Data Flag (ADF), followed by up to 1024 words.

All Data Identification (DID), Secondary Data Identification (SDID), Data Count (DC), user data, and checksum words are written into the device memory.

The device detects ancillary data packet DID's placed anywhere in the video data stream, including the active picture area.

In HD and 3G mode, ancillary data from the Y channel or Data Stream 1 is placed in the Least Significant Word (LSW) of the FIFO, allocated to the lower 8 bits of each FIFO address.

Ancillary data from the C channel or Data Stream 2 is placed in the Most Significant Word (MSW) (upper 8 bits) of each FIFO address.

Note: Please contact Semtech FAE for further information about ANC insertion and extraction and an example of ancillary data extraction using the GS3470.

In SD mode, ancillary data is placed in the LSW of the FIFO. The MSW is set to zero.

If the **ANC_TYPE_[5:1]_DS[2:1]** (Address 81E_h to 827_h) registers are all set to zero, the device extracts all types of ancillary data. If programmable ancillary data extraction is required, then up to five types of ancillary data to be extracted can be programmed in the **ANC_TYPE_[5:1]_DS[2:1]** registers (see [Section 4.17.1](#)).

Additionally, the lines from which the packets are to be extracted can be programmed into the **ANC_LINEA** (Address 818_h, bit [10:0]) and **ANC_LINEB** (Address 819_h, bit [10:0]) registers, allowing ancillary data from a maximum of two lines per frame to be extracted. If only one line number register is programmed (with the other set to zero), ancillary data packets are extracted from one line per frame only. When both registers are set to zero, the device extracts packets from all lines.

To start Ancillary Data Extraction, **ANC_DATA_EXT_MASK** (Address 800_h, bit 8) in the **IOPROC_1** register must be set LOW. Ancillary data packet extraction begins in the following frame (see [Figure 4-25: Ancillary Data Extraction - Step A](#)).

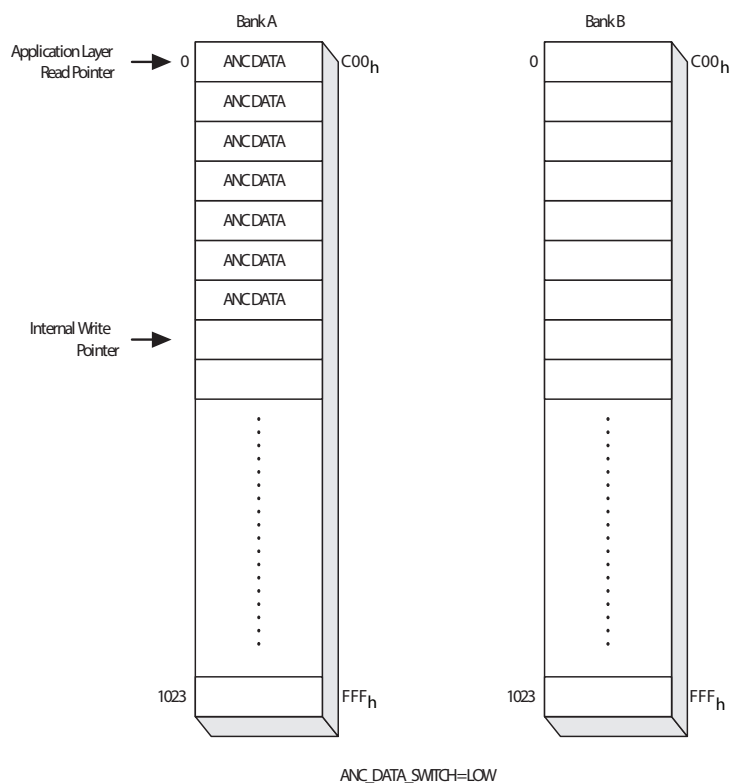


Figure 4-25: Ancillary Data Extraction - Step A

Ancillary data is written into Bank A until full. The Y/1ANC and C/2ANC output flags can be used to determine the length of the ancillary data extracted and when to begin reading the extracted data from memory.

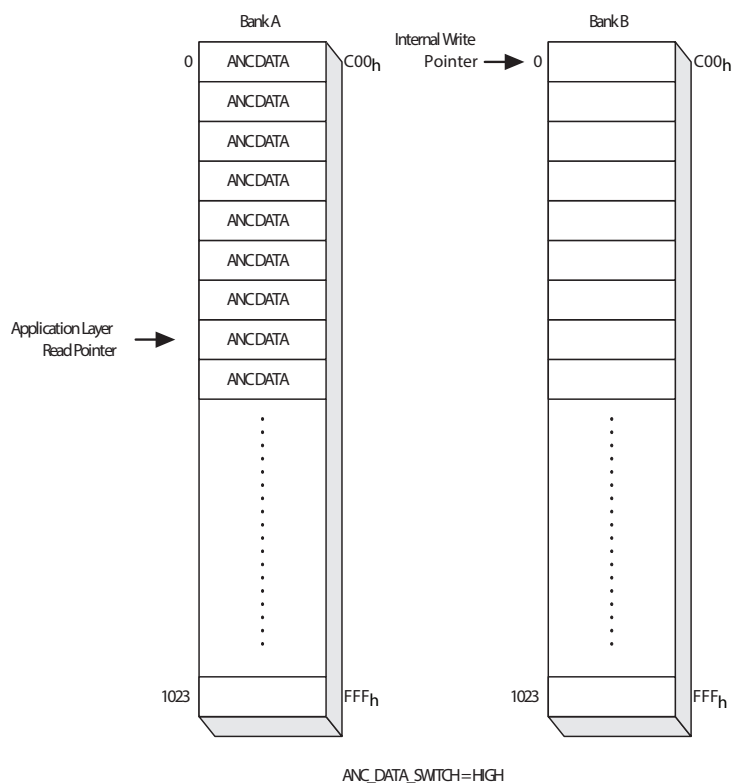


Figure 4-26: Ancillary Data Extraction - Step B

While the **ANC_DATA_EXT_MASK** bit is set LOW, the **ANC_DATA_SWITCH** (Address 817_h, bit 3) can be set HIGH during or after reading the extracted data. New data is then written into Bank B (up to 1024 x 16-bit words), at the corresponding host interface addresses (see [Figure 4-26: Ancillary Data Extraction - Step B](#)).

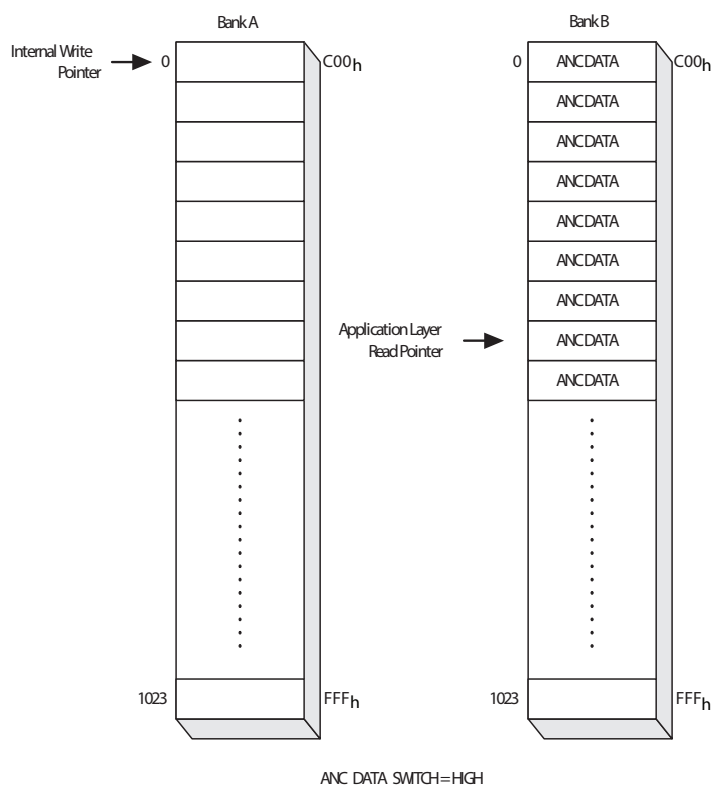


Figure 4-28: Ancillary Data Extraction - Step D

If the **ANC_DATA_SWITCH** bit is not toggled, extracted data is written into Bank B until full. To continue extraction in Bank A, the **ANC_DATA_SWITCH** bit must be toggled HIGH (see [Figure 4-28: Ancillary Data Extraction - Step D](#)).

Toggling the **ANC_DATA_SWITCH** bit LOW returns the process to step A ([Figure 4-25](#)).

Note 1: Toggling the **ANC_DATA_SWITCH** must occur at a time when no extraction is taking place, i.e. when the both the Y/1ANC and C/2ANC signals are LOW.

To turn extraction off, the **ANC_DATA_EXT_MASK** bit must be set HIGH.

In HD mode, the device can detect ancillary data packets in the Luma video data only, Chroma video data only, or both. By default (at power-up or after a system reset) the device extracts ancillary data packets from the Luma channel only.

In 3G mode Level A, the device can detect ancillary data packets in Luma video (Data Stream 1) only, Chroma video (Data Stream 2) only, or both. By default (at power-up or after a system reset) the device extracts ancillary data packets from Data Stream 1 only.

In 3G mode Level B, the device can detect ancillary data packets in Luma video only, Chroma video only, or both from either Link A or Link B. Selection of Link A or Link B for ANC data extraction is done via the host interface.

In 3G Level B the device may be programmed via the user interface so that the YANC data from Link A is extracted as the YANC data and the YANC data from Link B is extracted as CANC data. By default, or at reset or power-up, this is the setup for 3G level B input signals.

To extract packets from the Chroma/Data Stream 2 channel only, **HD_ANC_C2** (Address 817_h, bit 0) must be set HIGH. To extract packets from both Luma/Data Stream 1 and Chroma/Data Stream 2 video data, the **HD_ANC_Y1_C2** bit must be set HIGH (the setting of the **HD_ANC_C2** bit is ignored).

The default setting: **HD_ANC_C2** = LOW and **HD_ANC_Y1_C2** = LOW is the setting to extract ANC packet from Luma/Data Stream 1 when the device is configured for HG/3G video standards. The setting of these bits is ignored when the device is configured for SD video standards.

After extraction, the ancillary data may be deleted from the video stream by setting **ANC_DATA_DEL** (Address 817_h, bit 2) HIGH. When set HIGH, all existing ancillary data is removed and replaced with blanking values on the parallel data outputs. If any of the **ANC_TYPE_[5:1]_DS[2:1]** registers are programmed with a DID and/or SDID, only the ancillary data packets with the matching IDs are deleted from the video stream.

Ancillary data packet extraction and deletion is disabled when the **IOPROC_EN** bit is set LOW.

Note 2: After the ancillary data determined by the **ANC_TYPE_[5:1]_DS[2:1]** registers has been deleted, other existing ancillary data may not be contiguous. The device does not concatenate the remaining ancillary data.

Note 3: Reading extracted ancillary data from the host interface must be performed while there is a valid video signal present at the serial input and the device is locked (LOCKED signal is HIGH).

This allows a software only ANC data extraction application, which is only interfaced to the device via the host interface, and does not have access to the Y/1ANC and C/2ANC output flags, to determine when ANC data has been detected by the device.

4.19 Audio De-Embedder

Up to eight channels of audio may be extracted from the received serial digital video stream. The output signal formats supported by the device include AES/EBU, I²S (default), and industry standard serial digital formats.

The audio de-embedder is enabled by default. It can be disabled by setting **AUD_EXT_MASK** (Address 800_h, bit 7) HIGH.

16, 20, and 24-bit audio bit depths are supported for 48kHz synchronous audio for SD data rates. For HD and 3G data rates, 16, 20, and 24-bit audio bit depths are supported for 48kHz audio. The audio may be synchronous or asynchronous to the video.

In 3G mode:

- In Level A mode, all Audio Control Packets are extracted from Data Stream 1 and all Audio Data Packets are extracted from Data Stream 2, in accordance with SMPTE ST 425. This is similar to HD, in which Audio Control Packets are embedded in the Luma channel and Audio Data Packets in the Chroma channel
- In Level B mode, extraction of audio packets from Link A (default) or Link B is selectable via the **AUDIO_SEL_DS2_DS1** bit in the host interface.
 - ♦ **AUDIO_SEL_DS2_DS1** = 0: Extract audio packet from Link A
 - ♦ **AUDIO_SEL_DS2_DS1** = 1: Extract audio packet from Link B

Additional audio processing features include audio mute on loss of lock, de-embed and delete, group selection, audio output re-mapping, ECC error detection and correction (HD/3G modes only), and audio channel status extraction.

4.19.1 Serial Audio Data I/O Signals

The Serial Audio Data I/O pins are listed in [Table 4-15: Serial Audio Pin Descriptions](#).

Table 4-15: Serial Audio Pin Descriptions

Pin Name	Description
AUDIO_EN/ $\overline{DIS}$	Enable Input for Audio Processing
AOUT_1_2	Serial Audio Output; Channels 1 and 2
AOUT_3_4	Serial Audio Output; Channels 3 and 4
AOUT_5_6	Serial Audio Output; Channels 5 and 6
AOUT_7_8	Serial Audio Output; Channels 7 and 8
ACLK	64fs Audio Bit Clock
WCLK	Word Clock
AMCLK	Audio Master Clock, selectable 128fs, 256fs, or 512fs

The timing of the serial audio out signals, the *WCLK* output signal and the *ACLK* output is shown in [Figure 4-29: ACLK to Data Signal Output Timing](#).

When *AUDIO_EN/ $\overline{DIS}$* pin is set HIGH, audio extraction is enabled and the audio output signals are extracted from the video data stream. When set LOW, the serial audio outputs, *ACLK*, and *WCLK* outputs are set LOW.

In addition, all functional logic associated with audio extraction is disabled to reduce power consumption.

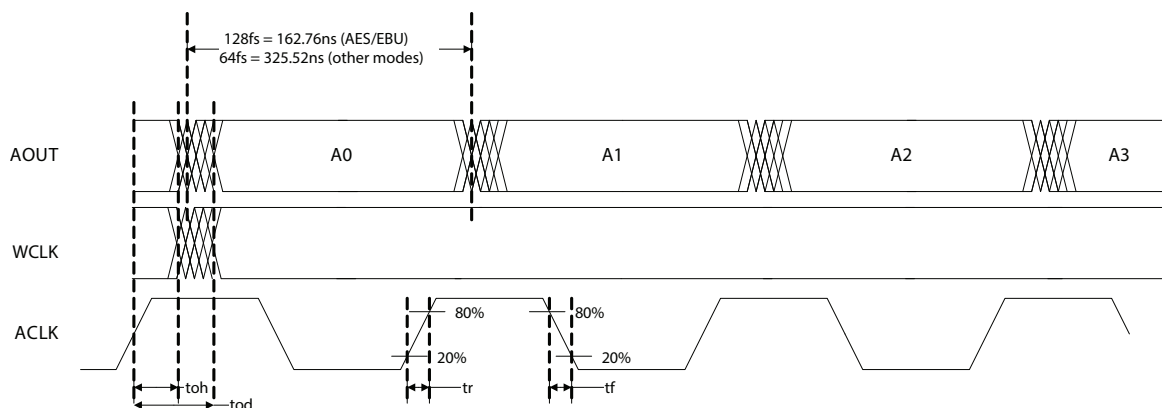


Figure 4-29: ACLK to Data Signal Output Timing

4.19.2 Serial Audio Data Format Support

The GS3470 supports the following serial audio data formats:

- I²S (default)
- AES/EBU
- Serial Audio Left Justified, MSB First
- Serial Audio Right Justified, MSB First (supported for HD and 3G only)

By default (at power up or after system reset) I²S is selected. The other data formats are selectable via **AMA/AMB[1:0]** of **CFG_AUD**(HD/3G) (Address A01_h, bit[7:6]/bit[9:8]) or selectable via **AMA/AMB/AMC/AMD[1:0]** of **CFG_OUTPUT**(SD) (Address B0A_h, bit[1:0]/bit[3:2]/bit[5:4]/bit[7:6]).

Table 4-16: Audio Output Formats

AMA/AMB[1:0] (HD/3G) or AMA/AMB/AMC/ AMD[3:0] (SD)	Audio Output Format
00	AES/EBU audio output
01	Serial audio output: Left Justified; MSB first
10	Serial audio output: Right Justified; MSB first (supported for HD and 3G only)
11	I ² S (Default)

When I²S format is desired, both groups must be set to I²S (i.e. AMA = AMB = 11). This is because they share the same *WCLK*.

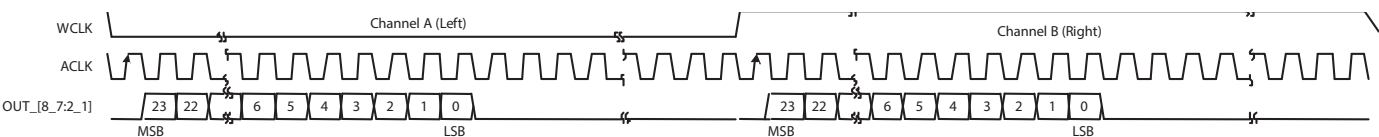


Figure 4-30: I²S Audio Output Format

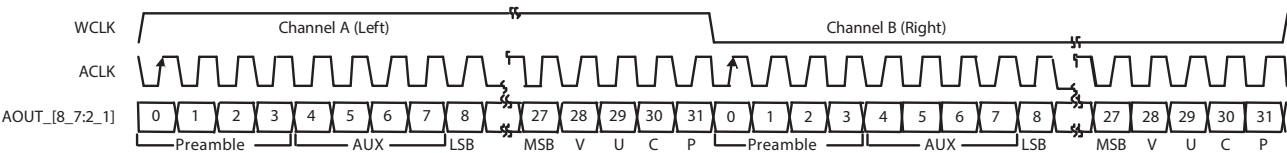


Figure 4-31: AES/EBU Audio Output Format

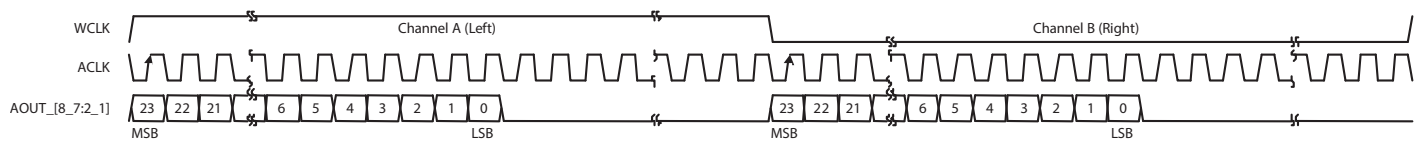


Figure 4-32: Serial Audio, Left Justified, MSB First

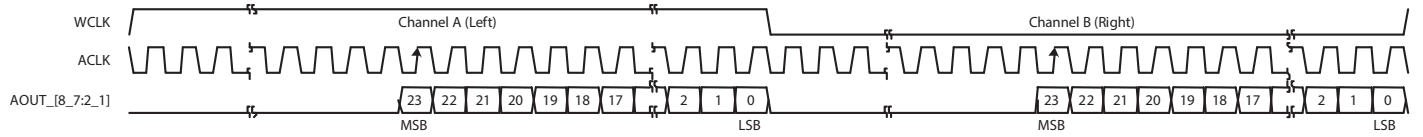


Figure 4-33: Serial Audio, Right Justified, MSB First

4.19.2.1 AES/EBU Mode

In AES/EBU output mode, the audio de-embedder uses a 128fs (6.144MHz audio bit clock) clock as shown in [Figure 4-34](#).

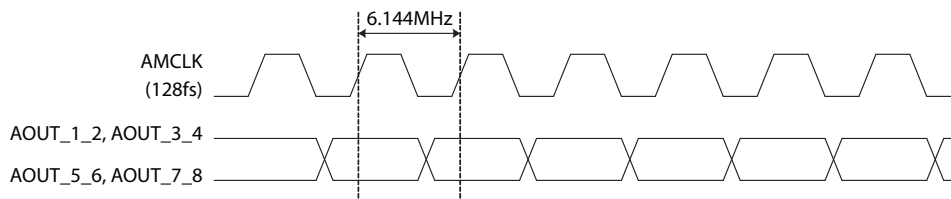


Figure 4-34: AES/EBU Audio Output to Bit Clock Timing

4.19.2.2 Audio Data Packet Extraction Block

The audio de-embedder looks for audio data packets on every line of the incoming video.

The audio data must be embedded according to SMPTE ST 272 (SD) or SMPTE ST 299/299-1/299-2 (HD or 3G).

In 3G Level A signals, the audio data packets must be embedded only in Data Stream 2.

In 3G Level B signals, the audio data packets must be embedded in the Data Stream 2 of either Link A or Link B and have a data rate of 74.25MHz.

The Audio Group Detect registers are set HIGH when audio data packets with a corresponding group DID are detected in the input video stream. The host interface reports the individual audio groups detected.

Table 4-17: Audio Data Packet Detect Register

Name	Description	Default
ADPG8_DET	Audio Group Eight Data Packet Detection (1: Detected) (HD/3G only)	0
ADPG7_DET	Audio Group Seven Data Packet Detection (1: Detected) (HD/3G only)	0
ADPG6_DET	Audio Group Six Data Packet Detection (1: Detected) (HD/3G only)	0
ADPG5_DET	Audio Group Five Data Packet Detection (1: Detected) (HD/3G only)	0
ADPG4_DET	Audio Group Four Data Packet Detection (1: Detected)	0
ADPG3_DET	Audio Group Three Data Packet Detection (1: Detected)	0
ADPG2_DET	Audio Group Two Data Packet Detection (1: Detected)	0
ADPG1_DET	Audio Group One Data Packet Detection (1: Detected)	0

For SD, when an audio data packet with a DID set in **IDA[1:0]** and **IDB[1:0]** in **CFG_AUD** of SD Audio Core CSR (Address B01_h, bit[1:0] and bit[3:2]) is detected, the audio sample information is extracted and written into the audio FIFO. For HD and 3G, **EXTEND_IDA** and **EXTEND_IDB** in **CFG_AUD** of HD and 3G Audio Core CSR (Address A01_h, bit 2 and bit 5) are used in addition to **IDA[1:0]** and **IDB[1:0]** in **CFG_AUD** of HD and 3G Audio Core CSR (Address A01_h, bit[1:0] and bit[4:3]) to select one of 8 audio groups.

The embedded audio group selected by **IDA[1:0]** is described henceforth in this document as Group A or Primary Group. The embedded audio group selected by **IDB[1:0]** is described henceforth in this document as Group B or Secondary Group.

4.19.2.3 Audio Control Packets

The audio de-embedder automatically detects the presence of audio control packets in the video stream. When audio control packets for audio Group A are detected, **CTRA_DET** is set HIGH. When audio control packets for audio Group B are detected, **CTRB_DET** is set HIGH. For HD/3G, **CTRA_DET** (Address A03_h, bit 4) and **CTRB_DET** (Address A03_h, bit 5). For SD, **CTRA_DET** (Address B03_h, bit 4) and **CTRB_DET** (Address B03_h, bit 5).

The audio control packet data is accessible via the host interface.

The audio control packets must be embedded according to SMPTE ST 272 (SD) or SMPTE ST 299 (HD and 3G). In 3G Level A signals, the audio control packets must be embedded only in Data Stream 1. In 3G Level B signals the audio control packets must be embedded in the Luma streams of each link that carries audio.

Note 1: In SD, HD, and 3G, the audio control packet reporting via the host interface updates automatically when a new control packet is detected.

Note 2: If there is an HD audio packet checksum error, no audio is extracted. The audio packet is not recognized, and the audio stays in the video stream. If only the CLK phase parity bit is incorrect, the audio can still be extracted.

4.19.2.4 Setting Packet DID

Table 4-19 below, shows the 2-bit host interface setting for the audio group DID's.

For 24-bit audio support in SD mode, extended audio packets for Group A must have the same group DID set in **IDA[1:0]** in SD Audio Core CSR (Address B01_h, bit[1:0]). Extended audio packets for Group B must have the same group DID set in **IDB[1:0]** in SD Audio Core CSR (Address B01_h, bit[3:2]).

The audio de-embedder automatically detects the presence of extended audio packets. When detected, the audio output format is set to 24-bit audio sample word length.

The audio de-embedder defaults to audio Groups One and Two, where Group A is extracted from packets with audio Group One DID, and Group B from packets with audio Group Two DID.

Table 4-18: Audio Group DID Host Interface Settings

Audio Group	SD Data DID	SD Extended DID	SD Control DID	HD Data DID	HD Control DID	Host interface Register (2-bit)	Extended_ID Register Bit
1	2FF _h	1FE _h	1EF _h	2E7 _h	2E3 _h	00 _b	0
2	1FD _h	2FC _h	2EE _h	1E6 _h	2E2 _h	01 _b	0
3	1FB _h	2FA _h	2ED _h	1E5 _h	2E1 _h	10 _b	0
4	2F9 _h	1F8 _h	1EC _h	2E4 _h	1E0 _h	11 _b	0
5	—	—	—	1A7 _h	2A3 _h	00 _b	1
6	—	—	—	2A6 _h	1A2 _h	01 _b	1
7	—	—	—	2A5 _h	1A1 _h	10 _b	1
8	—	—	—	1A4 _h	2A0 _h	11 _b	1

Table 4-19: Audio Data and Control Packet DID Setting Register

Name	Description	Default
IDA[1:0]	Group A Audio data and control packet DID setting	00 _b
IDB[1:0]	Group B Audio data and control packet DID setting	01 _b
EXTEND_IDA	Group A Audio data and control packet DID setting (HD/3G only)	0 _b
EXTEND_IDB	Group B Audio data and control packet DID setting (HD/3G only)	0 _b

4.19.2.5 Audio Packet Delete Block

To delete all ancillary data with a group DID shown in Table 4-18, **ALL_DEL** (HD/3G: Address A02_h, bit2 and SD: Address B01_h, bit 13) must be set HIGH.

The device does not reorder packets when packets are deleted, so any remaining packets will not be adjusted to realign with EAV.

Note: Low latency mode option is not available when this feature is enabled.

4.19.2.6 ECC Error Detection & Correction Block (HD Mode Only)

The audio de-embedder performs BCH(31, 25) forward error detection and correction, as described in SMPTE ST 299-1. The error correction for all embedded audio data packets is activated when the host interface **ECC_OFF** bit is set LOW (default LOW). The audio de-embedder corrects any errors in both the audio output and the embedded packet.

When a one-bit error is detected in a bit array of the ECC protected region of the audio data packet with audio group DID set in **IDA[1:0]**, the **ECCA_ERROR** flag is set HIGH.

When a one-bit error is detected in the ECC protected region of the audio data packet with audio group DID set in **IDB[1:0]**, the **ECCB_ERROR** flag is set HIGH.

Figure 4-35 shows examples of error correction and detection. Up to 8 bits in error can be corrected, providing each bit error is in a different bit array (shown below). When there are two or more bits in error in the same 24-bit array, the errors are detected, but not corrected.

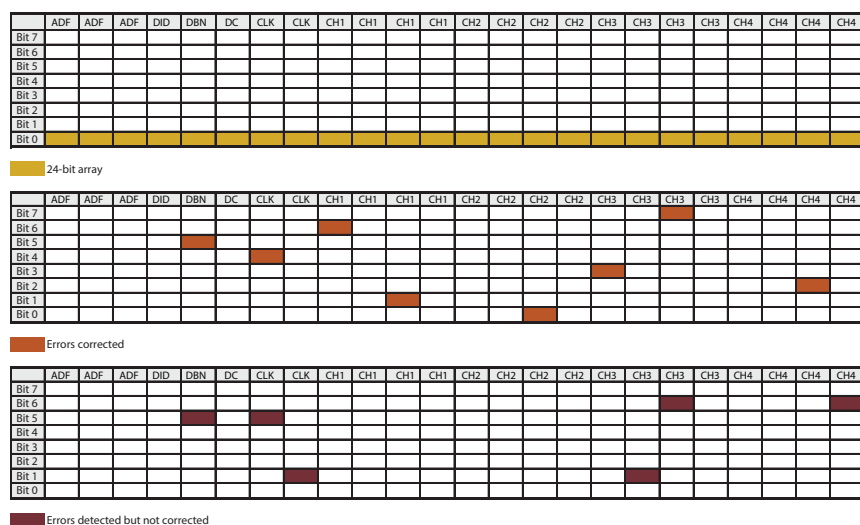


Figure 4-35: ECC 24-bit Array and Examples

4.19.3 Audio Processing

4.19.3.1 Audio Clock Generation

For SD and HD/3G audio, a single set of audio frequencies is generated for all audio channels.

For Mapping structure one signals (1080p 50, 59.94, 60 or 2048p 50, 59.94, 60), the pixel clock is 148.5/(M)MHz, and the phase data are based on this rate. An Audio Master Clock (AMCLK) is also generated. The frequency is selectable via the host interface as:

- $f_s \times 128$
- $f_s \times 256$
- $f_s \times 512$

In SD mode, audio clocks are derived from the *PCLK*.

In HD/3G modes, audio clocks are derived from the two embedded audio clock phase words in the audio data packet.

The audio de-embedder also includes a Flywheel block to overcome any inconsistencies in the embedded audio clock phase information.

If the audio phase data is not present or incorrect, the

INVALID_EMBEDDED_PHASE[B:A] (Address A0C_h, bit [3:2]) bits will be asserted and audio will be de-embedded using a clock derived from *PCLK* and the M value.

Alternatively, when the **IGNORE_PHASE** (Address A0B_h, bit 5) in the host interface is set HIGH, the M value can be programmed via the host interface. This can be done by setting the **FORCE_M** (Address A0B_h, bit 7) HIGH and programming the desired value into **FORCE_MEQ1001** (Address A0B_h, bit 6). The correct value can be obtained by reading the M bit from the Video Core Registers.

If the audio digital PLL is locked to phase data and audio data packets are lost or corrupted, the Clock Generator will flywheel for up to four audio data packets.

If the **IGNORE_PHASE** bit in the host interface is HIGH, the clock will free-run based on the video format, the *PCLK* and the M value, independent of the **INVALID_EMBEDDED_PHASE[B:A]** bits.

In the 720p/24 video format, the total line length is 4125 pixels, which requires a resolution of 13 bits for the audio clock phase words in the embedded audio data packets. SMPTE ST 299 only specifies a maximum of 12 bits resolution. The revised versions of SMPTE ST 299 require using bit 5 of UDW1 in the audio data packet as the MSB (ck12) for the audio clock phase data, providing 13 bits resolution.

Some audio encoders may hold the clock phase value at a maximum value when reached, until reset at the end of the line. This produces a small amount of audio phase jitter for the period of one sample. To overcome this issue, the audio de-embedder checks for all cases. On detection of the maximum value, a comparison is made between previous clock phases and the correct position interpolated. If the clock phase data value starts to decrease, the de-embedder checks to see if bit 5 (ck12) of UDW1 in the audio data packet is asserted. If ck12 is asserted, the correct value is used. If ck12 is not set, the correct position is interpolated.

4.19.3.2 Detect 5-Frame Sequence Block

Five-frame sequence detection is required for 525-line based video formats only. Figure 4-36 shows the number of samples per frame over a five-frame sequence.

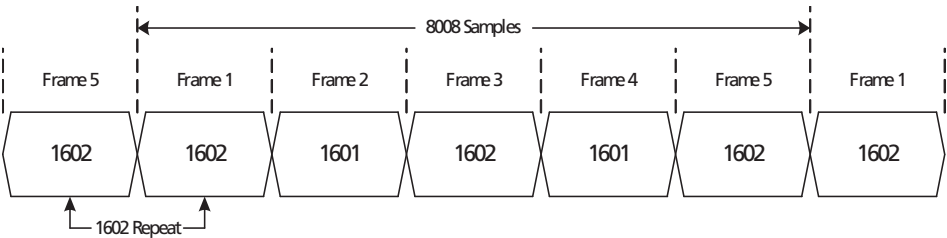


Figure 4-36: Sample Distribution over 5 Video Frames (525-line Systems)

When the audio inputs are asynchronously switched or disrupted, the audio de-embedder continues to write audio samples into the audio buffer, based on the current five-frame sequence. The de-embedder then re-locks to the new 5-frame sequence, at which point a sample may be lost.

Note: In SD, all four channel pairs must follow the same five-frame sequence.

4.19.3.3 SD Audio FIFO Block

The function of the FIFO block is to change the audio data word rate from the ANC rate multiplexed with the video signal to the 48kHz audio output rate.

The audio FIFO block contains the audio sample buffers; one per audio channel. Each buffer is 36 audio samples deep. At power up or reset, the read pointer is held at the zero position until 26 samples have been written into the FIFO (allows for 6 lines per frame with no audio samples; a maximum of 4 samples per line). See Figure 4-37.

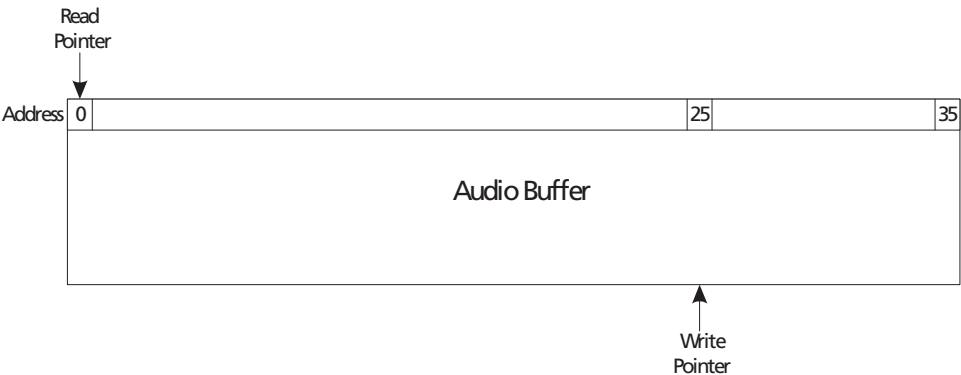


Figure 4-37: Audio Buffer After Initial 26 Sample Write

The position of the write pointer with respect to the read pointer is monitored continuously. If the write pointer is less than 6 samples ahead of the read pointer (point A in Figure 4-38), a sample is repeated from the read-side of the FIFO. If the write pointer is less than 6 samples behind the read pointer (point B in Figure 4-38), a sample is dropped. This avoids buffer underflow/overflow conditions.

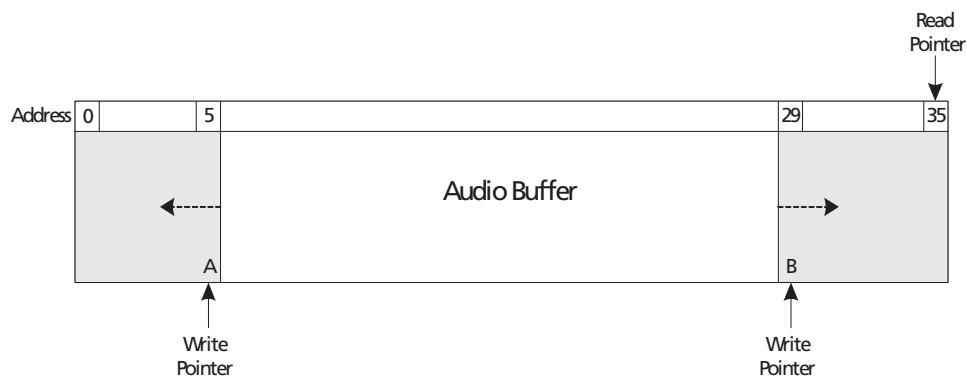


Figure 4-38: Audio Buffer Pointer Boundary Checking

The repeat or drop sample operation is performed a maximum of 28 consecutive times, after which the audio outputs are muted (all sample data set to zero). In SD mode, 26 samples are required to be written into the FIFO prior to starting the read operation again.

The audio buffer pointer offset may be reduced from 26 samples to 12 or 6 samples using the **OS_SEL[1:0]** (Address B01_h, bit [9:8]) of **CFG_AUD** of SD Audio Core CSR. The default setting is 26 samples (see Table 4-20).

When the **OS_SEL[1:0]** bits are set for 6-sample pointer offset, no boundary-checking is performed.

Table 4-20: Audio Buffer Pointer Offset Settings

OS_SEL[1:0]	Buffer Pointer Offset	FIFO Size
00 _b	26 samples (default)	36
01 _b	12 samples	22
10 _b	6 samples	16

4.19.3.4 Audio Crosspoint Block

The Audio Crosspoint is used for audio output channel re-mapping. This feature allows any of the selected audio channels in Group A or Group B to be output on any of the eight output channels. The default setting is for one to one mapping, where *AOUT_1_2* is extracted from Group A CH1 and CH2, *AOUT_3_4* is extracted from Group A CH3 and CH4, and so on.

Note: If audio samples from embedded audio packets with the group set in **IDA[1:0]** are to be paired with samples from the group set in **IDB[1:0]**, all of the channels must have been derived from the same Word Clock and must be synchronous.

The output channel is set in the **OP[8:1]_SRC** parameter of the **OUTPUT_SEL_[2:1]** (Address A0D_h/A0E_h for HD/3G, B0B_h/B0C_h for SD) registers. [Table 4-21](#) lists the 3-bit address for audio channel mapping.

Table 4-21: Audio Channel Mapping Codes

Audio Output Channel	3-bit Host Interface Source Address	Audio Group
1	000 _b	Primary audio group channel 1
2	001 _b	Primary audio group channel 2
3	010 _b	Primary audio group channel 3
4	011 _b	Primary audio group channel 4
5	100 _b	Secondary audio group channel 1
6	101 _b	Secondary audio group channel 2
7	110 _b	Secondary audio group channel 3
8	111 _b	Secondary audio group channel 4

4.19.3.5 Serial Audio Output Word Length

The audio output, in Serial mode, has a selectable 24, 20, or 16-bit sample word length. The **ASWL[B:A]** (Address A01_h for HD/3G) and **ASWL[D:A]** (Address B0A_h for SD) parameters are used to configure the audio output sample word length. [Table 4-22](#) shows the host interface 2-bit code for setting the audio sample word length. When the presence of extended audio packets is detected in SD mode, the audio de-embedder defaults to 24-bit audio sample word length.

Table 4-22: Audio Sample Word Lengths

ASWL[D:A]	Audio Sample Word Length (SD)	Audio Sample Word Length (HD)
00 _b	24-bit	24-bit (default)
01 _b	20-bit	20-bit
10 _b	16-bit	16-bit
11 _b	Auto 24/20-bit (default)	Reserved

4.19.3.6 Audio Channel Status

The GS3470 detects the AES/EBU Audio Channel Status (ACS) block information for each of the selected channel pairs.

ACS data detection is indicated by corresponding **ACS_DET[1_2:3_4]A/B** flag bits in the **AUDIO_STATUS** for SD (Address B03_h) and **ACS_DET** for HD/3G (Address A03_h) registers. The flag is cleared by writing to the same location.

4.19.3.7 Audio Channel Status Read

AES/EBU ACS data is available separately for each of the channels in a stereo pair. The GS3470 defaults to reading the first channel of each pair. There are 184 bits in each ACS packet, which are written to twelve 16-bit right-justified registers

ACSR[1_2:3_4]A/B_BYTE[0_1:22](Address A40_h to A7B_h for HD/3G, B40_h to B7B_h for SD).

The **ACS_USE_SECOND** in **CFG_AUD1** for HD/3G(Address A02_h, bit 0) and **CFG_AUD** for SD (Address B01_h, bit 11) (default LOW) selects the second channel in each audio pair when set HIGH.

Once all of the ACS data for a channel has been acquired, the corresponding **ACS_DET** bit is asserted, and acquisition stops. The ACS data is overwritten with new data when the **ACS_DET** bit is cleared in the system.

4.19.3.7.1 Audio Channel Status Regeneration

When the **ACS_REGEN** in **REGEN** (Address A06_h, bit 0 for HD-3G, B04_h, bit 0 for SD) is set HIGH, the audio de-embedder embeds the 24 bytes of the Audio Channel Status information programmed in the **ACSR[183:0]** registers into the 'C' bit of the AES/EBU outputs. The same Audio Channel Status information is used for all output channels.

In order to apply ACSR data:

1. Set the **ACS_REGEN** bit to logic HIGH
2. Write the desired ACSR data to the **ACSR** registers
3. Set the **ACS_APPLY** bit to HIGH

At the next status boundary, the device outputs the contents of the **ACSR** registers as ACS data. This event may occur at a different time for each of the output channels. While waiting for the status boundary, the device sets the appropriate flag:

ACS_APPLY_WAIT for HD/3G (Address A04_h, bit 0) or **ACS_APPLY_WAIT[D:A]** for SD (Address B05_h, bit [3:0]).

Table 4-23 shows the host interface default settings for the Audio Channel Status block. The audio de-embedder automatically generates the CRC word.

Table 4-23: Audio Channel Status Information Registers

Name	Description	Default
ACSR1	Audio channel status block byte 0 set. Used when ACS_REGEN is set HIGH	85 _h
ACSR2	Audio channel status block byte 1 set. Used when ACS_REGEN is set HIGH	08 _h
ACSR3	Audio channel status block byte 2 set. Used when ACS_REGEN is set HIGH	28 _h (SD) 2C _h (HD)
ACSR[4:22]	Audio channel status block data for bytes 3 to 22. Used when ACS_REGEN is set HIGH	00 _h
ACS_REGEN	Audio channel status regenerate	0
ACS_APPLY	Apply new ACSR data	0
ACS_APPLY_WAIT[D:A]	Waiting to apply new ACSR data	0
ACSR[1_2:3_4][B:A]_BYTE[0_1:22]	Audio channel status block data for bytes 0 to 22	00 _h : 00 _h

Table 4-24: Audio Channel Status Block for Regenerate Mode Default Settings

Name	Byte	Bit	Default	Mode
PRO	0	0	1 _b	Professional use of channel status block
Emphasis	0	4:2	100 _b	100 _b None. Rec. manual override disabled
Sample Frequency	0	7:6	01 _b	48kHz. Manual override or auto disabled
Channel Mode	1	3:0	0001 _b	Two channels. Manual override disabled
AUX	2	2:0	000 _b	SD Mode: Maximum audio word length is 20 bits
			001 _b	HD Mode: Maximum audio word length is 24 bits
Source Word Length	2	5:3	101 _b	Maximum word length (based on AUX setting). 24-bit for HD Mode; 20-bit for SD Mode
All other bits set to zero				

4.19.3.8 Audio Mute

When **MUTE[B:A]** in **CH_MUTE** (Address A07_h) for HD/3G and **MUTE** in **CH_MUTE** (Address B07_h) for SD are set HIGH, the audio outputs are muted (all audio sample bits are set to zero). To set all the audio output channels to mute, set **MUTE_ALL** (Address A02_h, bit 1 for HD/SD and Address B01_h, bit 12 for SD) HIGH.

Table 4-25: Audio Mute Control Bits

Data Rate	Parameter Name	Description	Default
HD/3G	MUTEB	Mute Secondary output channels 4 to 1; bits 3:0 = channel 4:1 0 = Normal 1 = Muted	0
HD/3G	MUTEA	Mute Primary output channels 4 to 1; bits 3:0 = channel 4:1 0 = Normal 1 = Muted	0
SD	MUTE	Mute output channel 8 to 1; bits 7:0 = channel 8:1 (channel 8:5 = Secondary, channel 4:1 = Primary) 0 = Normal 1 = Muted	0

4.19.3.9 Mute On Loss Of Lock

When the GS3470 loses lock (LOCKED signal is LOW), the audio de-embedder sets all audio outputs LOW (no audio formatting is performed). The *ACLK*, *WCLK*, and *AMCLK* outputs are also forced LOW.

4.19.4 Error Reporting

4.19.4.1 Data Block Number Error

When the 1 to 255_d count sequence in the Data Block Number (DBN) word of Primary/Secondary audio data packets is discontinuous, the **DBN[B:A]_ERR** bit in **ACS_DET**(HD/3G) or **DBN_ERR**(SD) register is set HIGH.

The **DBNA_ERR** and **DBNB_ERR** flags also have associated **AUDIO_INT_ENABLE[2]** register flags for configuration of error reporting in the Receiver. The **DBNA_ERR** and **DBNB_ERR** flags remains set until cleared by writing to these locations.

4.19.4.2 ECC Error

The GS3470 monitors the ECC error status of the two selected audio groups, as described in [Section 4.19.2.6](#).

The **ECC[B:A]_ERROR** flags also have associated **AUD_DET2** register flags for configuration of error reporting in the Receiver. The **ECC[B:A]_ERROR** flags remain set until written via the host interface.

Note: Low latency mode option is not available when ECC error is enabled.

4.20 GSPI - HOST Interface

The GS3470 is configured via the Gennum Serial Peripheral Interface (GSPI).

The GSPI host interface is comprised of a serial data input signal (*SDIN* pin), serial data output signal (*SDOUT* pin), an active-LOW chip select ($\overline{CS}$ pin) and a burst clock (*SCLK* pin).

The GS3470 is a slave device, so the *SCLK*, *SDIN* and $\overline{CS}$ signals must be sourced by the application host processor.

All read and write access to the device is initiated and terminated by the application host processor.

4.20.1 $\overline{CS}$ Pin

The Chip Select pin ($\overline{CS}$) is an active-LOW signal provided by the host processor to the GS3470.

The HIGH-to-LOW transition of this pin marks the start of serial communication to the GS3470.

The LOW-to-HIGH transition of this pin marks the end of serial communication to the GS3470.

Each device may use its own separate Chip Select signal from the host processor or up to 32 devices may be connected to a single Chip Select when making use of the Unit Address feature.

Only those devices whose Unit Address matches the UNIT ADDRESS in the GSPI Command Word 1 will respond to communication from the host processor (unless the B'CAST ALL bit in the GSPI Command Word is set to 1).

4.20.2 *SDIN* Pin

The *SDIN* pin is the GSPI serial data input pin of the GS3470.

The 32-bit Command and 16-bit Data Words from the host processor or from the *SDOUT* pin of other devices are shifted into the device on the rising edge of *SCLK* when the $\overline{CS}$ pin is LOW.

4.20.3 SDOUT Pin

The *SDOUT* pin is the GSPI serial data output of the GS3470.

All data transfers out of the GS3470 to the host processor or to the *SDIN* pin of other connected devices occur from this pin.

By default at power up or after system reset, the *SDOUT* pin provides a non-clocked path directly from the *SDIN* pin, regardless of the $\overline{CS}$ pin state, except during the GSPI Data Word portion for read operations from the device. This allows multiple devices to be connected in Loop-Through configuration.

For read operations, the *SDOUT* pin is used to output data read from an internal Configuration and Status Register (CSR) when $\overline{CS}$ is LOW. Data is shifted out of the device on the falling edge of *SCLK*, so that it can be read by the host processor or other downstream connected device on the subsequent *SCLK* rising edge.

4.20.3.1 GSPI Link Disable Operation

It is possible to disable the direct *SDIN* to *SDOUT* (Loop-Through) connection by writing a value of 1 to the **GSPI_LINK_DISABLE** bit in **CONTROL_REG**. When disabled, any data appearing at the *SDIN* pin will not appear at the *SDOUT* pin and the *SDOUT* pin is HIGH.

Note: Disabling the Loop-Through operation is temporarily required when initializing the Unit Address for up to 32 connected devices.

The time required to enable/disable the Loop-Through operation from assertion of the register bit is less than the GSPI configuration command delay as defined by the parameter $t_{cmd_GSPI_config}$ (4 *SCLK* cycles).

Table 4-26: GSPI_LINK_DISABLE Bit Operation

Bit State	Description
0	SDIN pin is looped through to the SDOUT pin
1	Data appearing at SDIN does not appear at SDOUT, and SDOUT pin is HIGH.

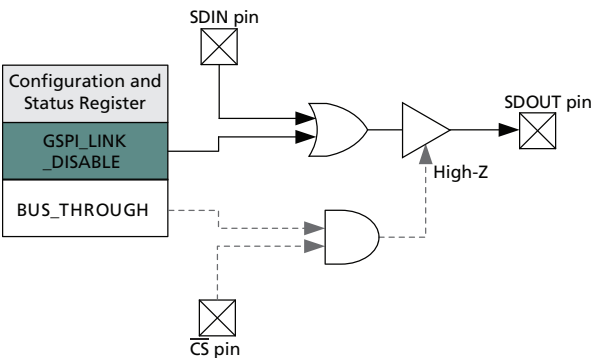


Figure 4-39: GSPI_LINK_DISABLE Operation

4.20.3.2 GSPI Bus-Through Operation

Using GSPI Bus-Through operation, the GS3470 can share a common PCB trace with other GSPI devices for *SDOUT* output.

When configured for Bus-Through operation, by setting

GSPI_BUS_THROUGH_ENABLE bit to 1, the *SDOUT* pin will be high-impedance when the $\overline{CS}$ pin is HIGH.

When the $\overline{CS}$ pin is LOW, the *SDOUT* pin will be driven and will follow regular read and write operation as described in [Section 4.20.3](#).

Multiple chains of GS3470 devices can share a single *SDOUT* bus connection to host by configuring the devices for Bus-Through operation. In such configuration, each chain requires a separate Chip Select ($\overline{CS}$).

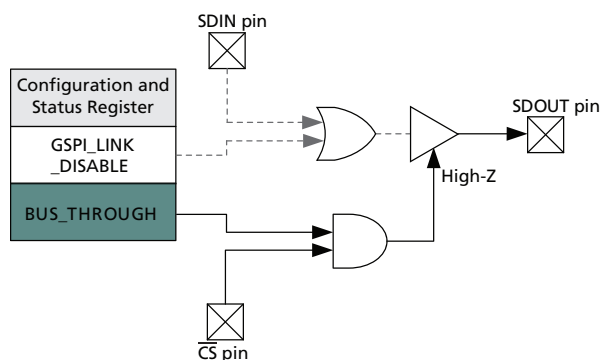


Figure 4-40: GSPI_BUS_THROUGH_ENABLE Operation

4.20.4 SCLK Pin

The *SCLK* pin is the GSPI serial data shift clock input to the device, and must be provided by the host processor.

Serial data is clocked into the GS3470 *SDIN* pin on the rising edge of *SCLK*. Serial data is clocked out of the device from the *SDOUT* pin on the falling edge of *SCLK* (read operation). *SCLK* is ignored when $\overline{CS}$ is HIGH.

4.20.5 Command Word 1 Description

All GSPI accesses are a minimum of 48 bits in length (two 16-bit Command Words followed by a 16-bit Data Word) and the start of each access is indicated by the HIGH-to-LOW transition of the chip select ($\overline{CS}$) pin of the GS3470.

The format of the Command Words and Data Word are shown in [Figure 4-41](#).

Data received immediately following this HIGH-to-LOW transition will be interpreted as a new Command Word.

4.20.5.1 R/ $\overline{W}$ bit - B15 Command Word 1

This bit indicates a read or write operation.

When $R/\overline{W}$ is set to 1, a read operation is indicated, and data is read from the register specified by the ADDRESS field of the Command Word.

When $R/\overline{W}$ is set to 0, a write operation is indicated, and data is written to the register specified by the ADDRESS field of the Command Word.

4.20.5.2 B'CAST ALL - B14 Command Word 1

This bit is used in write operations to configure all devices connected in Loop-Through and Bus-Through configuration with a single command.

When B'CAST ALL is set to 1, the following Data Word (AUTOINC = 0) or Data Words (AUTOINC = 1) are written to the register specified by the ADDRESS field of the Command Words (and subsequent addresses when AUTOINC = 1), regardless of the setting of the UNIT ADDRESS(es).

When B'CAST ALL is set to 0, a normal write operation is indicated. Only those devices that have a Unit Address matching the UNIT ADDRESS field of Command Word 1 write the Data Word to the register specified by the ADDRESS field of the Command Words.

4.20.5.3 EMEM - B13 Command Word 1

The EMEM bit must be set to 1 in Command Word 1. When EMEM is set to 1, a 23-bit address split between Command Word 1 and Command Word 2 is used to access the registers in this device.

4.20.5.4 AUTOINC - B12 Command Word 1

When **AUTOINC** is set to 1, Auto-Increment read or write access is enabled.

In Auto-Increment Mode, the device automatically increments the register address for each contiguous read or write access, starting from the address defined in the ADDRESS field of the Command Word.

The internal address is incremented for each 16-bit read or write access until a LOW-to-HIGH transition on the $\overline{CS}$ pin is detected.

When **AUTOINC** is set to 0, single read or write access is required.

Auto-Increment write must not be used to update values in **CONTROL_REG**.

4.20.5.5 UNIT ADDRESS - B11:B7 Command Word 1

The 5 bits of the UNIT ADDRESS field of the Command Word are used to select one of 32 devices connected on a single chip select in Loop-Through or Bus-Through configurations.

Read and write accesses are only accepted if the UNIT ADDRESS field matches the programmed **DEVICE_UNIT_ADDRESS** in **CONTROL_REG**.

By default at power-up or after a device reset, the **DEVICE_UNIT_ADDRESS** is set to 00_h

4.20.5.6 ADDRESS — B6:B0 Command Word 1 and B15:B0 Command Word 2

The Command and Data Word formats are shown in Figure 4-41 and Figure 4-42. As an example of the command word structure, reading register 90_h from a device with unit address 3, that has AUTOINC = 0, and B'CAST ALL = 0 would be structured as follows:

- Command word 1: 1010 0001 1000 0000 (A180_h)
- Command word 2: 0000 0000 1001 0000 (90_h)

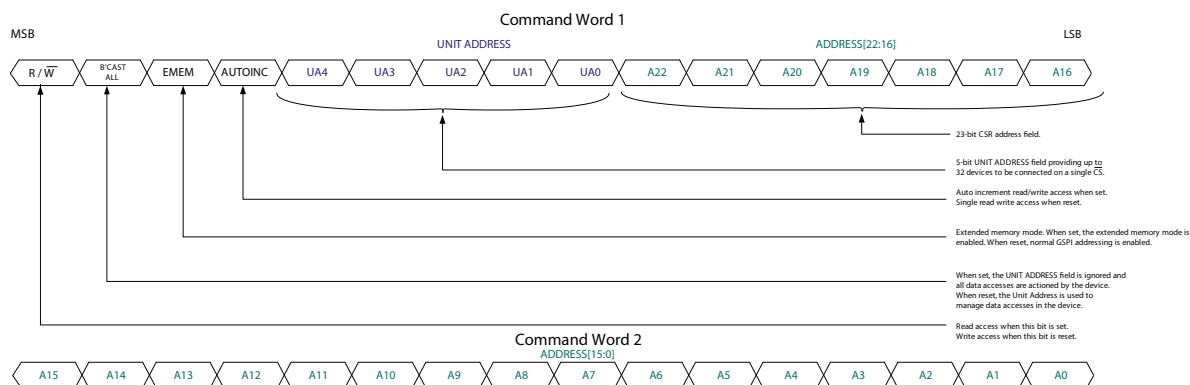


Figure 4-41: Command Word 1 and Command Word 2 Details

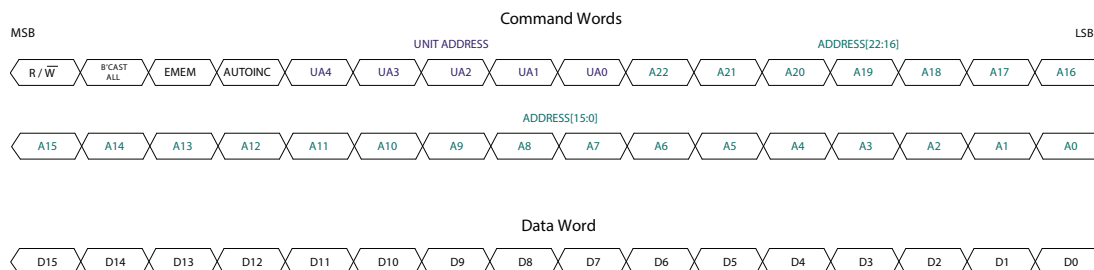


Figure 4-42: Command and Data Word Format

Note: Please see Section 4.20.5.6 for an example of the command word structure.

4.20.6 GSPI Transaction Timing

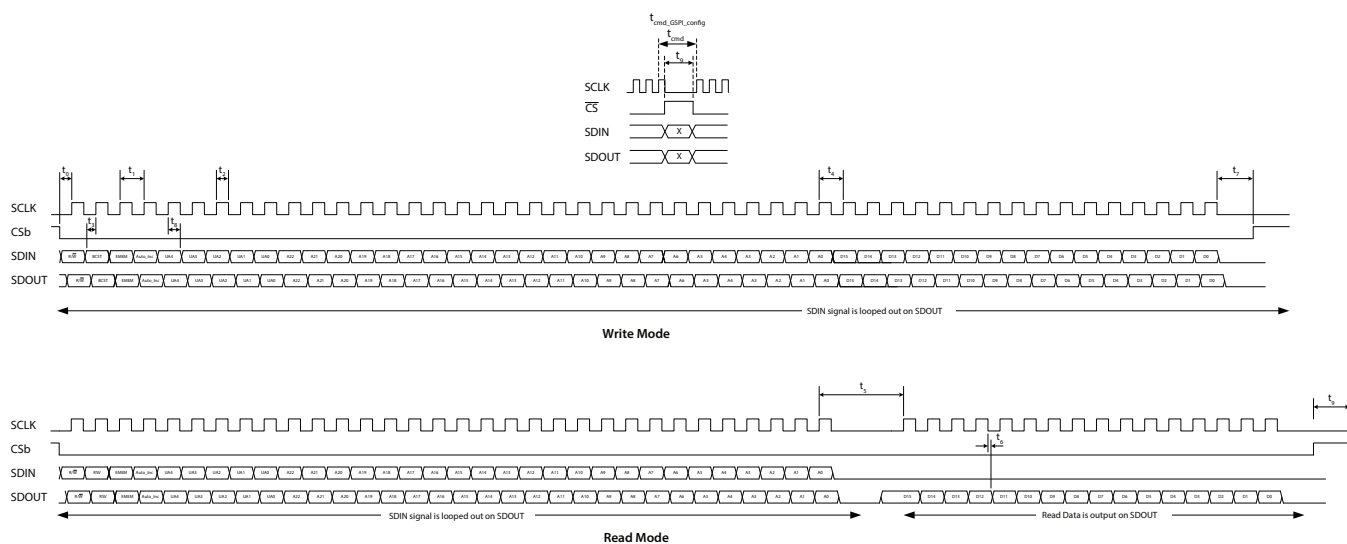


Figure 4-43: GSPI External Interface Timing

Table 4-27: GSPI Timing Parameters

Parameter	Symbol	Equivalent SCLK Cycles	Min	Typ	Max	Units
Time to GSPI ready after power up/reset	t_GSPI_ready		600	—	—	μs
$\overline{CS}$ low before SCLK rising edge	t ₀		2.7	—	—	ns
SCLK frequency			—	—	22	MHz
SCLK period	t ₁		45	—	—	ns
SCLK duty cycle	t ₂		40	50	60	%
Input data setup time	t ₃		1.8	—	—	ns
SCLK idle time - write	t ₄	1	45	—	—	ns
SCLK idle time - read	t ₅	4	161	—	—	ns
Inter-command delay time	t _{cmd}	4	161	—	—	ns
SDOUT after SCLK falling edge	t ₆		—	—	9.1	ns
CS high after final SCLK falling edge	t ₇		0.0	—	—	ns
Input data hold time	t ₈		1.1	—	—	ns
$\overline{CS}$ high time	t ₉		45	—	—	ns
SDIN to SDOUT combinational delay			—	—	7.4	ns
Max. chips daisy chained at max SCLK frequency	When host clocks in SDOUT data on rising edge of SCLK		—	—	1	GS3470 chips
Max. frequency for 32 daisy-chained devices			—	—	1.5	MHz
Max. chips daisy-chained at max. SCLK frequency	When host clocks in SDOUT data on falling edge of SCLK		—	—	5	GS3470 chips
Max. frequency for 32 daisy-chained devices			—	—	3.5	MHz

1. t_{cmd_GSPI_conf} inter-command delay must be used whenever modifying CONTROL_REG register at address 0x00.

4.20.7 Single Read/Write Access

Single read/write access timing for the GSPI interface is shown in Figure 4-44 to Figure 4-48.

When performing a single read or write access, one Data Word is read from/written to the device per access. Each access is a minimum of 48-bits long, consisting of two Command Words and a single Data Word. The read or write cycle begins with a HIGH-to-LOW transition of the $\overline{CS}$ pin. The read or write access is terminated by a LOW-to-HIGH transition of the $\overline{CS}$ pin.

The inter-command delay time indicated in the figures as t_{cmd} , is a minimum of 4 SCLK clock cycles. After modifying values in **CONTROL_REG**, the inter-command delay time, $t_{cmd_GSPI_config}$, is a minimum of 4 SCLK clock cycles.

For read access, the time from the last bit of the Command Word 2 to the start of the data output, as defined by t_5 , corresponds to no less than 4 SCLK clock cycles.

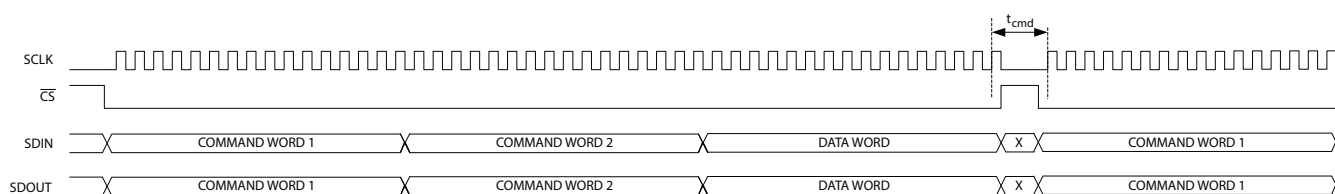


Figure 4-44: GSPI Write Timing – Single Write Access with Loop-Through Operation (default)

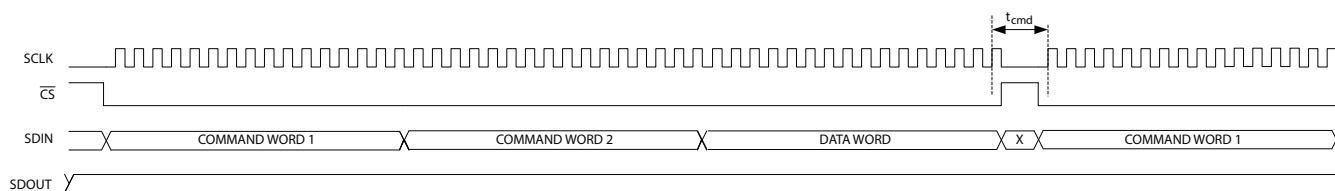


Figure 4-45: GSPI Write Timing – Single Write Access with GSPI Link-Disable Operation

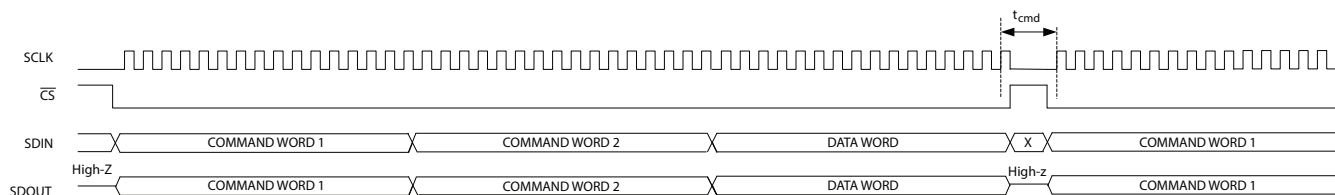


Figure 4-46: GSPI Write Timing – Single Write Access with Bus-Through Operation

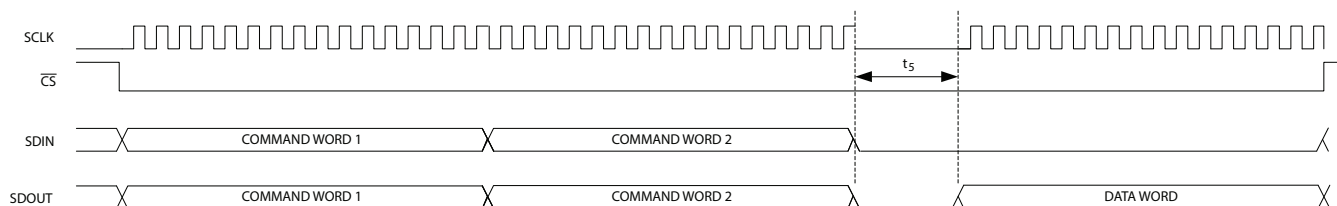


Figure 4-47: GSPI Read Timing – Single Read Access with Loop-Through Operation (default)

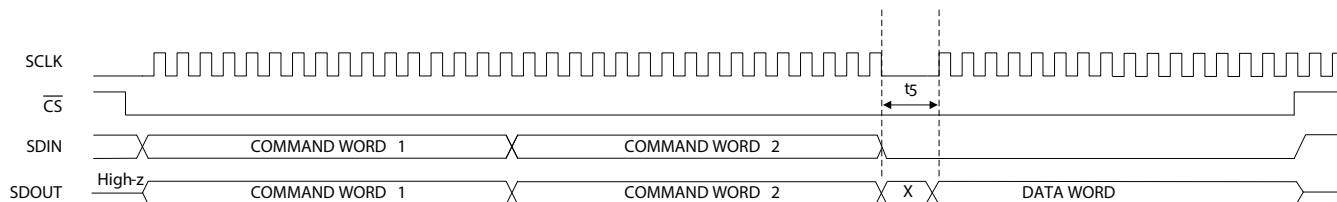


Figure 4-48: GSPI Read Timing – Single Read Access with Bus-Through Operation

4.20.8 Auto-Increment Read/Write Access

Auto-increment read/write access timing for the GSPI interface is shown in [Figure 4-49](#) to [Figure 4-53](#).

Auto-increment mode is enabled by setting the **AUTOINC** bit of Command Word 1.

In this mode, multiple Data Words can be read from/written to the device using only one starting address. Each access is initiated by a HIGH-to-LOW transition of the $\overline{CS}$ pin, and consists of a Command Word and one or more Data Words. The internal address is automatically incremented after the first read or write Data Word, and continues to increment until the read or write access is terminated by a LOW-to-HIGH transition of the $\overline{CS}$ pin.

Note: Writing to **CONTROL_REG** using Auto-increment access is not allowed.

For read access, the time from the last bit of the second Command Word to the start of the data output of the first Data Word as defined by t_5 , will be no less than 4 SCLK cycles. All subsequent read data accesses will not be subject to this delay during an Auto-Increment read.

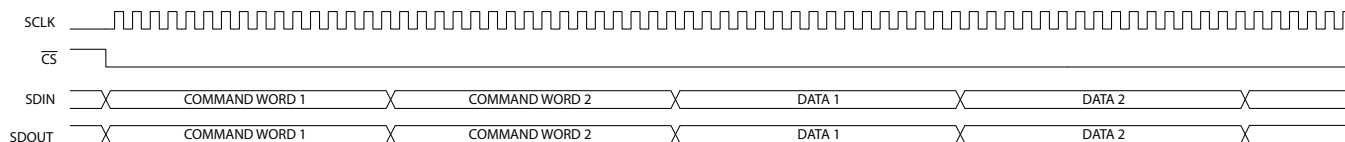


Figure 4-49: GSPI Write Timing – Auto-Increment with Loop-Through Operation (default)

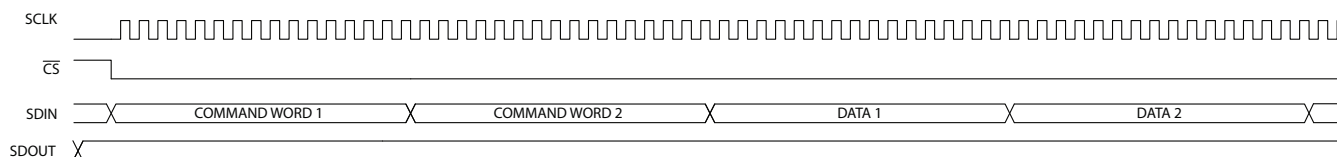


Figure 4-50: GSPI Write Timing – Auto-Increment with GSPI Link Disable Operation

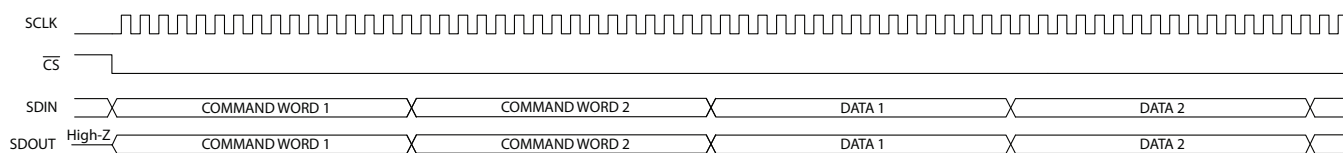


Figure 4-51: GSPI Write Timing – Auto-Increment with Bus-Through Operation

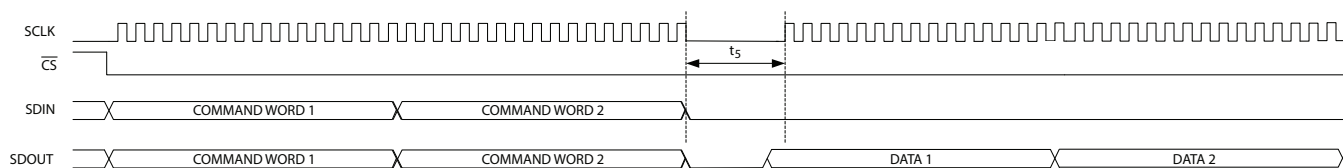


Figure 4-52: GSPI Read Timing – Auto-Increment Read with Loop-Through Operation (default)

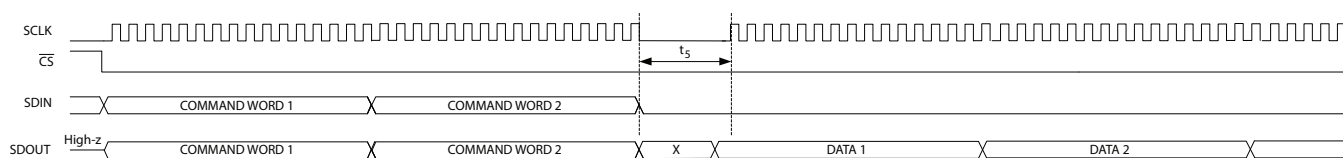


Figure 4-53: GSPI Read Timing – Auto-Increment Read with Bus-through Operation

4.20.9 Setting a Device Unit Address

Multiple (up to 32) GS3470 devices can be connected to a common Chip Select ($\overline{CS}$) in Loop-Through or Bus-Through operation.

To ensure that each device selected by a common $\overline{CS}$ can be separately addressed, a unique Unit Address must be programmed by the host processor at start-up as part of system initialization or following a device reset.

Note: By default at power up or after a device reset, the **DEVICE_UNIT_ADDRESS** of each device is set to 0_H and the $SDIN \rightarrow SDOUT$ non-clocked loop-through for each device is enabled.

These are the steps required to set the **DEVICE_UNIT_ADDRESS** of devices in a chain to values other than 0:

1. Write to Unit Address 0 selecting **CONTROL_REG** (ADDRESS = 0), with the **GSPI_LINK_DISABLE** bit set to 1 and the **DEVICE_UNIT_ADDRESS** field set to 0. This disables the direct $SDIN \rightarrow SDOUT$ non-clocked path for all devices on chip select.
2. Write to Unit Address 0 selecting **CONTROL_REG** (ADDRESS = 0), with the **GSPI_LINK_DISABLE** bit set to 0 and the **DEVICE_UNIT_ADDRESS** field set to a unique Unit Address. This configures **DEVICE_UNIT_ADDRESS** for the first device in the chain. Each subsequent such write to Unit Address 0 will configure the next device in the chain. If there are 32 devices in a chain, the last (32nd) device in the chain must use **DEVICE_UNIT_ADDRESS** value 0.
3. Repeat step 2 using new, unique values for the **DEVICE_UNIT_ADDRESS** field in **CONTROL_REG** until all devices in the chain have been configured with their own unique Unit Address value.

Note: $t_{cmd_GSPI_conf}$ delay must be observed after every write that modifies **CONTROL_REG**.

All connected devices receive this command (by default the Unit Address of all devices is 0), and the Loop-Through operation will be re-established for all connected devices.

Once configured, each device will only respond to Command Words with a UNIT ADDRESS field matching the **DEVICE_UNIT_ADDRESS** in **CONTROL_REG**.

Note: Although the Loop-Through and Bus-Through configurations are compatible with previous generation GSPI enabled devices (backward compatibility), only devices supporting Unit Addressing can share a chip select. All devices on any single chip select must be connected in a contiguous chain with only the last device's *SDOUT* connected to the application host processor. Multiple chains configured in Bus-Through mode can have their final *SDOUT* outputs connected to a single application host processor input.

4.20.10 Default GSPI Operation

By default at power up or after a device reset, the GS3470 is set for Loop-Through Operation and the internal **DEVICE_UNIT_ADDRESS** field of the device is set to 0.

Figure 4-54 shows a functional block diagram of the Configuration and Status Register (CSR) map in the GS3470.

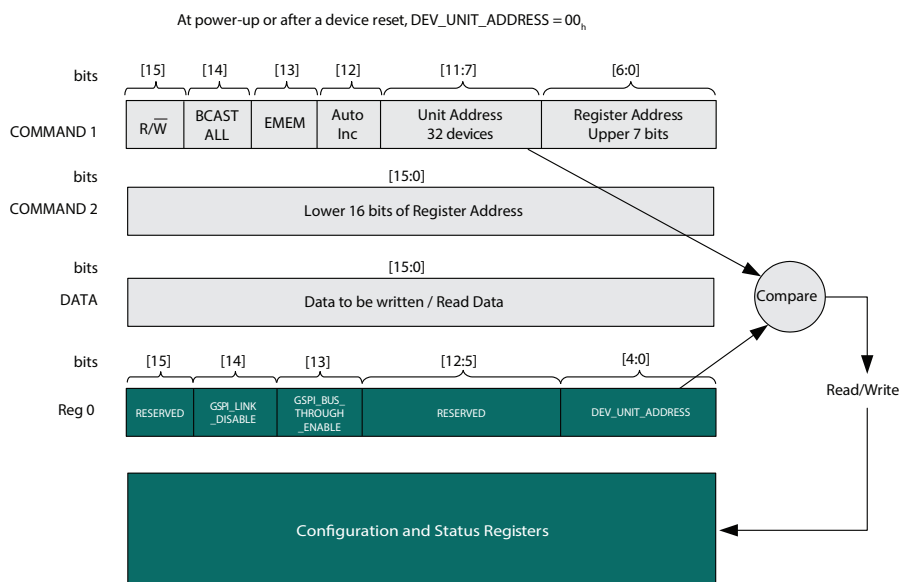


Figure 4-54: Internal Register Map Functional Block Diagram

The steps required for the application host processor to write to the Configuration and Status Registers via the GSPI, are as follows:

1. Set Command Word 1 for write access ($R/\overline{W} = 0$); set Auto Increment; set the Unit Address field in Command Word 1 to match the configured **DEVICE_UNIT_ADDRESS** which will be zero after power-up or reset. Set the Register Address bits in Command Word 1 to match the upper 7 bits of the register address to be accessed. Set the bits in Command Word 2 to match the lower 16 bits of the register address to be accessed. Write Command Word 1 and Command Word 2.

-
2. Write the Data Word to be written to the first register.
 3. Write the Data Word to be written to the next register in Auto Increment mode, etc.

Read access is the same as the above with the exception of step 1, where the Command Word is set for read access ($R/\overline{W} = 1$).

Note: The UNIT ADDRESS field of the Command Word must always match **DEVICE_UNIT_ADDRESS** for an access to be accepted by the device. Changing **DEVICE_UNIT_ADDRESS** to a value other than 0 is only required if multiple devices are connected to a single chip select (in Loop-Through or Bus-Through configuration.)

4.21 JTAG Test Operation

When the $JTAG_EN/\overline{DIS}$ pin of the GS3470 is set HIGH, JTAG test mode is enabled.

The JTAG can be used as a stand-alone in-circuit ATE (Automatic Test Equipment) during PCB assembly.

Scan coverage is limited to digital pins only. There is no scan coverage for analog pins $DDO/\overline{DDO}$, $RBIAS$, LF , $CT0$, and $CT1$.

The $JTAG_EN/\overline{DIS}$ pin must be held LOW during scan and therefore has no scan coverage.

Please contact your Semtech representative to obtain the BSDL model for the GS3470.

5. Register Map

5.1 Control Registers

Table 5-1: Host Configuration Register

Function	GSPI Address _h	Register Name
Host Configuration Register	0	CONTROL_REG

Table 5-2: Video Core Registers

Function	GSPI Address _h	Register Name
IO Processing	800	IOPROC_1
	801	IOPROC_2
Flywheel Switch Line	802 to 803	FWSL_LINE1_FE1 to FWSL_LINE2_FE1
	804 to 805	FWSL_LINE1_FE2 to FWSL_LINE2_FE2
Sticky Error Status	806	ERROR_STAT_1_STICKY
	808	ERROR_STAT_2_STICKY
Error Status	807	ERROR_STAT_1
	809	ERROR_STAT_2
Error Detected Flag	80A	EDH_FLAG_IN
	80B	EDH_FLAG_OUT
Data Format	80C	DATA_FORMAT_DS1
	80E	DATA_FORMAT_DS2
Video Standard Detection	80D	VID_STD
Power Down	811	POWER_DOWN
IO Configuration	812	IO_CONFIG
	813	IO_CONFIG_2
ANC Control	817	ANC_CONTROL

Table 5-2: Video Core Registers (Continued)

Function	GSPI Address _h	Register Name
ANC Extraction Line	818	ANC_LINEA
	819	ANC_LINEB
ANC Type	81E to 822	ANC_TYPE_1_DS1 to ANC_TYPE_5_DS1
	823 to 827	ANC_TYPE_1_DS2 to ANC_TYPE_5_DS2
Video Format	828 to 82B	VIDEO_FORMAT_352_A_1 to VIDEO_FORMAT_352_D_1
	82C to 82F	VIDEO_FORMAT_352_A_2 to VIDEO_FORMAT_352_D_2
Raster Information	830 to 833	RASTER_STRUC_1_DS1 to RASTER_STRUC_4_DS1
	8CF to 8D2	RASTER_STRUC_1_DS2 to RASTER_STRUC_4_DS2
Flywheel Status	834	FLYWHEEL_STATUS
Rate Selection	835	RATE_SEL
CEA-861-D Format	836	TIM_861_FORMAT
CEA-861-D Configuration	837	TIM_861_CFG
Error Mask	848 to 84A	ERROR_MASK_1 to ERROR_MASK_3
Audio Clock Control	84B	ACG_CTRL
Input Configuration	84D	INPUT_CONFIG
PLL Control	868	PLL_CTRL_0
	869 - 86A	RSVD
	86B	PLL_STAT
PLL Status Sticky	86C	PLL_STICKY_STAT
LOS Control	86F	LOS_CTRL
Delay Line Control	870	DELAY_LINE_CTRL_1
Delay Line Control	871	DELAY_LINE_CTRL_2
Clock Generation	874	CLK_GEN
PIN/CSR Selector	877	PIN_CSR_SELECT
IO Drive Strength	878	IO_DRIVE_STRENGTH

Table 5-2: Video Core Registers (Continued)

Function	GSPI Address _h	Register Name
Output Buffer Control	87B	OUTPUT_BUFFER_CTRL_1
	87C	OUTPUT_BUFFER_CTRL_2
M Detection Tolerance	87E	M_DETECTION_TOLERANCE_DS1
	880	M_DETECTION_TOLERANCE_DS2
Analog Mute Control	883	DDO_MUTE_CTRL
General Status	8CB	GENERAL_STATUS
General Control	8D3	GENERAL_CONTROL

Table 5-3: HD and 3G Audio Core Registers

Function	GSPI Address _h	Register Name
Audio Configuration	A01	CFG_AUD
	A02	CFG_AUD_1
	A0B	CFG_AUD_2
	A0C	CFG_AUD_3
Audio Channel Status Detection	A03	ACS_DET
Audio Detection	A04	AUD_DET1
	A05	AUD_DET2
Audio Channel Status Regeneration	A06	REGEN
Channel Mute	A07	CH_MUTE
Channel Validity	A08	CH_VALID
Audio Interrupt Enable	A09	AUDIO_INT_ENABLE
	A0A	AUDIO_INT_ENABLE2
Output Channel Selection	A0D	OUTPUT_SEL_1
	A0E	OUTPUT_SEL_2
Primary Audio Frame Number	A20	AFNA

Table 5-3: HD and 3G Audio Core Registers (Continued)

Function	GSPI Address _h	Register Name
Primary Audio Sampling Frequency	A21	RATEA
Primary Active Channels Indicator	A22	ACTA
Primary Audio Group Delay	A23 to A28	PRIM_AUD_DELAY_1 to PRIM_AUD_DELAY_6
Secondary Audio Frame Number	A30	AFNB
Secondary Audio Sampling Frequency	A31	RATEB
Secondary Active Channels Indicator	A32	ACTB
Secondary Audio Group Delay	A33 to A38	SEC_AUD_DELAY_1 to SEC_AUD_DELAY_6
Audio Group A Channel Status	A40 to A4A	ACSR1_2A_BYTE0_1 to ACSR1_2A_BYTE20_21
	A4B	ACSR1_2A_BYTE22
	A50 to A5A	ACSR3_4A_BYTE0_1 to ACSR3_4A_BYTE20_21
	A5B	ACSR3_4A_BYTE22
Audio Group B Channel Status	A60 to A6A	ACSR1_2B_BYTE0_1 to ACSR1_2B_BYTE20_21
	A6B	ACSR1_2B_BYTE22
	A70 to A7A	ACSR3_4B_BYTE0_1 to ACSR3_4B_BYTE20_21
	A7B	ACSR3_4B_BYTE22
Audio Channel Status Regeneration Bytes	A80 to A96	ACSR_BYTE_0 to ACSR_BYTE_22

Table 5-4: SD Audio Core Registers

Function	GSPI Address _h	Register Name
Audio Configuration	B01	CFG_AUD
Audio Status Error	B03	AUDIO_STATUS
Audio Channel Status Regeneration	B04	REGEN

Table 5-4: SD Audio Core Registers (Continued)

Function	GSPI Address _h	Register Name
Audio Detection	B05	AUD_DET
Checksum Error Detection	B06	CSUM_ERR_DET
Channel Mute	B07	CH_MUTE
Channel Validity	B08	CH_VALID
Audio Interrupt Enable	B09	AUDIO_INT_ENABLE
Output Configuration	B0A	CFG_OUTPUT
Output Channel Selection	B0B	OUTPUT_SEL_1
	B0C	OUTPUT_SEL_2
Primary Audio Frame Number	B20	AFNA12
	B21	AFNA34
Primary Audio Sampling Frequency	B22	RATEA
Primary Active Channels Indicator	B23	ACT_A
Primary Audio Group Delay	B24 to B2F	PRIM_AUD_DELAY_1 to PRIM_AUD_DELAY_12
Secondary Audio Frame Number	B30	AFNB12
	B31	AFNB34
Secondary Audio Sampling Frequency	B32	RATEB
Secondary Active Channels Indicator	B33	ACT_B
Secondary Audio Group Delay	B34 to B3F	SEC_AUD_DELAY_1 to SEC_AUD_DELAY_12
Audio Group A Channel Status	B40 to B4A	ACSR1_2A_BYTE0_1 to ACSR1_2A_BYTE20_21
	B4B	ACSR1_2A_BYTE22
	B50 to B5A	ACSR3_4A_BYTE0_1 to ACSR3_4A_BYTE20_21
	B5B	ACSR3_4A_BYTE22

Table 5-4: SD Audio Core Registers (Continued)

Function	GSPI Address _h	Register Name
Audio Group B Channel Status	B60 to B6A	ACSR1_2B_BYTE0_1 to ACSR1_2B_BYTE20_21
	B6B	ACSR1_2B_BYTE22
	B70 to B7A	ACSR3_4B_BYTE0_1 to ACSR3_4B_BYTE20_21
	B7B	ACSR3_4B_BYTE22
Audio Channel Status Regeneration Bytes	B80 to B96	ACSR_BYTE_0 to ACSR_BYTE_22

Table 5-5: ANC Extraction FIFO Access Registers

Function	GSPI Address _h	Register Name
ANC FIFO	C00 to FFF	ANC_FIFO_0 to ANC_FIFO_1023

Table 5-6: Glossary of Terms

Term	Description
RW	Read-write
RO	Read-only
WO	Write-only
ROCW ¹	Read-only, clear on write
ROCR	Read-only, clear on read
RSVD ²	Read, modify, write

Note:

1) For each bit in the register that needs to be cleared, write 1.
If all bits in the register need to be cleared, write FFFF_h.

2) To modify a bit in a register that has other bits marked RSVD, read the existing value, modify only the non-reserved bits and then write the final value. For example, to change bit 9 in register 989_h, a read would produce 0000_h, modify to 0200_h and write 0200_h.

Table 5-7: Host Configuration Register Descriptions

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
0	CONTROL_REG	RSVD	15:15	—	—	Reserved.
		GSPI_LINK_DISABLE	14:14	RW	0	GSPI loop-through disable.
		GSPI_BUS_THROUGH_ENABLE	13:13	RW	0	GSPI bus-through enable.
		RSVD	12:5	—	—	Reserved.
		DEVICE_UNIT_ADDRESS	4:0	RW	0	Device address programmed by application.

Table 5-8: Video Core CSR Descriptions

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
800	IOPROC_1	RSVD	15:15	—	—	Reserved.
		LOW_LATENCY_BYPASS	14:14	RW	0	0 = Takes the low latency bypass path through the audio and ancillary data extraction when possible (when not deleting or updating the packets). 1 = Takes the higher latency path through the audio and ANC extraction regardless of what specific features are enabled/disabled.
		FW_SWITCH_LINE_OVERRIDE_FE1	13:13	RW	0	When HIGH, flywheel uses FW_SWITCH_LINE values instead of the indicated values in standards.
		TRS_WORD_REMAP_FE1_DISABLE_MASK	12:12	RW	0	When HIGH, disables TRS word remap.
		EDH_FLAG_UPDATE_MASK	11:11	RW	0	When HIGH, disables update for EDH error flags.
		EDH_CRC_INS_MASK	10:10	RW	0	When HIGH, disables EDH CRC error correction and insertion.
		H_CONFIG	9:9	RW	0	Selects the H blanking indication: 0 = Active picture timing 1 = SMPTE H timing
		ANC_DATA_EXT_MASK	8:8	RW	0	When HIGH, disables ancillary data extraction FIFO.
		AUD_EXT_MASK	7:7	RW	0	When HIGH, disables audio extraction.
		TIMING_861	6:6	RW	0	0 = Selects digital FVH timing output 1 = Selects 861 timing output
		RSVD	5:5	RW	0	Reserved.
		ILLEGAL_WORD_REMAP_DS1_MASK	4:4	RW	0	When HIGH, disables illegal word remapping.
		ANC_CHECKSUM_INSERTION_DS1_MASK	3:3	RW	0	When HIGH, disables insertion of ancillary data checksums.
		CRC_INS_DS1_MASK	2:2	RW	0	When HIGH, disables insertion of HD/3G CRC words.
		LNUM_INS_DS1_MASK	1:1	RW	0	When HIGH, disables insertion of HD/3G line numbers.
		TRS_INS_DS1_MASK	0:0	RW	0	When HIGH, disables insertion of TRS words.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
801	IOPROC_2	RSVD	15:15	—	—	Reserved.
		NONINV	14:14	RW	1	0 = Forces inverted MPEG-2 decoding 1 = Forces non-inverted MPEG-2 decoding
		RSVD	13:13	RW	1	Reserved.
		FW_SWITCH_LINE_OVERRIDE_FE2	12:12	RW	0	When HIGH, flywheel uses FW_SWITCH_LINE values instead of the indicated values in standards.
		TRS_WORD_REMAP_FE2_DISABLE_MASK	11:11	RW	0	When HIGH, disables TRS word remapping.
		REGEN_352_MASK	10:10	RW	0	When HIGH, disables regeneration of the SMPTE 352 packet for 3G Level B data. Note: This bit needs to be enabled via the host interface to disable SMPTE ST 352 packet regeneration. For a 3G Level B signal, setting REGEN_352_MASK LOW will replace byte 1 of the SMPTE 352M packet such that it identifies the outgoing stream as a 1.5Gb/s stream, instead of a 3Gb/s stream. Byte 1 is updated based on the input video format, as identified by byte 1 of the incoming SMPTE 352M packet as shown in Table 5-12 . Note: When receiving a 3G Level B source, it may be necessary to disable the REGEN_352 function when re-serializing the GS3470 output back to 3G using older generation SDI transmitter devices (for example, the GS2972).
		DS_SWAP_3G	9:9	RW	0	Swaps DS1 and DS2 at the output in 3G mode.
		ANC_EXT_SEL_DS1_Y_DS2_Y	8:8	RW	1	For 3G Level B, enables a mode to select which inputs to extract: 0 = Selects Luma and Chroma inputs from DS1 or DS2 based on ANC_EXT_SEL_DS2_DS1 1 = Selects Luma inputs from both DS1 and DS2
		ANC_EXT_SEL_DS2_DS1	7:7	RW	0	For 3G Level B, selects data stream to extract ANC data from.
		AUDIO_SEL_DS2_DS1	6:6	RW	0	Selects data stream to be sent to audio core.
		RSVD	5:5	RW	0	Reserved.
		ILLEGAL_WORD_REMAP_DS2_MASK	4:4	RW	0	When HIGH, disables illegal word remapping.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
801 (Continued)	IOPROC_2 (Continued)	ANC_CHECKSUM_INSERTION_DS2_MASK	3:3	RW	0	When HIGH, disables insertion of ancillary data checksums.
		CRC_INS_DS2_MASK	2:2	RW	0	When HIGH, disables insertion of HD/3G CRC words.
		LNUM_INS_DS2_MASK	1:1	RW	0	When HIGH, disables insertion of HD/3G line numbers.
		TRS_INS_DS2_MASK	0:0	RW	0	When HIGH, disables insertion of TRS words.
802	FWSL_LINE1_FE1	RSVD	15:11	—	—	Reserved.
		FWSL_LINE1_FE1	10:0	RW	0	Value of flywheel line number to override for switch line 1 when FW_SWITCH_LINE_OVERRIDE_FE1 asserted.
803	FWSL_LINE2_FE1	RSVD	15:11	—	—	Reserved.
		FWSL_LINE2_FE1	10:0	RW	0	Value of flywheel line number to override for switch line 2 when FW_SWITCH_LINE_OVERRIDE_FE1 asserted.
804	FWSL_LINE1_FE2	RSVD	15:11	—	—	Reserved.
		FWSL_SWITCH_LINE1_FE2	10:0	RW	0	Value of flywheel line number to override for switch line 1 when FW_SWITCH_LINE_OVERRIDE_FE2 asserted.
805	FWSL_LINE2_FE2	RSVD	15:11	—	—	Reserved.
		FWSL_SWITCH_LINE2_FE2	10:0	RW	0	Value of flywheel line number to override for switch line 2 when FW_SWITCH_LINE_OVERRIDE_FE2 asserted.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
806	ERROR_STAT_1_STICKY	RSVD	15:10	—	—	Reserved.
		FF_CRC_ERR_STICKY	9:9	ROCR	0	EDH full frame CRC error indication.
		AP_CRC_ERR_STICKY	8:8	ROCR	0	EDH active picture CRC error indication – sticky, clear on read.
		LOCK_ERR_STICKY	7:7	ROCR	0	Lock error indication – sticky, clear on read.
		CCS_ERR_1_STICKY	6:6	ROCR	0	Chroma/DS2 ancillary data checksum error indication – sticky, clear on read.
		YCS_ERR_1_STICKY	5:5	ROCR	0	Luma/DS1 ancillary data checksum error indication - sticky, clear on read.
		CCRC_ERR_1_STICKY	4:4	ROCR	0	Chroma/DS2 CRC error indication (HD/3G only) – sticky, clear on read.
		YCRC_ERR_1_STICKY	3:3	ROCR	0	Luma/DS1 CRC error indication (HD/3G only) – sticky, clear on read.
		LNUM_ERR_1_STICKY	2:2	ROCR	0	Line number error indication (HD/3G only) – sticky, clear on read.
		SAV_ERR_1_STICKY	1:1	ROCR	0	SAV error indication – sticky, clear on read.
		EAV_ERR_1_STICKY	0:0	ROCR	0	EAV error indication – sticky, clear on read.
807	ERROR_STAT_1	RSVD	15:10	—	—	Reserved.
		FF_CRC_ERR	9:9	ROCW	0	EDH full frame CRC error indication.
		AP_CRC_ERR	8:8	ROCW	0	EDH active picture CRC error indication.
		LOCK_ERR	7:7	ROCW	0	Lock error indication.
		CCS_ERR_1	6:6	ROCW	0	Chroma/DS2 ancillary data checksum error indication.
		YCS_ERR_1	5:5	ROCW	0	Luma/DS1 ancillary data checksum error indication.
		CCRC_ERR_1	4:4	ROCW	0	Chroma/DS2 CRC error indication (HD/3G only).
		YCRC_ERR_1	3:3	ROCW	0	Luma/DS1 CRC error indication (HD/3G only).
		LNUM_ERR_1	2:2	ROCW	0	Line number error indication (HD/3G only).
		SAV_ERR_1	1:1	ROCW	0	SAV error indication.
		EAV_ERR_1	0:0	ROCW	0	EAV error indication.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
808	ERROR_STAT_2_STICKY	RSVD	15:7	—	—	Reserved.
		CCS_ERR_2_STICKY	6:6	ROCR	0	Chroma ancillary data checksum error indication for the second stream of a 3G level B input – sticky, clear on read.
		YCS_ERR_2_STICKY	5:5	ROCR	0	Luma ancillary data checksum error indication for the second stream of a 3G level B input – sticky, clear on read.
		CCRC_ERR_2_STICKY	4:4	ROCR	0	Chroma CRC error indication for the second stream of a 3G level B input – sticky, clear on read.
		YCRC_ERR_2_STICKY	3:3	ROCR	0	Luma CRC error indication for the second stream of a 3G level B input – sticky, clear on read.
		LNUM_ERR_2_STICKY	2:2	ROCR	0	Line number error indication for the second stream of a 3G level B input – sticky, clear on read.
		SAV_ERR_2_STICKY	1:1	ROCR	0	SAV error indication for the second stream of a 3G level B input – sticky, clear on read.
		EAV_ERR_2_STICKY	0:0	ROCR	0	EAV error indication for the second stream of a 3G level B input – sticky, clear on read.
809	ERROR_STAT_2	RSVD	15:7	—	—	Reserved.
		CCS_ERR_2	6:6	ROCW	0	Chroma ancillary data checksum error indication for the second stream of a 3G level B input.
		YCS_ERR_2	5:5	ROCW	0	Luma ancillary data checksum error indication for the second stream of a 3G level B input.
		CCRC_ERR_2	4:4	ROCW	0	Chroma CRC error indication for the second stream of a 3G level B input.
		YCRC_ERR_2	3:3	ROCW	0	Luma CRC error indication for the second stream of a 3G level B input.
		LNUM_ERR_2	2:2	ROCW	0	Line number error indication for the second stream of a 3G level B input.
		SAV_ERR_2	1:1	ROCW	0	SAV error indication for the second stream of a 3G level B input
		EAV_ERR_2	0:0	ROCW	0	EAV error indication for the second stream of a 3G level B input

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
80A	EDH_FLAG_IN	EDH_DETECT	15:15	RO	0	Embedded EDH packet detected.
		ANC_UES_IN	14:14	RO	0	Ancillary data – unknown error status flag.
		ANC_IDA_IN	13:13	RO	0	Ancillary data – internal error detected already flag.
		ANC_IDH_IN	12:12	RO	0	Ancillary data – internal error detected here flag.
		ANC_EDA_IN	11:11	RO	0	Ancillary data – error detected already flag.
		ANC_EDH_IN	10:10	RO	0	Ancillary data – error detected here flag.
		FF_UES_IN	9:9	RO	0	EDH Full Field – unknown error status flag.
		FF_IDA_IN	8:8	RO	0	EDH Full Field – internal error detected already flag.
		FF_IDH_IN	7:7	RO	0	EDH Full Field – internal error detected here flag.
		FF_EDA_IN	6:6	RO	0	EDH Full Field – error detected already flag.
		FF_EDH_IN	5:5	RO	0	EDH Full Field – error detected here flag.
		AP_UES_IN	4:4	RO	0	EDH Active Picture – unknown error status flag.
		AP_IDA_IN	3:3	RO	0	EDH Active Picture – internal error detected already flag.
		AP_IDH_IN	2:2	RO	0	EDH Active Picture – internal error detected here flag.
		AP_EDA_IN	1:1	RO	0	EDH Active Picture – error detected already flag.
		AP_EDH_IN	0:0	RO	0	EDH Active Picture – error detected here flag.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
80B	EDH_FLAG_OUT	RSVD	15:15	—	—	Reserved.
		ANC_UES	14:14	RO	1	Ancillary data – unknown error status flag.
		ANC_IDA	13:13	RO	0	Ancillary data – internal error detected already flag.
		ANC_IDH	12:12	RO	0	Ancillary data – internal error detected here flag.
		ANC_EDA	11:11	RO	0	Ancillary data – error detected already flag.
		ANC_EDH	10:10	RO	0	Ancillary data – error detected here flag.
		FF_UES	9:9	RO	1	EDH Full Field – unknown error status flag.
		FF_IDA	8:8	RO	0	EDH Full Field – internal error detected already flag.
		FF_IDH	7:7	RO	0	EDH Full Field – internal error detected here flag.
		FF_EDA	6:6	RO	0	EDH Full Field – error detected already flag.
		FF_EDH	5:5	RO	0	EDH Full Field – error detected here flag.
		AP_UES	4:4	RO	1	EDH Active Picture – unknown error status flag.
		AP_IDA	3:3	RO	0	EDH Active Picture – internal error detected already flag.
		AP_IDH	2:2	RO	0	EDH Active Picture – internal error detected here flag.
		AP_EDA	1:1	RO	0	EDH Active Picture – error detected already flag.
		AP_EDH	0:0	RO	0	EDH Active Picture – error detected here flag.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
80C	DATA_FORMAT_DS1	RSVD	15:8	—	—	Reserved.
		CDATA_FORMAT_AP1	7:4	RO	F	Data format as indicated in Chroma channel of the second stream of a 3G level B input: 0000 _b = DVC Pro without ECC 0001 _b = DVC Pro with ECC 0010 _b = DV CAM 0011 _b = SDTi CP 0100 _b = Other fixed blocksize 0101 _b = Other variable blocksize 0110 _b = Other SDI 0111 _b = Reserved 1000 _b = TDM Video 1001 _b = HD SDTi 1010 _b to 1110 _b = Reserved 1111 _b = Unknown
		YDATA_FORMAT_AP1	3:0	RO	F	Data format as indicated in Luma channel of the second stream of a 3G level B input: 0000 _b = DVC Pro without ECC 0001 _b = DVC Pro with ECC 0010 _b = DV CAM 0011 _b = SDTi CP 0100 _b = Other fixed blocksize 0101 _b = Other variable blocksize 0110 _b = Other SDI 0111 _b = Reserved 1000 _b = TDM Video 1001 _b = HD SDTi 1010 _b to 1110 _b = Reserved 1111 _b = Unknown
		RSVD	15:11	—	—	Reserved.
80D	VID_STD	LEVEL_B_DETECTED	10:10	RO	0	Level B detection as reported by the GS3470: 0 = SD, HD, or 3G Level A signal detected 1 = 3G Level B signal detected
		LINK_DS	9:6	RO	0	Link indication: bits [7:4] of byte 4 of SMPTE 352 packet.
		VD_STD_DS	5:0	RO	1D	Detected video standard. See Table 4-10 for video standard number.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
		RSVD	15:8	—	—	Reserved.
80E	DATA_FORMAT_DS2	CDATA_FORMAT_AP2	7:4	RO	F	Data format as indicated in Chroma channel of the second stream of a 3G level B input: 0000 _b = DVC Pro without ECC 0001 _b = DVC Pro with ECC 0010 _b = DV CAM 0011 _b = SDTi CP 0100 _b = Other fixed blocksize 0101 _b = Other variable blocksize 0110 _b = Other SDI 0111 _b = Reserved 1000 _b = TDM Video 1001 _b = HD SDTi 1010 _b to 1110 _b = Reserved 1111 _b = Unknown
		YDATA_FORMAT_AP2	3:0	RO	F	Data format as indicated in Luma channel of the second stream of a 3G level B input: 0000 _b = DVC Pro without ECC 0001 _b = DVC Pro with ECC 0010 _b = DV CAM 0011 _b = SDTi CP 0100 _b = Other fixed blocksize 0101 _b = Other variable blocksize 0110 _b = Other SDI 0111 _b = Reserved 1000 _b = TDM Video 1001 _b = HD SDTi 1010 _b to 1110 _b = Reserved 1111 _b = Unknown
80F to 810	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
811	POWER_DOWN	RSVD	15:4	—	—	Reserved.
		PD_CSR_ACCESS	3:3	RW	0	CSR access selection during power-down: 0 = Selects no CSR access during power-down when PWR_DWN is asserted 1 = Selects CSR access during power-down when PWR_DWN is asserted
		RC_BYP	2:2	RW	0	Retimer bypass selection: 0 = Selects retimed data (when DDO is enabled) 1 = Selects non-retimed data (when DDO is enabled)
		SERIAL_LOOPBACK_EN	1:1	RW	0	Serial loopback enable: 0 = The serial digital output signals DDO and $\overline{DDO}$ are disabled 1 = The serial digital output signals DDO and $\overline{DDO}$ are enabled
		PD_PCLK_ENABLE	0:0	RW	0	PCLK enabled during power-down: 0 = Selects PCLK to be shut off when PWR_DWN is asserted 1 = Selects PCLK to be output when PWR_DWN is asserted
812	IO_CONFIG	RSVD	15:15	—	—	Reserved.
		STAT2_CONFIG	14:10	RW	2	Configures STAT2 output pin: 00000 _b = H/HSYNC 00001 _b = V/VSYNC 00010 _b = F/DE 00011 _b = LOCKED 00100 _b = Y/1ANC 00101 _b = C/2ANC 00110 _b = $\overline{\text{DATA_ERROR}}$ 00111 _b = $\overline{\text{VIDEO_ERROR}}$ 01000 _b = $\overline{\text{AUDIO_ERROR}}$ 01001 _b = $\overline{\text{EDH_DETECT}}$ 01010 _b = $\overline{\text{CARRIER_DETECT}}$ (active LOW) 01011 _b = $\overline{\text{SD/HD}}$ 01100 _b = $\overline{\text{3G/HD}}$ 01101 _b to 11111 _b = Reserved
		STAT1_CONFIG	9:5	RW	1	Configure STAT1 output pin. Refer to STAT2_CONFIG for decoding.
		STAT0_CONFIG	4:0	RW	0	Configure STAT0 output pin. Refer to STAT2_CONFIG for decoding.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
813	IO_CONFIG_2	RSVD	15:15	—	—	Reserved.
		STAT5_CONFIG	14:10	RW	6	Configure STAT5 output pin. Refer to STAT2_CONFIG for decoding.
		STAT4_CONFIG	9:5	RW	B	Configure STAT4 output pin. Refer to STAT2_CONFIG for decoding.
		STAT3_CONFIG	4:0	RW	3	Configure STAT3 output pin. Refer to STAT2_CONFIG for decoding.
814 to 816	RSVD	RSVD	15:0	—	—	Reserved.
817	ANC_CONTROL	RSVD	15:4	—	—	Reserved.
		ANC_DATA_SWITCH	3:3	RW	0	Switches between FIFO memories.
		ANC_DATA_DEL	2:2	RW	0	Removes ancillary data from output video stream.
		HD_ANC_Y1_C2	1:1	RW	0	Extracts ancillary data from both Y (DS1) and C (DS2). Extract Ancillary data from Luma and Chroma channels (HD inputs) Extract Ancillary data from Data Stream 1 and Data Stream 2 (3G Level A inputs)
						Extract Ancillary data from Luma and Chroma channels of Data Stream 1 (3G Level B inputs, when ANC_EXT_SEL_DS2_DS1 = 0)
						Extract Ancillary data from Luma and Chroma channels of Data Stream 2 (3G Level B inputs, when ANC_EXT_SEL_DS2_DS1 = 1)
		HD_ANC_C2	0:0	RW	0	Extracts ancillary data from C (DS2) only. Extract Ancillary data only from Chroma channel (HD inputs) Extract Ancillary data only from Data Stream 2 (3G Level A inputs)
						Extract Ancillary data only from Chroma channel of Data Stream 1 (3G Level B inputs, when ANC_EXT_SEL_DS2_DS1 = 0)
						Extract Ancillary data only from Chroma channel of Data Stream 2 (3G Level B inputs, when ANC_EXT_SEL_DS2_DS1 = 1)

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
818	ANC_LINEA	RSVD	15:11	—	—	Reserved.
		ANC_LINE_A	10:0	RW	0	Video line number to extract ancillary data from.
819	ANC_LINEB	RSVD	15:11	—	—	Reserved.
		ANC_LINE_B	10:0	RW	0	Second video line number to extract ancillary data from.
81A to 81D	RSVD	RSVD	15:0	—	—	Reserved.
81E	ANC_TYPE_1_DS1	ANC_TYPE1_DS1	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
81F	ANC_TYPE_2_DS1	ANC_TYPE2_DS1	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
820	ANC_TYPE_3_DS1	ANC_TYPE3_DS1	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
821	ANC_TYPE_4_DS1	ANC_TYPE4_DS1	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
822	ANC_TYPE_5_DS1	ANC_TYPE5_DS1	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
823	ANC_TYPE_1_DS2	ANC_TYPE1_DS2	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
824	ANC_TYPE_2_DS2	ANC_TYPE2_DS2	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
825	ANC_TYPE_3_DS2	ANC_TYPE3_DS2	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
826	ANC_TYPE_4_DS2	ANC_TYPE4_DS2	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID
827	ANC_TYPE_5_DS2	ANC_TYPE5_DS2	15:0	RW	0	Programmable DID/SDID pair to extract: [15:8] = DID [7:0] = SDID

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
828	VIDEO_FORMAT_352_A_1	VIDEO_FORMAT_2_DS1_LUMA	15:8	RO	0	SMPTE 352 embedded packet Luma – byte 2.
		VIDEO_FORMAT_1_DS1_LUMA	7:0	RO	0	SMPTE 352 embedded packet Luma – byte 1: [7] = VERSION_352M [6:0] = Video Payload Identifier
829	VIDEO_FORMAT_352_B_1	VIDEO_FORMAT_4_DS1_LUMA	15:8	RO	0	SMPTE 352 embedded packet Luma – byte 4.
		VIDEO_FORMAT_3_DS1_LUMA	7:0	RO	0	SMPTE 352 embedded packet Luma – byte 3.
82A	VIDEO_FORMAT_352_C_1	VIDEO_FORMAT_2_DS1_CHROMA	15:8	RO	0	SMPTE 352 embedded packet Chroma – byte 2.
		VIDEO_FORMAT_1_DS1_CHROMA	7:0	RO	0	SMPTE 352 embedded packet Chroma – byte 1.
82B	VIDEO_FORMAT_352_D_1	VIDEO_FORMAT_4_DS1_CHROMA	15:8	RO	0	SMPTE 352 embedded packet Chroma – byte 4.
		VIDEO_FORMAT_3_DS1_CHROMA	7:0	RO	0	SMPTE 352 embedded packet Chroma – byte 3.
82C	VIDEO_FORMAT_352_A_2	VIDEO_FORMAT_2_DS2_LUMA	15:8	RO	0	SMPTE 352 embedded packet – byte 2 (3G mode - second data stream).
		VIDEO_FORMAT_1_DS2_LUMA	7:0	RO	0	SMPTE 352 embedded packet – byte 1 (3G mode - second data stream): [7] = VERSION_352M [6:0] = Video Payload Identifier
82D	VIDEO_FORMAT_352_B_2	VIDEO_FORMAT_4_DS2_LUMA	15:8	RO	0	SMPTE 352 embedded packet – byte 4 (3G mode - second data stream).
		VIDEO_FORMAT_3_DS2_LUMA	7:0	RO	0	SMPTE 352 embedded packet – byte 3 (3G mode - second data stream).
82E	VIDEO_FORMAT_352_C_2	VIDEO_FORMAT_2_DS2_CHROMA	15:8	RO	0	SMPTE 352 embedded packet Chroma – byte 2 (3G mode - second data stream).
		VIDEO_FORMAT_1_DS2_CHROMA	7:0	RO	0	SMPTE 352 embedded packet Chroma – byte 1 (3G mode - second data stream).
82F	VIDEO_FORMAT_352_D_2	VIDEO_FORMAT_4_DS2_CHROMA	15:8	RO	0	SMPTE 352 embedded packet Chroma – byte 4 (3G mode - second data stream).
		VIDEO_FORMAT_3_DS2_CHROMA	7:0	RO	0	SMPTE 352 embedded packet Chroma – byte 3 (3G mode - second data stream).
830	RASTER_STRUC_1_DS1	RSVD	15:14	—	—	Reserved.
		WORDS_PER_ACTLINE_DS1	13:0	RO	0	Words per active line.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
831	RASTER_STRUC_2_DS1	RSVD	15:14	—	—	Reserved.
		WORDS_PER_LINE_DS1	13:0	RO	0	Total words per line.
832	RASTER_STRUC_3_DS1	RSVD	15:11	—	—	Reserved.
		LINES_PER_FRAME_DS1	10:0	RO	0	Total lines per frame.
833	RASTER_STRUC_4_DS1	RATE_DET	15:14	RO	0	Detected rate: 00 = HD 01, 11 = SD 10 = 3G
		M_DS1	13:13	RO	0	Specifies M value: 0 = 1.000 1 = 1.001
		STD_LOCK_DS1	12:12	RO	0	Video standard lock.
		INT_PROG_DS1	11:11	RO	0	Interlaced or progressive.
		ACTLINE_PER_FIELD_DS1	10:0	RO	0	Active lines per frame.
834	FLYWHEEL_STATUS	RSVD	15:2	—	—	Reserved.
		V_LOCK_FE	1:1	RO	0	Indicates that flywheel is locked to V timing.
		H_LOCK_FE	0:0	RO	0	Indicates that flywheel is locked to H timing.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
835	RATE_SEL	RSVD	15:5	—	—	Reserved.
		SMPTE_BYPASS	4:4	RW	1	When the AUTO_MAN bit is HIGH (Default), this bit is OUTPUT and indicates 1 when the device locks to a SMPTE compliant input. It indicates 0 when under all other conditions. When the AUTO_MAN bit is LOW, this bit is INPUT and set 1 to carry out SMPTE scrambling and I/O processing. If set 0, no SMPTE scrambling takes place and none of I/O processing features of the device are available.
		DVB_ASI	3:3	RW	0	Used to enable/disable DVB-ASI data extraction in manual mode. When the AUTO_MAN bit is LOW and SMPTE_BYPASS bit is LOW, this pin is INPUT and this bit is set HIGH, the device will carry out DVB-ASI data extraction and processing.
		AUTO_MAN	2:2	RW	1	Data rate detect mode: 0 = Manual mode 1 = Automatic mode
		RATE_SEL_TOP	1:0	RW	0	Programmable rate select in manual mode: 00 _b = HD 01 _b , 11 _b = SD 10 _b = 3G
836	TIM_861_FORMAT	RSVD	15:7	—	—	Reserved.
		FORMAT_ERR	6:6	RO	1	Indicates standard is not recognized for CEA861-D conversion.
		FORMAT_ID_861	5:0	RO	0	CEA-861-D format ID of input video stream.
837	TIM_861_CFG	RSVD	15:3	—	—	Reserved.
		VSNC_INVERT	2:2	RW	0	Invert V-sync pulse.
		HSYNC_INVERT	1:1	RW	0	Invert H-sync pulse.
		RSVD	0:0	RW	—	Reserved.
838 to 847	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
		RSVD	15:10	—	—	Reserved.
848	ERROR_MASK_1	ERROR_MASK_DS1_STICKY	9:0	RW	0	Error mask for global error vector - DS1 triggers (mask for sticky errors): [9] = EDH full field CRC error mask [8] = EDH active picture CRC error mask [7] = Lock error mask [6] = Chroma channel ancillary data checksum error mask [5] = Luma channel ancillary data checksum error mask [4] = Chroma channel CRC error mask [3] = Luma channel CRC error mask [2] = Line number error mask [1] = SAV error mask [0] = EAV error mask
		RSVD	15:10	—	—	Reserved.
849	ERROR_MASK_2	ERROR_MASK_DS1	9:0	RW	0	Error mask for global error vector - DS1 triggers (mask for non-sticky errors): [9] = EDH full field CRC error mask [8] = EDH active picture CRC error mask [7] = Lock error mask [6] = Chroma channel ancillary data checksum error mask [5] = Luma channel ancillary data checksum error mask [4] = Chroma channel CRC error mask [3] = Luma channel CRC error mask [2] = Line number error mask [1] = SAV error mask [0] = EAV error mask

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
84A	ERROR_MASK_3	RSVD	15:14	—	—	Reserved.
		ERROR_MASK_DS2	13:0	RW	0	Error mask for global error vector - DS2 triggers (sticky and non-sticky): [13] = Chroma channel ancillary data checksum error mask (non-sticky version) [12] = Luma channel ancillary data checksum error mask (non-sticky version) [11] = Chroma channel CRC error mask (non-sticky version) [10] = Luma channel CRC error mask (non-sticky version) [9] = Line number error mask (non-sticky version) [8] = SAV error mask (non-sticky version) [7] = EAV error mask (non-sticky version) [6] = Chroma channel ancillary data checksum error mask (sticky version) [5] = Luma channel ancillary data checksum error mask (sticky version) [4] = Chroma channel CRC error mask (sticky version) [3] = Luma channel CRC error mask (sticky version) [2] = Line number error mask (sticky version) [1] = SAV error mask (sticky version) [0] = EAV error mask (sticky version)
84B	ACG_CTRL	RSVD	15:6	—	—	Reserved.
		AUDIO_FREE_RUN	5:5	RW	0	0 = Audio digital PLL runs in acquisition mode and tries to lock to REF_CLK 1 = Audio digital PLL runs in free run mode and ignores the REF_CLK
		SCLK_INV	4:4	RW	0	When HIGH, inverts polarity of output serial audio clock.
		AMCLK_INV	3:3	RW	0	When HIGH, inverts polarity of output audio master clock.
		RSVD	2:2	RW	0	Reserved.
		AMCLK_SEL	1:0	RW	0	Audio Master Clock (AMCLK) select: 00 _b = 128fs 01 _b = 256fs 10 _b = 512fs
84C	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
		RSVD	15:4	—	—	Reserved.
84D	INPUT_CONFIG	SEL_INPUT	3:2	RW	3	Bit 0 selects the parallel retimed output and Bit 1 selects the DDO path: 00 _b = Selects DDIO/ $\overline{\text{DDI0}}$ parallel retimed output and DDO path 01 _b = Selects DD11/ $\overline{\text{DDI1}}$ parallel retimed output and DDIO/ $\overline{\text{DDI0}}$ for DDO path 10 _b = Selects DDIO/ $\overline{\text{DDI0}}$ parallel retimed output and DD11/ $\overline{\text{DDI1}}$ for DDO path 11 _b = Selects DD11/ $\overline{\text{DDI1}}$ parallel retimed output and DDO path
		RSVD	1:0	—	—	Reserved.
84E to 867	RSVD	RSVD	15:0	—	—	Reserved.
		RSVD	15:2	—	—	Reserved.
868	PLL_CTRL_0	CD_SELECT	1:0	RW	2	Controls the source of CARRIER_DETECT: 00 _b = CD_DIGITAL_FILT (edge detector) 01 _b = CD_ANALOG_FILT (LOS detector) 10 _b = CD_DIGITAL_FILT AND CD_ANALOG_FILT 11 _b = CD_DIGITAL_FILT OR CD_ANALOG_FILT
869 to 86A	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
86B	PLL_STAT	RSVD	15:10	—	—	Reserved.
		PLLSM_CDRSEL	9:9	RO	0	Unforced CDRSEL output: 0 = The device is using the clock as reference 1 = The device is using data as reference
		PLLSM_LOCK	8:8	RO	0	Unforced LOCK output.
		LOS	7:7	RO	0	Loss of signal indicator. Inverse of CARRIER_DETECT.
		PHASELOCK_ANALOG	6:6	RO	0	Output of analog lock detector.
		PLD_PHASELOCK_DIGITAL	5:5	RO	0	Output of digital phaselock detector.
		CD_CARRIER_DETECT	4:4	RO	0	Unforced CARRIER_DETECT signal (after MUX).
		CD_DIGIAL_FILT	3:3	RO	0	Filtered digital carrier detect status: 0 = The edge detection circuit is not detecting data 1 = The edge detection circuit is detecting data
		CD_ANALOG_FILT	2:2	RO	0	Filtered analog carrier detect status: 0 = When LOW, the LOS circuit is not detecting data 1 = When HIGH the LOS circuit is detecting data
86C	PLL_STICKY_STAT	RSVD	1:0	—	—	Reserved.
		RSVD	15:7	—	—	Reserved.
		LOCK_LOST_STICKY	6:6	ROCW	0	Sticky bit for loss of lock indicator. Inverse of LOCK output.
		PHASELOCK_LOST_STICK	5:5	ROCW	0	Sticky bit for loss of phaselock indicator. Inverse of phaselock output from PHASE_DETECT.
		LOS_STICKY	4:4	ROCW	0	Sticky bit for loss of signal indicator (after MUX). Inverse of CARRIER_DETECT.
		RSVD	3:0	ROCW	—	Reserved.
86D to 86E	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
86F	LOS_CTRL	RSVD	15:9	—	—	Reserved.
		LOS_SEL	8:8	RW	1	LOS select: 0 = Selects DDI0 (directly from pin) for LOS sensing 1 = Selects DDI1 (directly from pin) for LOS sensing
		LOS_CD_TIME_CONSTANT	7:6	RW	1	Controls the magnitude of the current that is used to discharge the hold capacitor present at the output of the voltage-to-current amplifier in the LOS circuit.
		LOS_CD_HYSTERESIS	5:4	RW	1	Carrier detect hysteresis. Controls the amount of Schmitt trigger hysteresis in the LOS circuit.
		RSVD	3:0	RW	5	Reserved.
870	DELAY_LINE_CTRL_1	RSVD	15:15	—	—	Reserved.
		PCLK_DELAY_SD	14:10	RW	0	Controls the offset for the delay line in SD: 00000 _b = No delay 11111 _b = Max delay Step Size = 0.006 UI
		PCLK_DELAY_HD	9:5	RW	0	Controls the offset for the delay line in HD: 00000 _b = No delay 11111 _b = Max delay Step Size = 0.031 UI
		PCLK_DELAY_3G	4:0	RW	0	Controls the offset for the delay line in 3G: 00000 _b = No delay 11111 _b = Max delay Step Size = 0.031 UI
871	DELAY_LINE_CTRL_2	RSVD	15:3	—	—	Reserved.
		PCLK_INVERT_SD	2:2	RW	0	Controls the inversion of PCLK in SD.
		PCLK_INVERT_HD	1:1	RW	0	Controls the inversion of PCLK in HD.
		PCLK_INVERT_3G	0:0	RW	0	Controls the inversion of PCLK in 3G.
872 to 873	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
874	CLK_GEN	RSVD	15:3	—	—	Reserved.
		DDR_PHASE_DET_INVERT	2:2	RW	1	Swaps the phase in DDR mode. User can choose whether to start data transmission from a rising edge or a falling edge of the DDR clock. By default, rising edge is DS1 (Luma) and falling edge is DS2 (Chroma)
		RSVD	1:1	RW	0	Reserved.
		SD_HD_DDR_SEL	0:0	RW	0	Enables DDR mode for SD, HD, and non-SMPTE data if DDR is supported for that specific rate.
875	RSVD	RSVD	15:0	RW	0	Reserved.
876	RSVD	RSVD	15:0	—	—	Reserved.
877	PIN_CSR_SELECT	RSVD	15:3	—	—	Reserved.
		20BIT/ $\overline{10BIT}$	2:2	RW	0	CSR value for 20BIT/ $\overline{10BIT}$ to be used when 20BIT/ $\overline{10BIT}$ _PIN = 1 0 = Output parallel data bus is 10 bits 1 = Output parallel data bus is 20 bits
		20BIT/ $\overline{10BIT}$ _PIN	1:1	RW	0	Override for 20BIT/ $\overline{10BIT}$ pin: 0 = Selects pin value 1 = Selects CSR value
		IOPROC_EN	0:0	RW	1	CSR value for IOPROC_EN: 0 = Disables all video and audio processing 1 = Enables all video and audio processing that are not masked in the IOPROC_1, IOPROC_2 registers

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
878	IO_DRIVE_STRENGTH	RSVD	15:9	—	—	Reserved.
		CSR_PCLK_HIZ	9:9	RW	0	In auto rate detection mode (AUTO_MAN = 1), when PCLK_UNLOCKED_HIZ_SEL is HIGH, 0 = PCLK is active, regardless of locked condition of device 1 = PCLK is HIZ when device is unlocked, active when device is locked
		MUTE_ON_NO_LOCK	8:8	RW	1	0 = Video data output bus is not muted when not locked 1 = Parallel video data output bus is muted (set to zero) when not locked
		IO_DS_CTRL	7:6	RW	1	Drive strength control for PCLK: 00 _b = 4mA 01 _b , 10 _b = 8mA 11 _b = 12mA
		IO_DS_CTRL_STAT	5:4	RW	1	Drive strength control for STAT[5:0]: 00 _b = 4mA 01 _b , 10 _b = 8mA 11 _b = 12mA
		IO_DS_CTRL_DOUT_LSBS	3:2	RW	1	Drive strength control for DOUT[9:0]: 00 _b = 4mA 01 _b , 10 _b = 8mA 11 _b = 12mA
		IO_DS_CTRL_DOUT_MSBS	1:0	RW	1	Drive strength control for DOUT[19:10]: 00 _b = 4mA 01 _b , 10 _b = 8mA 11 _b = 12mA
879 to 87A	RSVD	RSVD	15:0	—	—	Reserved.
87B	OUTPUT_BUFFER_CTRL_1	RSVD	15:12	—	—	Reserved.
		SWING_3G	11:8	RW	3	3G swing control. Step Size = 40mV.
		SWING_HD	7:4	RW	3	HD swing control. Step Size = 40mV.
		SWING_SD	3:0	RW	3	SD swing control. Step Size = 40mV.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
87C	OUTPUT_BUFFER_CTRL_2	RSVD	15:10	—	—	Reserved.
		DRIVER_POL_INV_D2A	9:9	RW	0	Input signal polarity invert within the driver block.
		DE_LEVEL_3G	8:6	RW	0	De-emphasis level control for 3G rate. 0 = 0 dB 1 = 0.6 dB 2 = 1.9 dB 3 = 3.5 dB 4 = 5.5 dB 5 = 8.0 dB 6 = 11.5 dB 7 = 17.2 dB
		DE_LEVEL_HD	5:3	RW	0	De-emphasis level control for HD rate. 0 = 0 dB 1 = 0.9 dB 2 = 2.1 dB 3 = 3.7 dB 4 = 5.6 dB 5 = 8.1 dB 6 = 11.6 dB 7 = 17.4 dB
		DE_LEVEL_SD	2:0	RW	0	De-emphasis level control for SD rate. 0 = 0 dB 1 = 0.8 dB 2 = 2.2 dB 3 = 3.8 dB 4 = 5.8 dB 5 = 8.2 dB 6 = 11.6 dB 7 = 17.3 dB
87D	RSVD	RSVD	15:0	—	—	Reserved.
87E	M_DETECTION_TOLERANCE_DS1	RSVD	15:4	—	—	Reserved.
		M_DETECTION_TOLERANCE_DS1	3:0	RW	2	16 line count tolerance window horizontal count from reference count.
87F	RSVD	RSVD	15:0	—	—	Reserved.
880	M_DETECTION_TOLERANCE_DS2	RSVD	15:4	—	—	Reserved.
		M_DETECTION_TOLERANCE_DS2	3:0	RW	2	16 line count tolerance window horizontal count from reference count.
881 to 882	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
883	DDO_MUTE_CTRL	RSVD	15:3	—	—	Reserved.
		FORCE_DDO_MUTE_VAL	2:2	RW	0	When FORCE_DDO_MUTE is HIGH: 0 = Sets DDO_MUTE_0 (does not mute DDO) 1 = Sets DDO_MUTE_1 (mutes DDO)
		FORCE_DDO_MUTE	1:1	RW	0	Forces the DDO output mute to the value in FORCE_DDO_MUTE_VAL.
		RSVD	0:0	—	—	Reserved.
884 to 8CA	RSVD	RSVD	15:0	—	—	Reserved.
8CB	GENERAL_STATUS	VERSION_ID	15:8	RO	0	Version ID.
		SMPTE_BYPASSB_STATUS	7:7	RO	0	Same as value that can be brought on STAT outputs, indicates if the device is in SMPTE_BYPASS mode: 0 = SMPTE_BYPASS mode 1 = Not in SMPTE_BYPASS mode
		DVB_ASI_STATUS	6:6	RO	0	Same as value that can be brought on STAT outputs, indicates if the device is locked to DVB_ASI: 0 = Not locked to DVB_ASI 1 = Locked to DVB_ASI
		RSVD	5:3	RO	—	Reserved.
		RSVD	2:0	RO	—	Reserved.
		RSVD	15:0	—	—	Reserved.
8CC to 8CE	RSVD	RSVD	15:0	—	—	Reserved.
8CF	RASTER_STRUC_1_DS2	RSVD	15:14	—	—	Reserved.
		WORDS_PER_ACTLINE_DS2	13:0	RO	0	Words per active line.
8D0	RASTER_STRUC_2_DS2	RSVD	15:14	—	—	Reserved.
		WORDS_PER_LINE_DS2	13:0	RO	0	Total words per line.
8D1	RASTER_STRUC_3_DS2	RSVD	15:11	—	—	Reserved.
		LINES_PER_FRAME_DS2	10:0	RO	0	Total lines per frame.

Table 5-8: Video Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit Slice	R/W	Reset Value _h	Description
8D2	RASTER_STRUC_4_DS2	RSVD	15:14	—	—	Reserved.
		M_DS2	13:13	RO	0	Specifies M value: 0 = 1.000 1 = 1.001
		STD_LOCK_DS2	12:12	RO	0	Video standard lock.
		INT_PROG_DS2	11:11	RO	0	Interlaced or progressive.
		ACTLINE_PER_FIELD_DS2	10:0	RO	0	Active lines per frame.
8D3	GENERAL_CONTROL	RSVD	15:5	—	—	Reserved.
		DVB_NOISE_IMMUNITY	4:4	RW	0	Enables extra noise immunity when in DVB_ASI mode: 0 = Exit DVB_ASI and enter SMPTE mode when 3 TRSs in 2 lines 1 = Exit DVB_ASI and enter SMPTE mode when 7 TRSs in 4 lines
		LOCK_NOISE_IMM_INCR	3:3	RW	0	Enables extra noise immunity on SMPTE detected lock when HIGH by forcing detection of 3 TRS words with the last 2 TRS words having the same alignment before locking to SMPTE.
		RSVD	2:0	RW	0	Reserved.
8D4 to 989	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-9: HD and 3G Audio Core CSR Descriptions

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A00	RSVD	RSVD	15:0	—	—	Reserved.
		RSVD	15:14	—	—	Reserved.
		ASWLB	13:12	RW	0	Secondary group output word length. 00 _b = 24 bits 01 _b = 20 bits 10 _b = 16 bits 11 _b = Invalid
		ASWLA	11:10	RW	0	Primary group output word length. 00 _b = 24 bits 01 _b = 20 bits 10 _b = 16 bits 11 _b = Invalid
		AMB	9:8	RW	3	Secondary group output format selector. 00 _b = AES/EBU audio output 01 _b = Serial audio output: left justified 10 _b = Serial audio output: right justified 11 _b = I ² S serial audio output
	CFG_AUD	AMA	7:6	RW	3	Primary group output format selector. 00 _b = AES/EBU audio output 01 _b = Serial audio output: left justified 10 _b = Serial audio output: right justified 11 _b = I ² S serial audio output
		EXTEND_IDB	5:5	RW	0	0 = Secondary audio group will extracted audio groups 1, 2, 3, or 4 1 = Secondary audio group will extracted audio groups 5, 6, 7, or 8
		IDB	4:3	RW	1	Along with EXTEND_IDB, specifies the Secondary audio group to extract. 00 _b = Audio group #1 or #5 01 _b = Audio group #2 or #6 10 _b = Audio group #3 or #7 11 _b = Audio group #4 or #8 Note: Should IDA & EXTEND_IDA and IDB & EXTEND_IDB be set to the same audio group, IDA and IDB automatically revert to their default values.
		EXTEND_IDA	2:2	RW	0	0 = Primary audio group will extracted audio groups 1, 2, 3 or 4. 1 = Primary audio group will extracted audio groups 5, 6, 7 or 8.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A01 (Continued)	CFG_AUD (Continued)	IDA	1:0	RW	0	Along with EXTEND_IDA, specifies the Primary audio group to extract. 00_b = Audio group #1 or #5 01_b = Audio group #2 or #6 10_b = Audio group #3 or #7 11_b = Audio group #4 or #8 Note: Should IDA & EXTEND_IDA and IDB & EXTEND_IDB be set to the same audio group, IDA and IDB automatically revert to their default values.
A02	CFG_AUD_1	RSVD	15:4	—	—	Reserved.
		ECC_OFF	3:3	RW	0	When HIGH, disables ECC error correction.
		ALL_DEL	2:2	RW	0	Selects deletion of all audio data and all audio control packets: 0 = Do not delete existing audio data and control packets 1 = Delete existing audio data and control packets
		MUTE_ALL	1:1	RW	0	Mute all output channels: 0 = Normal 1 = Muted
		ACS_USE_SECOND	0:0	RW	0	Extract Audio Channel Status from second channel pair.
A03	ACS_DET	RSVD	15:8	—	—	Reserved.
		DBNB_ERR	7:7	ROCW	0	Set when Secondary group audio Data Block Number sequence is discontinuous.
		DBNA_ERR	6:6	ROCW	0	Set when Primary group audio Data Block Number sequence is discontinuous.
		CTRB_DET	5:5	ROCW	0	Set when Secondary group audio control packet is detected.
		CTRA_DET	4:4	ROCW	0	Set when Primary group audio control packet is detected.
		ACS_DET3_4B	3:3	ROCW	0	Secondary group audio status detected for channels 3 and 4.
		ACS_DET1_2B	2:2	ROCW	0	Secondary group audio status detected for channels 1 and 2.
		ACS_DET3_4A	1:1	ROCW	0	Primary group audio status detected for channels 3 and 4.
		ACS_DET1_2A	0:0	ROCW	0	Primary group audio status detected for channels 1 and 2.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A04	AUD_DET1	RSVD	15:13	—	—	Reserved.
		IDB_READBACK	12:11	RO	1	Actual value of IDB in the hardware.
		IDA_READBACK	10:9	RO	0	Actual value of IDA in the hardware.
		ADPG8_DET	8:8	RO	0	Set to 1 and remains set while Group 8 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ADPG7_DET	7:7	RO	0	Set to 1 and remains set while Group 7 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ADPG6_DET	6:6	RO	0	Set to 1 and remains set while Group 6 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ADPG5_DET	5:5	RO	0	Set to 1 and remains set while Group 5 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ADPG4_DET	4:4	RO	0	Set to 1 and remains set while Group 4 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ADPG3_DET	3:3	RO	0	Set to 1 and remains set while Group 3 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ADPG2_DET	2:2	RO	0	Set to 1 and remains set while Group 2 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ADPG1_DET	1:1	RO	0	Set to 1 and remains set while Group 1 audio data packets are detected. Goes to 0 if none are detected for more than 1 frame.
		ACS_APPLY_WAIT	0:0	RO	0	Set while output channels 1 and 2 are waiting for a status boundary to apply the ACSR[183:0] data.
A05	AUD_DET2	RSVD	15:2	—	—	Reserved.
		ECCA_ERROR	1:1	ROCW	0	Primary group audio data packet error detected.
		ECCB_ERROR	0:0	ROCW	0	Secondary group audio data packet error detected.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A06	REGEN	RSVD	15:2	—	—	Reserved.
		ACS_APPLY	1:1	RW	0	When set to 1, this causes channel status data in ACSR[183:0] (ACSR_BYTE_* registers) to be transferred to the channel status replacement mechanism. The transfer shall not occur until the next status boundary.
		ACS_REGEN	0:0	RW	0	Specifies that Audio Channel Status of all channels should be replaced with ACSR[183:0] field (ACSR_BYTE_* registers). 0 = Do not replace Channel Status 1 = Replace Channel Status of all channels
A07	CH_MUTE	RSVD	15:8	—	—	Reserved.
		MUTEB	7:4	RW	0	Mute Secondary output channels 4 to 1; bits 3:0 = channel 4:1 0 = Normal 1 = Muted
		MUTEA	3:0	RW	0	Mute Primary output channels 4 to 1; bits 3:0 = channel 4:1 0 = Normal 1 = Muted
A08	CH_VALID	RSVD	15:8	—	—	Reserved.
		CH4_VALIDDB	7:7	RO	0	Secondary group channel 4 sample validity flag.
		CH3_VALIDDB	6:6	RO	0	Secondary group channel 3 sample validity flag.
		CH2_VALIDDB	5:5	RO	0	Secondary group channel 2 sample validity flag.
		CH1_VALIDDB	4:4	RO	0	Secondary group channel 1 sample validity flag.
		CH4_VALIDA	3:3	RO	0	Primary group channel 4 sample validity flag.
		CH3_VALIDA	2:2	RO	0	Primary group channel 3 sample validity flag.
		CH2_VALIDA	1:1	RO	0	Primary group channel 2 sample validity flag.
		CH1_VALIDA	0:0	RO	0	Primary group channel 1 sample validity flag.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A09	AUDIO_INT_ENABLE	RSVD	15:8	—	—	Reserved.
		EN_ADPG8_DET	7:7	RW	0	When enabled, interrupt will assert when ADPG8_DET is set.
		EN_ADPG7_DET	6:6	RW	0	When enabled, interrupt will assert when ADPG7_DET is set.
		EN_ADPG6_DET	5:5	RW	0	When enabled, interrupt will assert when ADPG6_DET is set.
		EN_ADPG5_DET	4:4	RW	0	When enabled, interrupt will assert when ADPG5_DET is set.
		EN_ADPG4_DET	3:3	RW	0	When enabled, interrupt will assert when ADPG4_DET is set.
		EN_ADPG3_DET	2:2	RW	0	When enabled, interrupt will assert when ADPG3_DET is set.
		EN_ADPG2_DET	1:1	RW	0	When enabled, interrupt will assert when ADPG2_DET is set.
		EN_ADPG1_DET	0:0	RW	0	When enabled, interrupt will assert when ADPG1_DET is set.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A0A	AUDIO_INT_ENABLE2	RSVD	15:11	—	—	Reserved.
		EN_MISSING_PHASE	10:10	RW	0	When enabled, interrupt will assert when chosen group's phase data is missing or invalid.
		EN_ACS_DET3_4B	9:9	RW	0	When enabled, interrupt will assert when ACS_DET3_4B is set.
		EN_ACS_DET1_2B	8:8	RW	0	When enabled, interrupt will assert when ACS_DET1_2B is set.
		EN_ACS_DET3_4A	7:7	RW	0	When enabled, interrupt will assert when ACS_DET3_4A is set.
		EN_ACS_DET1_2A	6:6	RW	0	When enabled, interrupt will assert when ACS_DET1_2A is set.
		EN_CTRB_DET	5:5	RW	0	When enabled, interrupt will assert when CTRB_DET is set.
		EN_CTRA_DET	4:4	RW	0	When enabled, interrupt will assert when CTRA_DET is set.
		EN_DBNB_ERR	3:3	RW	0	When enabled, interrupt will assert when DBNB_ERR is set.
		EN_DBNA_ERR	2:2	RW	0	When enabled, interrupt will assert when DBNA_ERR is set.
		EN_ECCB_ERR	1:1	RW	0	When enabled, interrupt will assert when ECCB_ERROR is set.
		EN_ECCA_ERR	0:0	RW	0	When enabled, interrupt will assert when ECCA_ERROR is set.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A0B	CFG_AUD_2	RSVD	15:12	—	—	Reserved.
		SEL_PHASE_SRC	11:11	RW	0	Selects between the Primary and Secondary embedded phase info when FORCE_PHASE_SRC set to 1. 0 = Use primary group phase information 1 = Use secondary group phase information
		FORCE_PHASE_SRC	10:10	RW	0	Phase source override: 0 = Auto phase source selection 1 = Manually set the phase source using SEL_PHASE_SRC
		RSVD	9:8	—	—	Reserved.
		FORCE_M	7:7	RW	0	Disables auto-detection of M value and forces it to the state specified by the FORCE_MEQ1001 signal.
		FORCE_MEQ1001	6:6	RW	0	Specifies M value when FORCE_M is set: 0 = M is 1.000 1 = M is 1.001
		IGNORE_PHASE	5:5	RW	0	When HIGH, causes the HD_DEMUX to ignore the embedded clock info in both the Primary and Secondary group audio data packets. Clock is generated based on the video format and M value.
		FORCE_ACLK128	4:4	RW	0	When HIGH, causes the HD_DEMUX to ignore embedded clock info and derive phase information from ACLK128.
		EN_NOT_LOCKED	3:3	RW	0	When enabled, interrupt will assert when locked is not asserted.
		EN_NO_VIDEO	2:2	RW	0	When enabled, interrupt will assert when the video format is unknown.
		EN_INVALID_EMBEDDED_PHASE_B	1:1	RW	0	When enabled, interrupt will assert when INVALID_EMBEDDED_PHASE_B is set.
		EN_INVALID_EMBEDDED_PHASE_A	0:0	RW	0	When enabled, interrupt will assert when INVALID_EMBEDDED_PHASE_A is set.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A0C	CFG_AUD_3	RSVD	15:6	—	—	Reserved.
		SELECTED_PHASE_SRC	5:5	RO	0	Indicates the source audio group for the embedded phase information: 0 = Using primary group phase information 1 = Using secondary group phase information
		MISSING_PHASE	4:4	RO	0	When HIGH, either INVALID_EMBEDDED_PHASE_A or INVALID_EMBEDDED_PHASE_B is asserted based on SELECTED_PHASE_SRC.
		INVALID_EMBEDDED_PHASE_B	3:3	RO	0	When HIGH, secondary group audio data packets have invalid embedded phase data.
		INVALID_EMBEDDED_PHASE_A	2:2	RO	0	When HIGH, primary group audio data packets have invalid embedded phase data.
		EMBEDDED_PHASE_IGNORED_B	1:1	RO	0	When HIGH, secondary group audio data packets have invalid embedded phase data or when IGNORE_PHASE or FORCE_ACLK128 are asserted.
		EMBEDDED_PHASE_IGNORED_A	0:0	RO	0	When HIGH, primary group audio data packets have invalid embedded phase data or when IGNORE_PHASE or FORCE_ACLK128 are asserted.
A0D	OUTPUT_SEL_1	RSVD	15:12	—	—	Reserved.
		OP4_SRC	11:9	RW	3	Output channel 4 source selector: 000 _b = Primary audio group channel 1 001 _b = Primary audio group channel 2 010 _b = Primary audio group channel 3 011 _b = Primary audio group channel 4 100 _b = Secondary audio group channel 1 101 _b = Secondary audio group channel 2 110 _b = Secondary audio group channel 3 111 _b = Secondary audio group channel 4
		OP3_SRC	8:6	RW	2	Output channel 3 source selector. See OP4_SRC for decoding.
		OP2_SRC	5:3	RW	1	Output channel 2 source selector. See OP4_SRC for decoding.
		OP1_SRC	2:0	RW	0	Output channel 1 source selected. See OP4_SRC for decoding.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A0E	OUTPUT_SEL_2	RSVD	15:12	—	—	Reserved.
		OP8_SRC	11:9	RW	7	Output channel 8 source selector. See OP4_SRC for decoding.
		OP7_SRC	8:6	RW	6	Output channel 7 source selector. See OP4_SRC for decoding.
		OP6_SRC	5:3	RW	5	Output channel 6 source selector. See OP4_SRC for decoding.
		OP5_SRC	2:0	RW	4	Output channel 5 source selector. See OP4_SRC for decoding.
A0F to A1F	RSVD	RSVD	15:0	—	—	Reserved.
A20	AFNA	RSVD	15:9	—	—	Reserved.
		AFNA	8:0	RO	0	Primary group audio frame number.
A21	RATEA	RSVD	15:4	—	—	Reserved.
		RATEA	3:1	RO	0	Primary group sampling frequency for channels 1 and 2.
		ASXA	0:0	RO	0	Primary group asynchronous mode for channels 1 and 2.
A22	ACTA	RSVD	15:4	—	—	Reserved.
		ACTA	3:0	RO	0	Primary group active channels.
A23	PRIM_AUD_DELAY_1	RSVD	15:9	—	—	Reserved.
		DEL1_2A_1	8:1	RO	0	Primary Audio group delay data for channels 1 and 2 [7:0].
		EBIT1_2A	0:0	RO	0	Primary Audio group delay data valid flag for channels 1 and 2.
A24	PRIM_AUD_DELAY_2	RSVD	15:9	—	—	Reserved.
		DEL1_2A_2	8:0	RO	0	Primary Audio group delay data for channels 1 and 2 [16:8].
A25	PRIM_AUD_DELAY_3	RSVD	15:9	—	—	Reserved.
		DEL1_2A_3	8:0	RO	0	Primary Audio group delay data for channels 1 and 2 [25:17].
A26	PRIM_AUD_DELAY_4	RSVD	15:9	—	—	Reserved.
		DEL3_4A_4	8:1	RO	0	Primary Audio group delay data for channels 3 and 4 [7:0].
		EBIT3_4A	0:0	RO	0	Primary Audio group delay data valid flag for channels 3 and 4.
A27	PRIM_AUD_DELAY_5	RSVD	15:9	—	—	Reserved.
		DEL3_4A_5	8:0	RO	0	Primary Audio group delay data for channels 3 and 4 [16:8].

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A28	PRIM_AUD_DELAY_6	RSVD	15:9	—	—	Reserved.
		DEL3_4A_6	8:0	RO	0	Primary Audio group delay data for channels 3 and 4 [25:17].
A29 to A2F	RSVD	RSVD	15:0	—	—	Reserved.
A30	AFNB	RSVD	15:9	—	—	Reserved.
		AFNB	8:0	RO	0	Secondary group audio frame number.
A31	RATEB	RSVD	15:4	—	—	Reserved.
		RATEB	3:1	RO	0	Secondary group sampling frequency for channels 1 and 2.
		ASXB	0:0	RO	0	Secondary group asynchronous mode for channels 1 and 2.
A32	ACTB	RSVD	15:4	—	—	Reserved.
		ACTB	3:0	RO	0	Secondary group active channels.
A33	SEC_AUD_DELAY_1	RSVD	15:9	—	—	Reserved.
		DEL1_2B_1	8:1	RO	0	Secondary Audio group delay data for channels 1 and 2 [7:0].
		EBIT1_2B	0:0	RO	0	Secondary Audio group delay data valid flag for channels 1 and 2.
A34	SEC_AUD_DELAY_2	RSVD	15:9	—	—	Reserved.
		DEL1_2B_2	8:0	RO	0	Secondary Audio group delay data for channels 1 and 2 [16:8].
A35	SEC_AUD_DELAY_3	RSVD	15:9	—	—	Reserved.
		DEL1_2B_3	8:0	RO	0	Secondary Audio group delay data for channels 1 and 2 [25:17].
A36	SEC_AUD_DELAY_4	RSVD	15:9	—	—	Reserved.
		DEL3_4B_4	8:1	RO	0	Secondary Audio group delay data for channels 3 and 4 [7:0].
		EBIT3_4B	0:0	RO	0	Secondary Audio group delay data valid flag for channels 3 and 4.
A37	SEC_AUD_DELAY_5	RSVD	15:9	—	—	Reserved.
		DEL3_4B_5	8:0	RO	0	Secondary Audio group delay data for channels 3 and 4 [16:8].
A38	SEC_AUD_DELAY_6	RSVD	15:9	—	—	Reserved.
		DEL3_4B_6	8:0	RO	0	Secondary Audio group delay data for channels 3 and 4 [25:17].
A39 to A3F	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A40	ACSR1_2A_BYTE0_1	ACSR1_2A_0	15:0	RO	0	Bytes 0 and 1 of audio group A channel status for channels 1 and 2.
A41	ACSR1_2A_BYTE2_3	ACSR1_2A_2	15:0	RO	0	Bytes 2 and 3 of audio group A channel status for channels 1 and 2.
A42	ACSR1_2A_BYTE4_5	ACSR1_2A_4	15:0	RO	0	Bytes 4 and 5 of audio group A channel status for channels 1 and 2.
A43	ACSR1_2A_BYTE6_7	ACSR1_2A_6	15:0	RO	0	Bytes 6 and 7 of audio group A channel status for channels 1 and 2.
A44	ACSR1_2A_BYTE8_9	ACSR1_2A_8	15:0	RO	0	Bytes 8 and 9 of audio group A channel status for channels 1 and 2.
A45	ACSR1_2A_BYTE10_11	ACSR1_2A_10	15:0	RO	0	Bytes 10 and 11 of audio group A channel status for channels 1 and 2.
A46	ACSR1_2A_BYTE12_13	ACSR1_2A_12	15:0	RO	0	Bytes 12 and 13 of audio group A channel status for channels 1 and 2.
A47	ACSR1_2A_BYTE14_15	ACSR1_2A_14	15:0	RO	0	Bytes 14 and 15 of audio group A channel status for channels 1 and 2.
A48	ACSR1_2A_BYTE16_17	ACSR1_2A_16	15:0	RO	0	Bytes 16 and 17 of audio group A channel status for channels 1 and 2.
A49	ACSR1_2A_BYTE18_19	ACSR1_2A_18	15:0	RO	0	Bytes 18 and 19 of audio group A channel status for channels 1 and 2.
A4A	ACSR1_2A_BYTE20_21	ACSR1_2A_20	15:0	RO	0	Bytes 20 and 21 of audio group A channel status for channels 1 and 2.
A4B	ACSR1_2A_BYTE22	RSVD	15:8	—	—	Reserved.
		ACSR1_2A_22	7:0	RO	0	Bytes 22 of audio group A channel status for channels 1 and 2.
A4C to A4F	RSVD	RSVD	15:0	—	—	Reserved.
A50	ACSR3_4A_BYTE0_1	ACSR3_4A_0	15:0	RO	0	Bytes 0 and 1 of audio group A channel status for channels 3 and 4.
A51	ACSR3_4A_BYTE2_3	ACSR3_4A_2	15:0	RO	0	Bytes 2 and 3 of audio group A channel status for channels 3 and 4.
A52	ACSR3_4A_BYTE4_5	ACSR3_4A_4	15:0	RO	0	Bytes 4 and 5 of audio group A channel status for channels 3 and 4.
A53	ACSR3_4A_BYTE6_7	ACSR3_4A_6	15:0	RO	0	Bytes 6 and 7 of audio group A channel status for channels 3 and 4.
A54	ACSR3_4A_BYTE8_9	ACSR3_4A_8	15:0	RO	0	Bytes 8 and 9 of audio group A channel status for channels 3 and 4.
A55	ACSR3_4A_BYTE10_11	ACSR3_4A_10	15:0	RO	0	Bytes 10 and 11 of audio group A channel status for channels 3 and 4.
A56	ACSR3_4A_BYTE12_13	ACSR3_4A_12	15:0	RO	0	Bytes 12 and 13 of audio group A channel status for channels 3 and 4.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A57	ACSR3_4A_BYTE14_15	ACSR3_4A_14	15:0	RO	0	Bytes 14 and 15 of audio group A channel status for channels 3 and 4.
A58	ACSR3_4A_BYTE16_17	ACSR3_4A_16	15:0	RO	0	Bytes 16 and 17 of audio group A channel status for channels 3 and 4.
A59	ACSR3_4A_BYTE18_19	ACSR3_4A_18	15:0	RO	0	Bytes 18 and 19 of audio group A channel status for channels 3 and 4.
A5A	ACSR3_4A_BYTE20_21	ACSR3_4A_20	15:0	RO	0	Bytes 20 and 21 of audio group A channel status for channels 3 and 4.
A5B	ACSR3_4A_BYTE22	RSVD	15:8	—	—	Reserved.
		ACSR3_4A_22	7:0	RO	0	Bytes 22 of audio group A channel status for channels 3 and 4.
A5C to A5F	RSVD	RSVD	15:0	—	—	Reserved.
A60	ACSR1_2B_BYTE0_1	ACSR1_2B_0	15:0	RO	0	Bytes 0 and 1 of audio group B channel status for channels 1 and 2.
A61	ACSR1_2B_BYTE2_3	ACSR1_2B_2	15:0	RO	0	Bytes 2 and 3 of audio group B channel status for channels 1 and 2.
A62	ACSR1_2B_BYTE4_5	ACSR1_2B_4	15:0	RO	0	Bytes 4 and 5 of audio group B channel status for channels 1 and 2.
A63	ACSR1_2B_BYTE6_7	ACSR1_2B_6	15:0	RO	0	Bytes 6 and 7 of audio group B channel status for channels 1 and 2.
A64	ACSR1_2B_BYTE8_9	ACSR1_2B_8	15:0	RO	0	Bytes 8 and 9 of audio group B channel status for channels 1 and 2.
A65	ACSR1_2B_BYTE10_11	ACSR1_2B_10	15:0	RO	0	Bytes 10 and 11 of audio group B channel status for channels 1 and 2.
A66	ACSR1_2B_BYTE12_13	ACSR1_2B_12	15:0	RO	0	Bytes 12 and 13 of audio group B channel status for channels 1 and 2.
A67	ACSR1_2B_BYTE14_15	ACSR1_2B_14	15:0	RO	0	Bytes 14 and 15 of audio group B channel status for channels 1 and 2.
A68	ACSR1_2B_BYTE16_17	ACSR1_2B_16	15:0	RO	0	Bytes 16 and 17 of audio group B channel status for channels 1 and 2.
A69	ACSR1_2B_BYTE18_19	ACSR1_2B_18	15:0	RO	0	Bytes 18 and 19 of audio group B channel status for channels 1 and 2.
A6A	ACSR1_2B_BYTE20_21	ACSR1_2B_20	15:0	RO	0	Bytes 20 and 21 of audio group B channel status for channels 1 and 2.
A6B	ACSR1_2B_BYTE22	RSVD	15:8	—	—	Reserved.
		ACSR1_2B_22	7:0	RO	0	Bytes 22 of audio group B channel status for channels 1 and 2.
A6C to A6F	RSVD	RSVD	15:0	—	—	Reserved.
A70	ACSR3_4B_BYTE0_1	ACSR3_4B_0	15:0	RO	0	Bytes 0 and 1 of audio group B channel status for channels 3 and 4.

Table 5-9: HD and 3G Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
A71	ACSR3_4B_BYTE2_3	ACSR3_4B_2	15:0	RO	0	Bytes 2 and 3 of audio group B channel status for channels 3 and 4.
A72	ACSR3_4B_BYTE4_5	ACSR3_4B_4	15:0	RO	0	Bytes 4 and 5 of audio group B channel status for channels 3 and 4.
A73	ACSR3_4B_BYTE6_7	ACSR3_4B_6	15:0	RO	0	Bytes 6 and 7 of audio group B channel status for channels 3 and 4.
A74	ACSR3_4B_BYTE8_9	ACSR3_4B_8	15:0	RO	0	Bytes 8 and 9 of audio group B channel status for channels 3 and 4.
A75	ACSR3_4B_BYTE10_11	ACSR3_4B_10	15:0	RO	0	Bytes 10 and 11 of audio group B channel status for channels 3 and 4.
A76	ACSR3_4B_BYTE12_13	ACSR3_4B_12	15:0	RO	0	Bytes 12 and 13 of audio group B channel status for channels 3 and 4.
A77	ACSR3_4B_BYTE14_15	ACSR3_4B_14	15:0	RO	0	Bytes 14 and 15 of audio group B channel status for channels 3 and 4.
A78	ACSR3_4B_BYTE16_17	ACSR3_4B_16	15:0	RO	0	Bytes 16 and 17 of audio group B channel status for channels 3 and 4.
A79	ACSR3_4B_BYTE18_19	ACSR3_4B_18	15:0	RO	0	Bytes 18 and 19 of audio group B channel status for channels 3 and 4.
A7A	ACSR3_4B_BYTE20_21	ACSR3_4B_20	15:0	RO	0	Bytes 20 and 21 of audio group B channel status for channels 3 and 4.
A7B	ACSR3_4B_BYTE22	RSVD	15:8	—	—	Reserved.
		ACSR3_4B_22	7:0	RO	0	Bytes 22 of audio group B channel status for channels 3 and 4.
A7C to A7F	RSVD	RSVD	15:0	—	—	Reserved.
A80 to A96	ACSR_BYTE_0 to ACSR_BYTE_22	RSVD	15:8	—	—	Reserved.
		ACSR0 to ACSR22	7:0	WO	0	Audio channel status to use when ACS_REGEN is set or when adding audio channel status to non-AES/EBU audio. 8 bits per register for 23 registers.
A97 to A99	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-10: SD Audio Core CSR Descriptions

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B00	RSVD	RSVD	15:0	—	—	Reserved.
		RSVD	15:14	—	—	Reserved.
		ALL_DEL	13:13	RW	0	Selects deletion of all audio data including audio control packets and data packets: 0 = Doesn't delete existing audio data 1 = Deletes existing audio data
		MUTE_ALL	12:12	RW	0	Mute all output channels: 0 = Normal 1 = Muted
		ACS_USE_SECOND	11:11	RW	0	Extract Audio Channel Status from second channel pair.
		CLEAR_AUDIO	10:10	RW	0	Clears all audio FIFO buffers and puts them in start-up state.
		OS_SEL	9:8	RW	0	Specifies the audio FIFO buffer size: 00 _b = 36 samples deep, 26 sample start-up count 01 _b = 22 samples deep, 12 sample start-up count 10 _b = 16 samples deep, 6 sample start-up count 11 _b = Reserved
		RSVD	7:4	—	—	Reserved.
		IDB	3:2	RW	1	Specifies the Secondary audio group to extract: 00 _b = Audio group #1 01 _b = Audio group #2 10 _b = Audio group #3 11 _b = Audio group #4 Note: Should IDA and IDB be set to the same value, they automatically revert to their default values
		IDA	1:0	RW	0	Specifies the Primary audio group to extract: 00 _b = Audio group #1 01 _b = Audio group #2 10 _b = Audio group #3 11 _b = Audio group #4 Note: Should IDA and IDB be set to the same value, they automatically revert to their default values
B01	CFG_AUD					
B02	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B03	AUDIO_STATUS	EXT_DET3_4B	15:15	ROCW	0	Set when Secondary group channels 3 and 4 have extended data. Write 1 to clear.
		EXT_DET1_2B	14:14	ROCW	0	Set when Secondary group channels 1 and 2 have extended data. Write 1 to clear.
		EXT_DET3_4A	13:13	ROCW	0	Set when Primary group channels 3 and 4 have extended data. Write 1 to clear
		EXT_DET1_2A	12:12	ROCW	0	Set when Primary group channels 1 and 2 have extended data. Write 1 to clear
		CTL_DBNB_ERR	11:11	ROCW	0	Set when Secondary group control packet Data Block Number sequence is discontinuous. Write 1 to clear
		CTL_DBNA_ERR	10:10	ROCW	0	Set when Primary group control packet Data Block Number sequence is discontinuous. Write 1 to clear.
		EXT_DBNB_ERR	9:9	ROCW	0	Set when Secondary group extended data packet Data Block Number sequence is discontinuous. Write 1 to clear.
		EXT_DBNA_ERR	8:8	ROCW	0	Set when Primary group extended data packet Data Block Number sequence is discontinuous. Write 1 to clear.
		DBNB_ERR	7:7	ROCW	0	Set when Secondary group data packet Data Block Number sequence is discontinuous. Write 1 to clear.
		DBNA_ERR	6:6	ROCW	0	Set when Primary group data packet Data Block Number sequence is discontinuous. Write 1 to clear.
		CTRB_DET	5:5	ROCW	0	Set when Secondary group audio control packet is detected. Write 1 to clear.
		CTRA_DET	4:4	ROCW	0	Set when Primary group audio control packet is detected. Write 1 to clear.
		ACS_DET3_4B	3:3	ROCW	0	Secondary group audio status detected for channels 3 and 4. Write 1 to clear.
		ACS_DET1_2B	2:2	ROCW	0	Secondary group audio status detected for channels 1 and 2. Write 1 to clear.
		ACS_DET3_4A	1:1	ROCW	0	Primary group audio status detected for channels 3 and 4. Write 1 to clear.
		ACS_DET1_2A	0:0	ROCW	0	Primary group audio status detected for channels 1 and 2. Write 1 to clear.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B04	REGEN	RSVD	15:2	—	—	Reserved.
		ACS_APPLY	1:1	RW	0	Causes channel status data in ACSR[183:0] to be transferred to the channel status replacement mechanism. The transfer does not occur until the next status boundary.
		ACS_REGEN	0:0	RW	0	Specifies that Audio Channel Status of all channels should be replaced with ACSR[183:0] field: 0 = Do not replace Channel Status 1 = Replace Channel Status of all channels
B05	AUD_DET	IDB_READBACK	15:14	RO	1	Actual value of IDB in the hardware.
		IDA_READBACK	13:12	RO	0	Actual value of IDA in the hardware.
		XDPG4_DET	11:11	RO	0	Set while embedded Group 4 audio extended packets are detected.
		XDPG3_DET	10:10	RO	0	Set while embedded Group 3 audio extended packets are detected.
		XDPG2_DET	9:9	RO	0	Set while embedded Group 2 audio extended packets are detected.
		XDPG1_DET	8:8	RO	0	Set while embedded Group 1 audio extended packets are detected.
		ADPG4_DET	7:7	RO	0	Set while Group 4 audio data packets are detected.
		ADPG3_DET	6:6	RO	0	Set while Group 3 audio data packets are detected.
		ADPG2_DET	5:5	RO	0	Set while Group 2 audio data packets are detected.
		ADPG1_DET	4:4	RO	0	Set while Group 1 audio data packets are detected.
		ACS_APPLY_WAITD	3:3	RO	0	Set while output channels 7 and 8 are waiting for a status boundary to apply the ACSR[183:0] data.
		ACS_APPLY_WAITC	2:2	RO	0	Set while output channels 5 and 6 are waiting for a status boundary to apply the ACSR[183:0] data
		ACS_APPLY_WAITB	1:1	RO	0	Set while output channels 3 and 4 are waiting for a status boundary to apply the ACSR[183:0] data
		ACS_APPLY_WAITA	0:0	RO	0	Set while output channels 1 and 2 are waiting for a status boundary to apply the ACSR[183:0] data

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B06	CSUM_ERR_DET	RSVD	15:1	—	—	Reserved.
		CSUM_ERROR	0:0	ROCW	0	Embedded packet checksum error detected. Write 1 to clear.
B07	CH_MUTE	RSVD	15:8	—	—	Reserved.
		MUTE	7:0	RW	0	Mutes output channels 8 to 1 Bits [7:0] = Channels 8:1. 0 = Normal 1 = Mute
B08	CH_VALID	RSVD	15:8	—	—	Reserved.
		CH4_VALIDB	7:7	RO	0	Secondary group channel 4 sample validity flag.
		CH3_VALIDB	6:6	RO	0	Secondary group channel 3 sample validity flag.
		CH2_VALIDB	5:5	RO	0	Secondary group channel 2 sample validity flag.
		CH1_VALIDB	4:4	RO	0	Secondary group channel 1 sample validity flag.
		CH4_VALIDA	3:3	RO	0	Primary group channel 4 sample validity flag.
		CH3_VALIDA	2:2	RO	0	Primary group channel 3 sample validity flag.
		CH2_VALIDA	1:1	RO	0	Primary group channel 2 sample validity flag.
		CH1_VALIDA	0:0	RO	0	Primary group channel 1 sample validity flag.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B09	AUDIO_INT_ENABLE	RSVD	15:15	—	—	Reserved.
		EN_NOT_LOCKED	14:14	RW	0	Asserts interrupt when LOCKED signal is not asserted.
		EN_NO_VIDEO	13:13	RW	0	Asserts interrupt when video format is unknown.
		EN_CSUM_ERROR	12:12	RW	0	Asserts interrupt when checksum error is detected.
		EN_ACS_DET3_4B	11:11	RW	0	Asserts interrupt when ACS_DET3_4B flag is set.
		EN_ACS_DET1_2B	10:10	RW	0	Asserts interrupt when ACS_DET1_2B flag is set.
		EN_ACS_DET3_4A	9:9	RW	0	Asserts interrupt when ACS_DET3_4A flag is set.
		EN_ACS_DET1_2A	8:8	RW	0	Asserts interrupt when ACS_DET1_2A flag is set.
		EN_CTRB_DET	7:7	RW	0	Asserts interrupt when CTRB_DET flag is set.
		EN_CTRA_DET	6:6	RW	0	Asserts interrupt when CTRA_DET flag is set.
		EN_DBNB_ERR	5:5	RW	0	Asserts interrupt when DBNB_ERR flag is set.
		EN_DBNA_ERR	4:4	RW	0	Asserts interrupt when DBNA_ERR flag is set.
		EN_ADPG4_DET	3:3	RW	0	Asserts interrupt when ADPG4_DET flag is set.
		EN_ADPG3_DET	2:2	RW	0	Asserts interrupt when ADPG3_DET flag is set.
		EN_ADPG2_DET	1:1	RW	0	Asserts interrupt when ADPG2_DET flag is set.
		EN_ADPG1_DET	0:0	RW	0	Asserts interrupt when ADPG1_DET flag is set.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
BOA	CFG_OUTPUT	ASWLD	15:14	RW	3	Output channels 7 and 8 word length: 00 _b = 24 bits 01 _b = 20 bits 10 _b = 16 bits 11 _b = Automatic 20-bit or 24-bit
		ASWLC	13:12	RW	3	Output channels 5 and 6 word length. See ASWLD for decoding.
		ASWLB	11:10	RW	3	Output channels 3 and 4 word length. See ASWLD for decoding.
		ASWLA	9:8	RW	3	Output channels 1 and 2 word length. See ASWLD for decoding.
		AMD	7:6	RW	3	Output channels 7 and 8 format selector: 00 _b = AES/EBU audio output 01 _b = Serial audio output. Left justified; MSB first 10 _b = Serial audio output. Right justified; MSB first 11 _b = I ² S serial audio output
		AMC	5:4	RW	3	Output channels 5 and 6 format selector. See AMD for decoding.
		AMB	3:2	RW	3	Output channels 3 and 4 format selector. See AMD for decoding.
		AMA	1:0	RW	3	Output channels 1 and 2 format selector. See AMD for decoding.
		RSVD	15:12	—	—	Reserved.
BOB	OUTPUT_SEL_1	OP4_SRC	11:9	RW	3	Output channel 4 source selector: 000 _b = Primary audio group channel 1 001 _b = Primary audio group channel 2 010 _b = Primary audio group channel 3 011 _b = Primary audio group channel 4 100 _b = Secondary audio group channel 1 101 _b = Secondary audio group channel 2 110 _b = Secondary audio group channel 3 111 _b = Secondary audio group channel 4
		OP3_SRC	8:6	RW	2	Output channel 3 source selector. See OP4_SRC for decoding.
		OP2_SRC	5:3	RW	1	Output channel 2 source selector. See OP4_SRC for decoding.
		OP1_SRC	2:0	RW	0	Output channel 1 source selector. See OP4_SRC for decoding.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B0C	OUTPUT_SEL_2	RSVD	15:12	—	—	Reserved.
		OP8_SRC	11:9	RW	7	Output channel 8 source selector. See OP4_SRC for decoding.
		OP7_SRC	8:6	RW	6	Output channel 7 source selector. See OP4_SRC for decoding.
		OP6_SRC	5:3	RW	5	Output channel 6 source selector. See OP4_SRC for decoding.
		OP5_SRC	2:0	RW	4	Output channel 5 source selector. See OP4_SRC for decoding.
B0D to B1F	RSVD	RSVD	15:0	—	—	Reserved.
B20	AFNA12	RSVD	15:9	—	—	Reserved.
		AFN1_2A	8:0	RO	0	Primary group audio frame number for channels 1 and 2.
B21	AFNA34	RSVD	15:9	—	—	Reserved.
		AFN3_4A	8:0	RO	0	Primary group audio frame number for channels 3 and 4.
B22	RATEA	RSVD	15:8	—	—	Reserved.
		RATE3_4A	7:5	RO	0	Primary group sampling frequency for channels 3 and 4.
		ASX3_4A	4:4	RO	0	Primary group asynchronous mode for channels 3 and 4.
		RATE1_2A	3:1	RO	0	Primary group sampling frequency for channels 1 and 2.
		ASX1_2A	0:0	RO	0	Primary group asynchronous mode for channels 1 and 2.
B23	ACT_A	RSVD	15:4	—	—	Reserved.
		ACTA	3:0	RO	0	Primary group active channels.
B24	PRIM_AUD_DELAY_1	RSVD	15:9	—	—	Reserved.
		DEL1A_1	8:1	RO	0	Primary Audio group delay data for channel 1[7:0].
		EBIT1A	0:0	RO	0	Primary Audio group delay data valid flag for channel 1.
B25	PRIM_AUD_DELAY_2	RSVD	15:9	—	—	Reserved.
		DEL1A_2	8:0	RO	0	Primary Audio group delay data for channel 1[16:8].
B26	PRIM_AUD_DELAY_3	RSVD	15:9	—	—	Reserved.
		DEL1A_3	8:0	RO	0	Primary Audio group delay data for channel 1[25:17].

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B27	PRIM_AUD_DELAY_4	RSVD	15:9	—	—	Reserved.
		DEL2A_4	8:1	RO	0	Primary Audio group delay data for channel 2[7:0].
		EBIT2A	0	RO	0	Primary Audio group delay data valid flag for channel 2.
B28	PRIM_AUD_DELAY_5	RSVD	15:9	—	—	Reserved.
		DEL2A_5	8:0	RO	0	Primary Audio group delay data for channel 2[16:8].
B29	PRIM_AUD_DELAY_6	RSVD	15:9	—	—	Reserved.
		DEL2A_6	8:0	RO	0	Primary Audio group delay data for channel 2[25:17].
B2A	PRIM_AUD_DELAY_7	RSVD	15:9	—	—	Reserved.
		DEL3A_7	8:1	RO	0	Primary Audio group delay data for channel 3[7:0].
		EBIT3A	0	RO	0	Primary Audio group delay data valid flag for channel 3.
B2B	PRIM_AUD_DELAY_8	RSVD	15:9	—	—	Reserved.
		DEL3A_8	8:0	RO	0	Primary Audio group delay data for channel 3[16:8].
B2C	PRIM_AUD_DELAY_9	RSVD	15:9	—	—	Reserved.
		DEL3A_9	8:0	RO	0	Primary Audio group delay data for channel 3[25:17].
B2D	PRIM_AUD_DELAY_10	RSVD	15:9	—	—	Reserved.
		DEL4A_10	8:1	RO	0	Primary Audio group delay data for channel 4[7:0].
		EBIT4A	0:0	RO	0	Primary Audio group delay data valid flag for channel 4.
B2E	PRIM_AUD_DELAY_11	RSVD	15:9	—	—	Reserved.
		DEL4A_11	8:0	RO	0	Primary Audio group delay data for channel 4[16:8].
B2F	PRIM_AUD_DELAY_12	RSVD	15:9	—	—	Reserved.
		DEL4A_12	8:0	RO	0	Primary Audio group delay data for channel 4[25:17].
B30	AFNB12	RSVD	15:9	—	—	Reserved.
		AFN1_2B	8:0	RO	0	Secondary group audio frame number for channels 1 and 2.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B31	AFNB34	RSVD	15:9	—	—	Reserved.
		AFN3_4B	8:0	RO	0	Secondary group audio frame number for channels 3 and 4.
B32	RATEB	RSVD	15:8	—	—	Reserved.
		RATE3_4B	7:5	RO	0	Secondary group sampling frequency for channels 3 and 4.
		ASX3_4B	4:4	RO	0	Secondary group asynchronous mode for channels 3 and 4.
		RATE1_2B	3:1	RO	0	Secondary group sampling frequency for channels 1 and 2.
		ASXB	0:0	RO	0	Secondary group asynchronous mode for channels 1 and 2.
B33	ACT_B	RSVD	15:4	—	—	Reserved.
		ACTB	3:0	RO	0	Secondary group active channels.
B34	SEC_AUD_DELAY_1	RSVD	15:9	—	—	Reserved.
		DEL1B_1	8:1	RO	0	Secondary Audio group delay data for channel 1[7:0].
		EBIT1B	0:0	RO	0	Secondary Audio group delay data valid flag for channel 1.
B35	SEC_AUD_DELAY_2	RSVD	15:9	—	—	Reserved.
		DEL1B_2	8:0	RO	0	Secondary Audio group delay data for channel 1[16:8].
B36	SEC_AUD_DELAY_3	RSVD	15:9	—	—	Reserved.
		DEL1B_3	8:0	RO	0	Secondary Audio group delay data for channel 1[25:17].
B37	SEC_AUD_DELAY_4	RSVD	15:9	—	—	Reserved.
		DEL2B_4	8:1	RO	0	Secondary Audio group delay data for channel 2[7:0].
		EBIT2B	0:0	RO	0	Secondary Audio group delay data valid flag for channel 2.
B38	SEC_AUD_DELAY_5	RSVD	15:9	—	—	Reserved.
		DEL2B_5	8:0	RO	0	Secondary Audio group delay data for channel 2[16:8].
B39	SEC_AUD_DELAY_6	RSVD	15:9	—	—	Reserved.
		DEL2B_6	8:0	RO	0	Secondary Audio group delay data for channel 2[25:17].

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B3A	SEC_AUD_DELAY_7	RSVD	15:9	—	—	Reserved.
		DEL3B_7	8:1	RO	0	Secondary Audio group delay data for channel 3[7:0].
		EBIT3B	0	RO	0	Secondary Audio group delay data valid flag for channel 3.
B3B	SEC_AUD_DELAY_8	RSVD	15:9	—	—	Reserved.
		DEL3B_8	8:0	RO	0	Secondary Audio group delay data for channel 3[16:8].
B3C	SEC_AUD_DELAY_9	RSVD	15:9	—	—	Reserved.
		DEL3B_9	8:0	RO	0	Secondary Audio group delay data for channel 3[25:17].
B3D	SEC_AUD_DELAY_10	RSVD	15:9	—	—	Reserved.
		DEL4B_10	8:1	RO	0	Secondary Audio group delay data for channel 4[7:0].
		EBIT4B	0	RO	0	Secondary Audio group delay data valid flag for channel 4.
B3E	SEC_AUD_DELAY_11	RSVD	15:9	—	—	Reserved.
		DEL4B_11	8:0	RO	0	Secondary Audio group delay data for channel 4[16:8].
B3F	SEC_AUD_DELAY_12	RSVD	15:9	—	—	Reserved.
		DEL4B_12	8:0	RO	0	Secondary Audio group delay data for channel [25:17].
B40	ACSR1_2A_BYTE0_1	ACSR1_2A_0	15:0	RO	0	Bytes 0 [7:0] and 1 [15:8] of audio group A channel status for channels 1 and 2.
B41	ACSR1_2A_BYTE2_3	ACSR1_2A_2	15:0	RO	0	Bytes 2 [7:0] and 3 [15:8] of audio group A channel status for channels 1 and 2.
B42	ACSR1_2A_BYTE4_5	ACSR1_2A_4	15:0	RO	0	Bytes 4 [7:0] and 5 [15:8] of audio group A channel status for channels 1 and 2.
B43	ACSR1_2A_BYTE6_7	ACSR1_2A_6	15:0	RO	0	Bytes 6 [7:0] and 7 [15:8] of audio group A channel status for channels 1 and 2.
B44	ACSR1_2A_BYTE8_9	ACSR1_2A_8	15:0	RO	0	Bytes 8 [7:0] and 9 [15:8] of audio group A channel status for channels 1 and 2.
B45	ACSR1_2A_BYTE10_11	ACSR1_2A_10	15:0	RO	0	Bytes 10 [7:0] and 11 [15:8] of audio group A channel status for channels 1 and 2.
B46	ACSR1_2A_BYTE12_13	ACSR1_2A_12	15:0	RO	0	Bytes 12 [7:0] and 13 [15:8] of audio group A channel status for channels 1 and 2.
B47	ACSR1_2A_BYTE14_15	ACSR1_2A_14	15:0	RO	0	Bytes 14 [7:0] and 15 [15:8] of audio group A channel status for channels 1 and 2.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B48	ACSR1_2A_BYTE16_17	ACSR1_2A_16	15:0	RO	0	Bytes 16 [7:0] and 17 [15:8] of audio group A channel status for channels 1 and 2.
B49	ACSR1_2A_BYTE18_19	ACSR1_2A_18	15:0	RO	0	Bytes 18 [7:0] and 19 [15:8] of audio group A channel status for channels 1 and 2.
B4A	ACSR1_2A_BYTE20_21	ACSR1_2A_20	15:0	RO	0	Bytes 20 [7:0] and 21 [15:8] of audio group A channel status for channels 1 and 2.
B4B	ACSR1_2A_BYTE22	RSVD	15:8	—	—	Reserved.
		ACSR1_2A_22	7:0	RO	0	Bytes 22 of audio group A channel status for channels 1 and 2.
B4C to B4F	RSVD	RSVD	15:0	—	—	Reserved.
B50	ACSR3_4A_BYTE0_1	ACSR3_4A_0	15:0	RO	0	Bytes 0 [7:0] and 1 [15:8] of audio group A channel status for channels 3 and 4.
B51	ACSR3_4A_BYTE2_3	ACSR3_4A_2	15:0	RO	0	Bytes 2 [7:0] and 3 [15:8] of audio group A channel status for channels 3 and 4.
B52	ACSR3_4A_BYTE4_5	ACSR3_4A_4	15:0	RO	0	Bytes 4 [7:0] and 5 [15:8] of audio group A channel status for channels 3 and 4.
B53	ACSR3_4A_BYTE6_7	ACSR3_4A_6	15:0	RO	0	Bytes 6 [7:0] and 7 [15:8] of audio group A channel status for channels 3 and 4.
B54	ACSR3_4A_BYTE8_9	ACSR3_4A_8	15:0	RO	0	Bytes 8 [7:0] and 9 [15:8] of audio group A channel status for channels 3 and 4.
B55	ACSR3_4A_BYTE10_11	ACSR3_4A_10	15:0	RO	0	Bytes 10 [7:0] and 11 [15:8] of audio group A channel status for channels 3 and 4.
B56	ACSR3_4A_BYTE12_13	ACSR3_4A_12	15:0	RO	0	Bytes 12 [7:0] and 13 [15:8] of audio group A channel status for channels 3 and 4.
B57	ACSR3_4A_BYTE14_15	ACSR3_4A_14	15:0	RO	0	Bytes 14 [7:0] and 15 [15:8] of audio group A channel status for channels 3 and 4.
B58	ACSR3_4A_BYTE16_17	ACSR3_4A_16	15:0	RO	0	Bytes 16 [7:0] and 17 [15:8] of audio group A channel status for channels 3 and 4.
B59	ACSR3_4A_BYTE18_19	ACSR3_4A_18	15:0	RO	0	Bytes 18 [7:0] and 19 [15:8] of audio group A channel status for channels 3 and 4.
B5A	ACSR3_4A_BYTE20_21	ACSR3_4A_20	15:0	RO	0	Bytes 20 [7:0] and 21 [15:8] of audio group A channel status for channels 3 and 4.
B5B	ACSR3_4A_BYTE22	RSVD	15:8	—	—	Reserved.
		ACSR3_4A_22	7:0	RO	0	Bytes 22 of audio group A channel status for channels 3 and 4.
B5C to B5F	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B60	ACSR1_2B_BYTE0_1	ACSR1_2B_0	15:0	RO	0	Bytes 0 [7:0] and 1 [15:8] of audio group B channel status for channels 1 and 2.
B61	ACSR1_2B_BYTE2_3	ACSR1_2B_2	15:0	RO	0	Bytes 2 [7:0] and 3 [15:8] of audio group B channel status for channels 1 and 2.
B62	ACSR1_2B_BYTE4_5	ACSR1_2B_4	15:0	RO	0	Bytes 4 [7:0] and 5 [15:8] of audio group B channel status for channels 1 and 2.
B63	ACSR1_2B_BYTE6_7	ACSR1_2B_6	15:0	RO	0	Bytes 6 [7:0] and 7 [15:8] of audio group B channel status for channels 1 and 2.
B64	ACSR1_2B_BYTE8_9	ACSR1_2B_8	15:0	RO	0	Bytes 8 [7:0] and 9 [15:8] of audio group B channel status for channels 1 and 2.
B65	ACSR1_2B_BYTE10_11	ACSR1_2B_10	15:0	RO	0	Bytes 10 [7:0] and 11 [15:8] of audio group B channel status for channels 1 and 2.
B66	ACSR1_2B_BYTE12_13	ACSR1_2B_12	15:0	RO	0	Bytes 12 [7:0] and 13 [15:8] of audio group B channel status for channels 1 and 2.
B67	ACSR1_2B_BYTE14_15	ACSR1_2B_14	15:0	RO	0	Bytes 14 [7:0] and 15 [15:8] of audio group B channel status for channels 1 and 2.
B68	ACSR1_2B_BYTE16_17	ACSR1_2B_16	15:0	RO	0	Bytes 16 [7:0] and 17 [15:8] of audio group B channel status for channels 1 and 2.
B69	ACSR1_2B_BYTE18_19	ACSR1_2B_18	15:0	RO	0	Bytes 18 [7:0] and 19 [15:8] of audio group B channel status for channels 1 and 2.
B6A	ACSR1_2B_BYTE20_21	ACSR1_2B_20	15:0	RO	0	Bytes 20 [7:0] and 21 [15:8] of audio group B channel status for channels 1 and 2.
B6B	ACSR1_2B_BYTE22	RSVD	15:8	—	—	Reserved.
		ACSR1_2B_22	7:0	RO	0	Byte 22 of audio group B channel status for channels 1 and 2.
B6C to B6F	RSVD	RSVD	15:0	—	—	Reserved.
B70	ACSR3_4B_BYTE0_1	ACSR3_4B_0	15:0	RO	0	Bytes 0 [7:0] and 1 [15:8] of audio group B channel status for channels 3 and 4.
B71	ACSR3_4B_BYTE2_3	ACSR3_4B_2	15:0	RO	0	Bytes 2 [7:0] and 3 [15:8] of audio group B channel status for channels 3 and 4.
B72	ACSR3_4B_BYTE4_5	ACSR3_4B_4	15:0	RO	0	Bytes 4 [7:0] and 5 [15:8] of audio group B channel status for channels 3 and 4.
B73	ACSR3_4B_BYTE6_7	ACSR3_4B_6	15:0	RO	0	Bytes 6 [7:0] and 7 [15:8] of audio group B channel status for channels 3 and 4.
B74	ACSR3_4B_BYTE8_9	ACSR3_4B_8	15:0	RO	0	Bytes 8 [7:0] and 9 [15:8] of audio group B channel status for channels 3 and 4.

Table 5-10: SD Audio Core CSR Descriptions (Continued)

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
B75	ACSR3_4B_BYTE10_11	ACSR3_4B_10	15:0	RO	0	Bytes 10 [7:0] and 11 [15:8] of audio group B channel status for channels 3 and 4.
B76	ACSR3_4B_BYTE12_13	ACSR3_4B_12	15:0	RO	0	Bytes 12 [7:0] and 13 [15:8] of audio group B channel status for channels 3 and 4.
B77	ACSR3_4B_BYTE14_15	ACSR3_4B_14	15:0	RO	0	Bytes 14 [7:0] and 15 [15:8] of audio group B channel status for channels 3 and 4.
B78	ACSR3_4B_BYTE16_17	ACSR3_4B_16	15:0	RO	0	Bytes 16 [7:0] and 17 [15:8] of audio group B channel status for channels 3 and 4.
B79	ACSR3_4B_BYTE18_19	ACSR3_4B_18	15:0	RO	0	Bytes 18 [7:0] and 19 [15:8] of audio group B channel status for channels 3 and 4.
B7A	ACSR3_4B_BYTE20_21	ACSR3_4B_20	15:0	RO	0	Bytes 20 [7:0] and 21 [15:8] of audio group B channel status for channels 3 and 4.
B7B	ACSR3_4B_BYTE22	ACSR3_4B_22	7:0	RO	0	Byte 22 of audio group B channel status for channels 3 and 4.
B7C to B7F	RSVD	RSVD	15:0	—	—	Reserved.
	RSVD	RSVD	15:8	—	—	Reserved.
B80 to B96	ACSR_BYTE_0 to ACSR_BYTE_22	ACSR0 to ACSR22	7:0	WO	0	Audio channel status to use when ACS_REGEN is set or when adding audio channel status to non-AES/EBU audio. 8 bits per register for 23 registers.
B97	RSVD	RSVD	15:0	—	—	Reserved.

Table 5-11: ANC Extraction FIFO Access CSR Descriptions

Address _h	Register Name	Parameter Name	Bit	R/W	Reset Value	Description
C00 to FFF	ANC_FIFO_0 to ANC_FIFO_1023	ANC_FIFO_0 to ANC_FIFO_1023	15:0	RO	0	Top of the ANC FIFO. Register addresses shared with the bottom of the ANC FIFO registers. ANC_DATA_SWITCH (817_h) is used to switch between the top and bottom registers of the ANC FIFO. Please refer to the 4.18.9 Ancillary Data Extraction.

Table 5-12: SMPTE 352M Packet Updates with REGEN_352_MASK is LOW

SMPTE 425M Video Format	Incoming SMPTE 352M Byte 1 Value for 3Gb/s stream_h	Outgoing SMPTE 352M Byte 1 Value for 1.5Gb/s stream_h
Level B Dual SMPTE 274M 1125 Line	8C _h	85 _h
Level B Dual SMPTE 296M 750 Line	8B _h	84 _h
Level B Dual SD 525/625 Line	8D _h	86 _h
Level B SMPTE 372M Dual Link	8A _h	87 _h

6.1 Typical Application Circuit

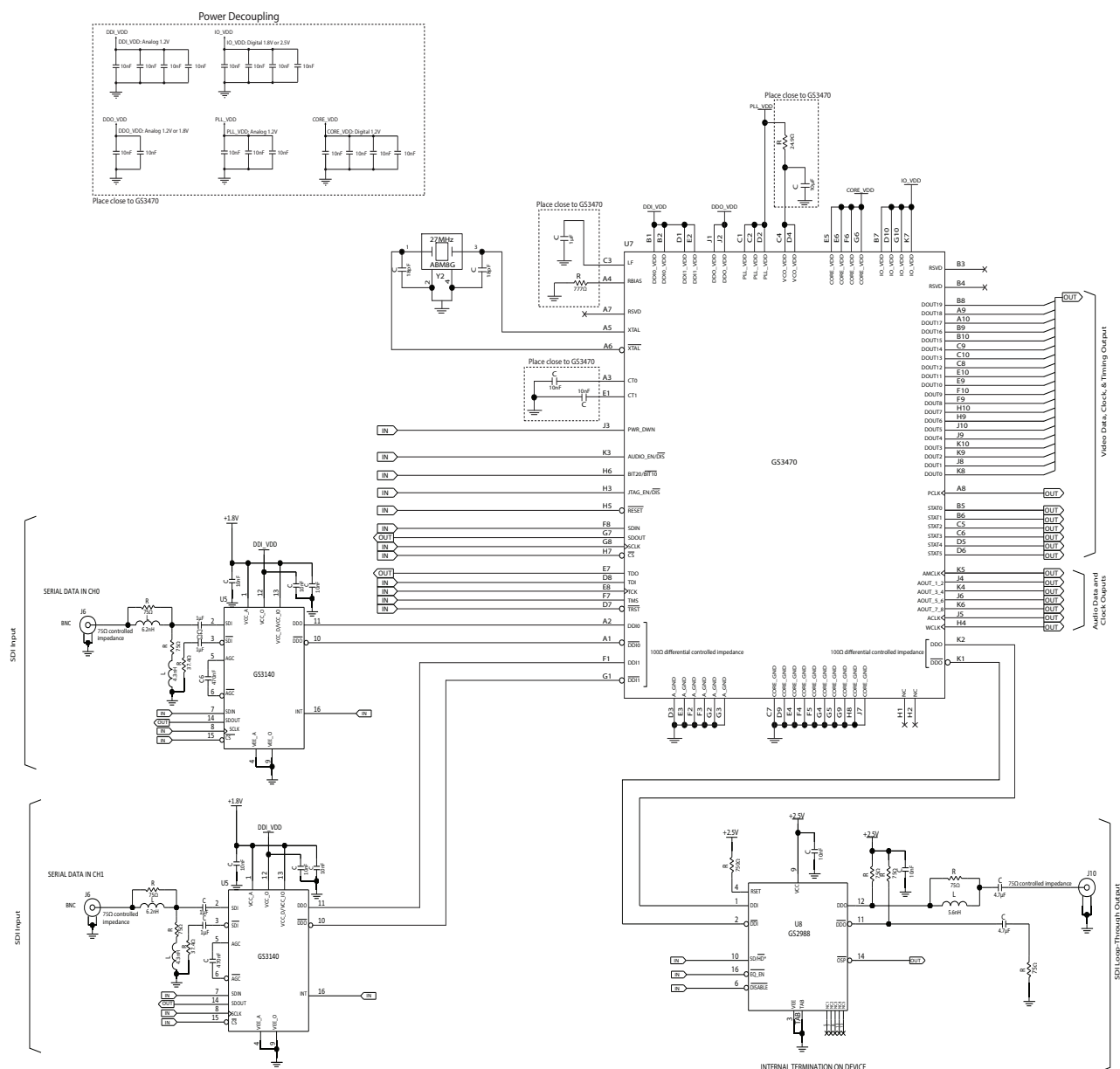


Figure 6-1: Typical Application Circuit

6.2 Layout Considerations

General best practices for high-speed board layout should be followed when designing with the GS3470. These general best practices can be found in a number of Semtech design and user guides including:

- GS3470 Evaluation Board User Guide (PDS-061161)
- Multi-Gigabit Serial Routing Products (PDS-060967)
- GS2989 Cable Driver Design Guide (GENDOC-052070) or most recent CD design guide available

The following layout considerations are specific to the GS3470:

- GS3470's analog supplies, particularly its 1.2V VCC supply, are sensitive to noise. For optimal performance, it is recommended that the power supplies are isolated to avoid external noise coupling. **DDI_VDD** and **DDO_VDD** may be combined if they are powered at the same supply level. However, **PLL_VDD**, **CORE_VDD** and **IO_VDD** should be routed separately.
- Linearly regulated power supplies are recommended wherever possible as they provide the cleanest power.
- Critical components such as power supply decoupling capacitors, LF capacitor, RBIAS resistor, CT0 and CT1 capacitors and VCO filter should be placed as close as possible to the GS3470.
- $\overline{DDI0}/DDI0$, $\overline{DDI1}/DDI1$, and $\overline{DDO}/DDO$ differential inputs and outputs should be routed using 100Ω differential controlled impedance transmission lines.
- When paired with a compatible cable equalizer or cable driver, it is possible to DC-couple the differential high-speed inputs and outputs. In cases requiring AC-coupling, anti-pads should be used underneath AC-coupling capacitors to minimize the parasitic capacitance associated with these components. Details on the use of anti-pads can be found in the Semtech video design guides referenced above.
- The parallel data outputs and timing signals should be electrically phase matched to within 1/20th of a wavelength. For 300MHz, this is approximately equivalent to 1" of FR4 micro-strip.
- Serial termination on the parallel data outputs and timing signals may be required for very long traces or traces going through connectors.

7. References & Relevant Standards

SMPTE ST 125	Component video signal 4:2:2 – bit parallel interface
SMPTE ST 259	10-bit 4:2:2 Component and 4fsc Composite Digital Signals - Serial Digital Interface
SMPTE ST 260	1125 / 60 high definition production system – digital representation and bit parallel interface
SMPTE ST 267	Bit parallel digital interface – component video signal 4:2:2 16 x 9 aspect ratio
SMPTE ST 272	Formatting AES/EBU Audio and Auxiliary Data into Digital Video Ancillary Data Space
SMPTE ST 274	1920 x 1080 scanning analog and parallel digital interfaces for multiple picture rates
SMPTE ST 291	Ancillary Data Packet and Space Formatting
SMPTE 292	Bit-Serial Digital Interface for High-Definition Television Systems
SMPTE ST 293	720 x 483 active line at 59.94Hz progressive scan production – digital representation
SMPTE ST 296	1280 x 720 scanning, analog and digital representation and analog interface
SMPTE ST 299	24-Bit Digital Audio Format for HDTV Bit-Serial Interface
SMPTE ST 352	Video Payload Identification for Digital Television Interfaces
SMPTE ST 424	Television - 3Gb/s Signal/Data Serial Interface
SMPTE ST 425	Television - 3Gb/s Signal/Data Serial Interface - Source Image Format Mapping
SMPTE RP165	Error Detection Checkwords and Status Flags for Use in Bit-Serial Digital Interfaces for Television
SMPTE RP168	Definition of Vertical Interval Switching Point for Synchronous Video Switching

8. Package & Ordering Information

8.1 Package Dimensions

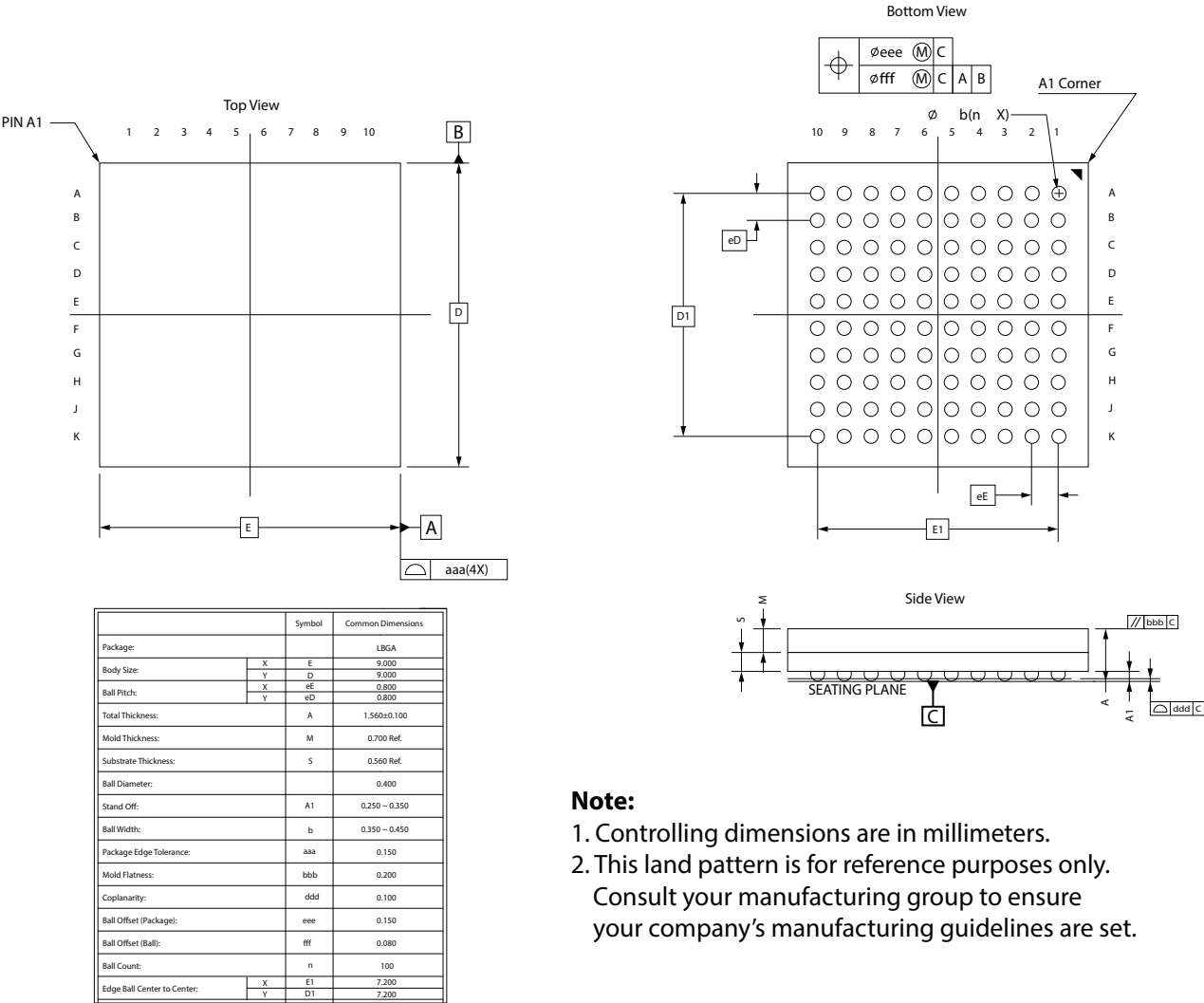


Figure 8-1: Package Dimensions

Note:

1. Controlling dimensions are in millimeters.
2. This land pattern is for reference purposes only. Consult your manufacturing group to ensure your company's manufacturing guidelines are set.

8.2 Recommended PCB Footprint

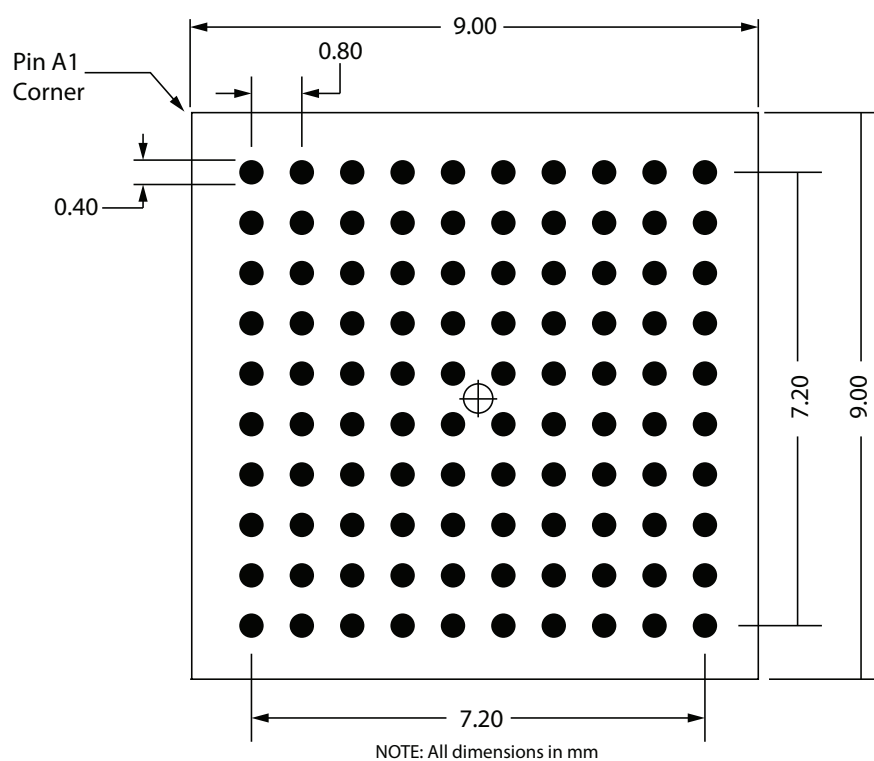


Figure 8-2: Recommended PCB Footprint

8.3 Packaging Data

Table 8-1: Packaging Data

Parameter	Value
Package Type	9mm x 9mm 100-ball LBGA package (0.80mm Ball Pitch)
Package Drawing Reference	JEDEC M0192 (with exceptions noted in Package Dimensions).
Moisture Sensitivity Level	3
Junction to Case Thermal Resistance, θ_{j-c}	28°C/W
Junction to Air Thermal Resistance, θ_{j-a} (at zero airflow)	48°C/W
Junction to Board Thermal Resistance, θ_{j-b}	41°C/W
Pb-free and RoHS Compliant	Yes

8.4 Marking Diagram



Figure 8-3: Marking Diagram

8.5 Solder Reflow Profile

The GS3470 is available in a Pb-free package. It is recommended that the Pb-free package be soldered with Pb-free paste using the reflow profile shown in Figure.

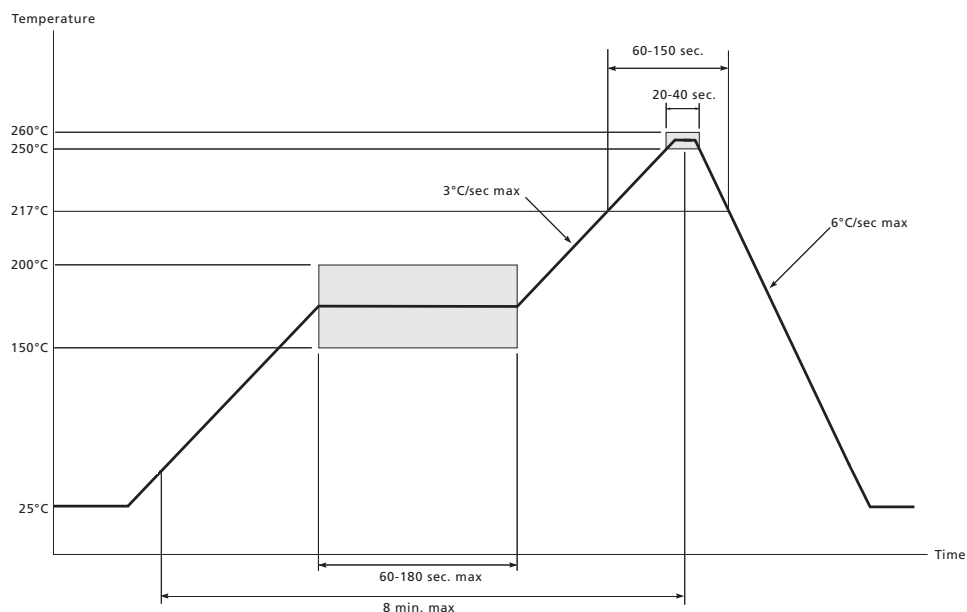


Figure 8-4: Pb-free Solder Reflow Profile

8.6 Ordering Information

Table 8-2: Ordering Information

Part Number	Minimum Order Quantity	Format
GS3470-IBE3	260	Tray
GS3470-IBTE3Z	2500	Tape and Reel



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